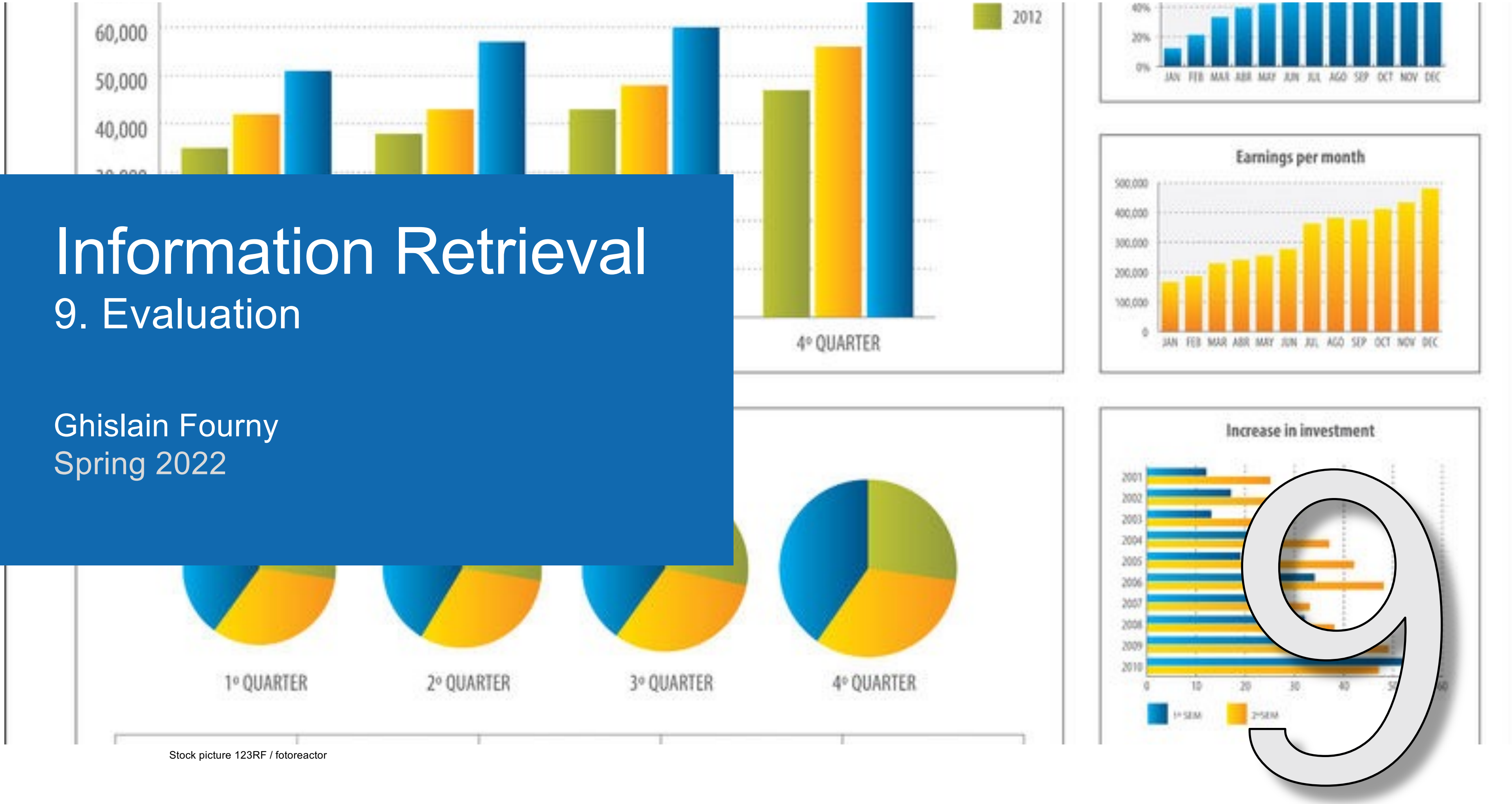
# Information Retrieval

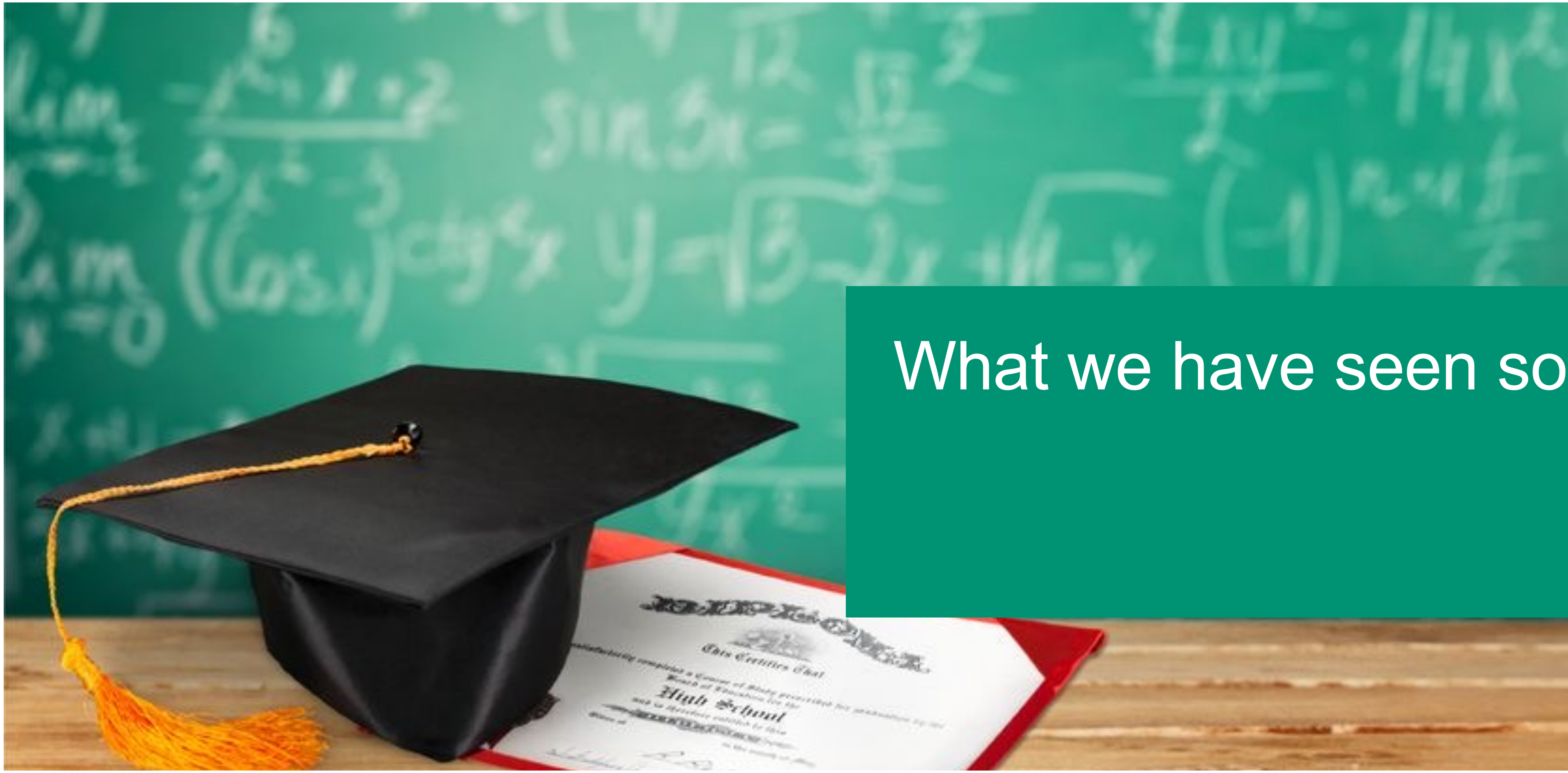
## 9. Evaluation

Ghislain Fourny
Spring 2022



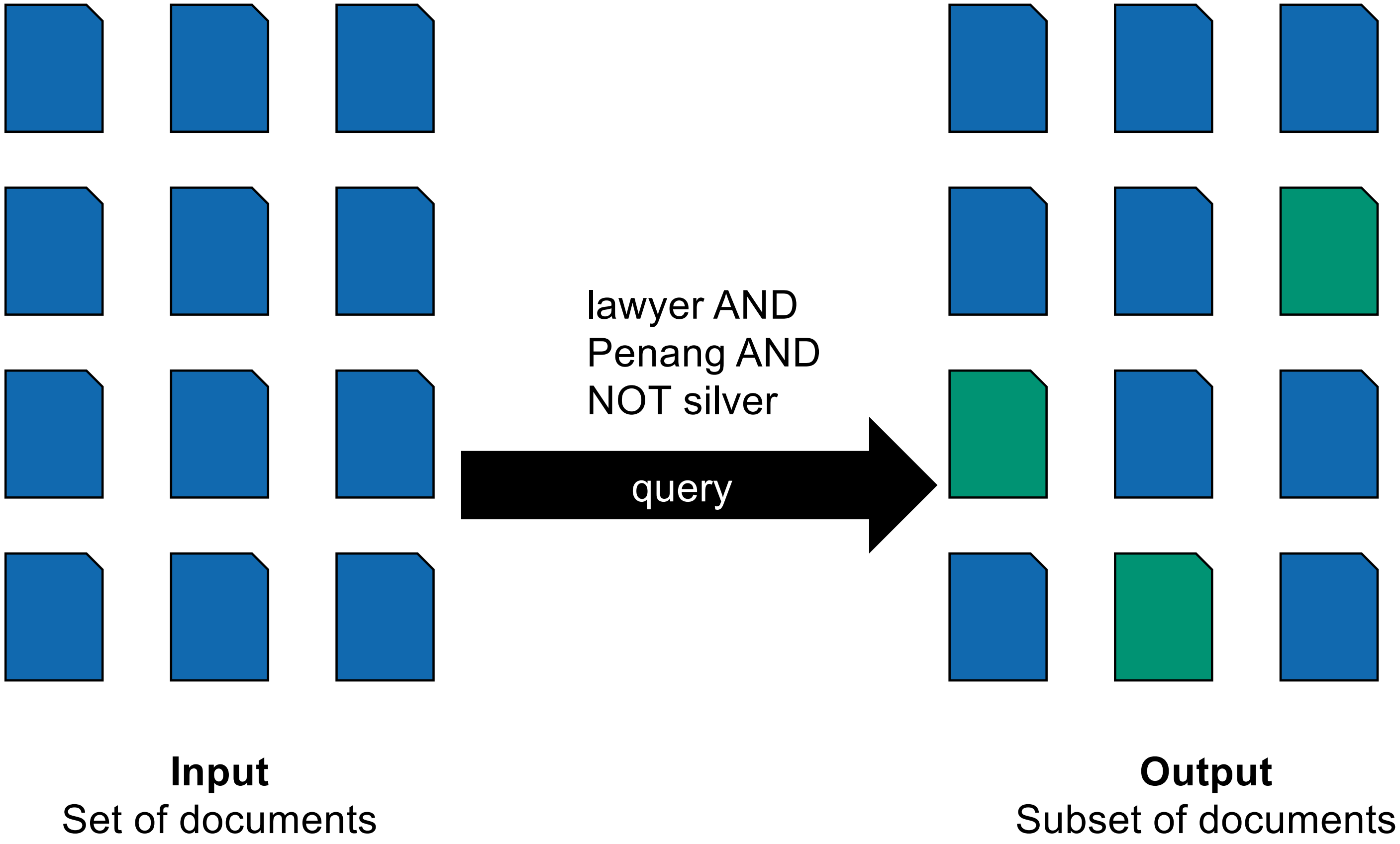

Stock picture 123RF / fotoreactor

ETH zürich

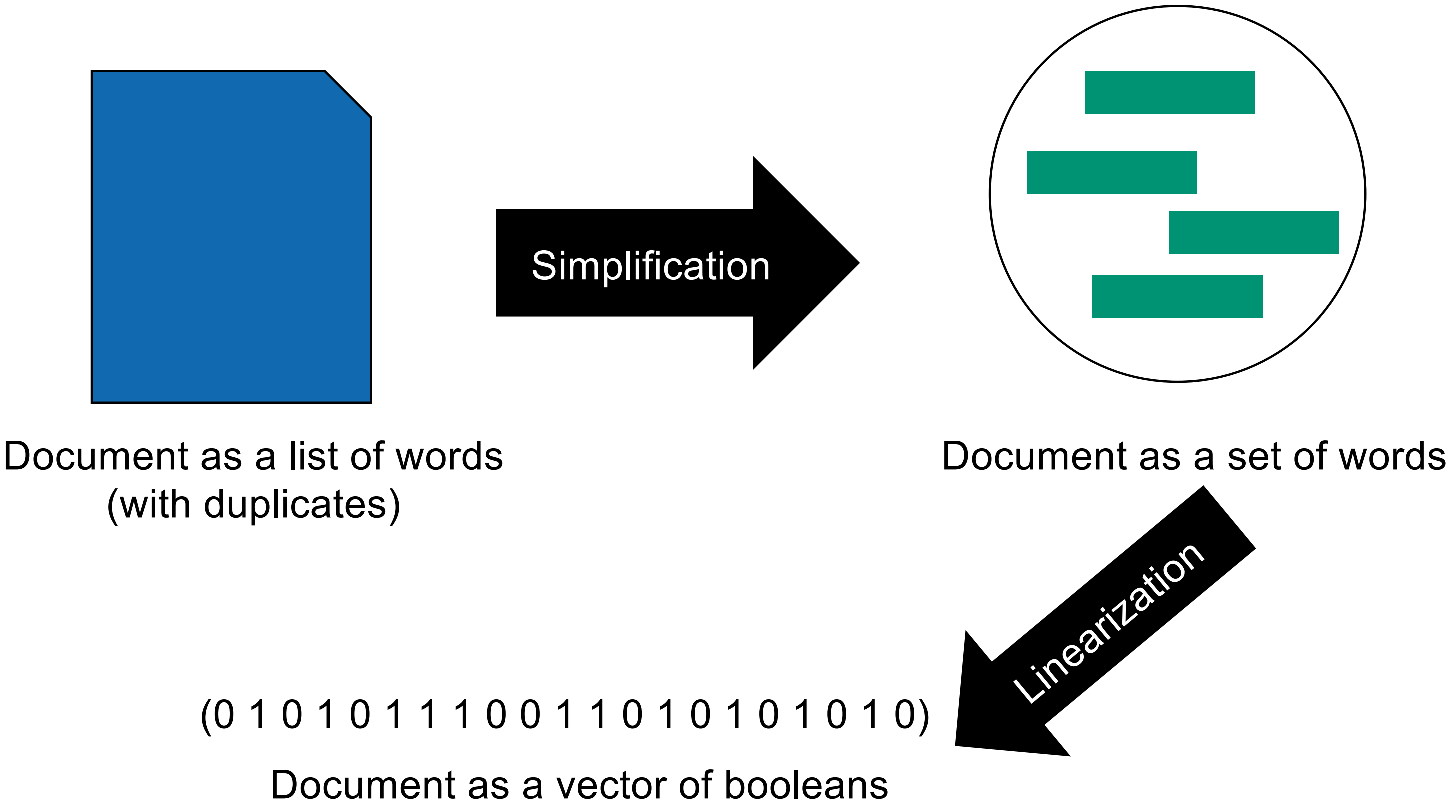


# What we have seen so far

# Boolean retrieval



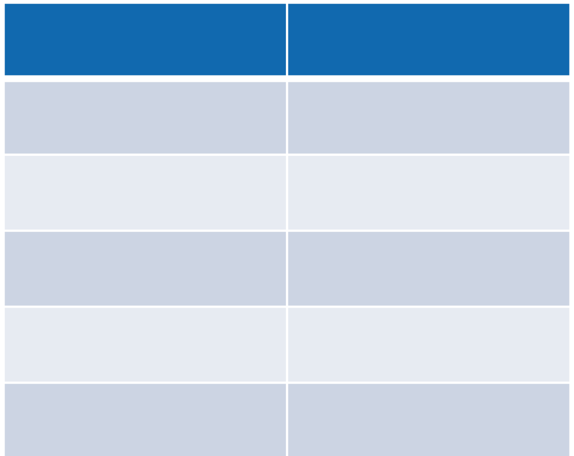
# Model and abstraction



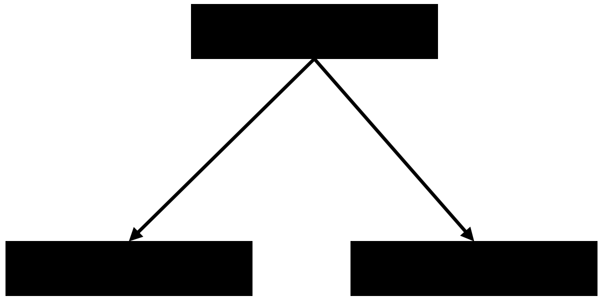
# Standard inverted index

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| ETH | 6 | 1 | 2 | 3 | 5 | 6 | 8 |
| Zürich | 5 | 3 | 4 | 7 | 8 | 9 | |
| computer | 5 | 1 | 2 | 4 | 5 | 7 | |
| data | 5 | 1 | 3 | 5 | 8 | 9 | |
| CPU | 4 | 2 | 3 | 4 | 7 | | |
| information | 6 | 1 | 2 | 4 | 5 | 8 | 9 |
| retrieval | 4 | 3 | 5 | 7 | 8 | | |

# Search structures

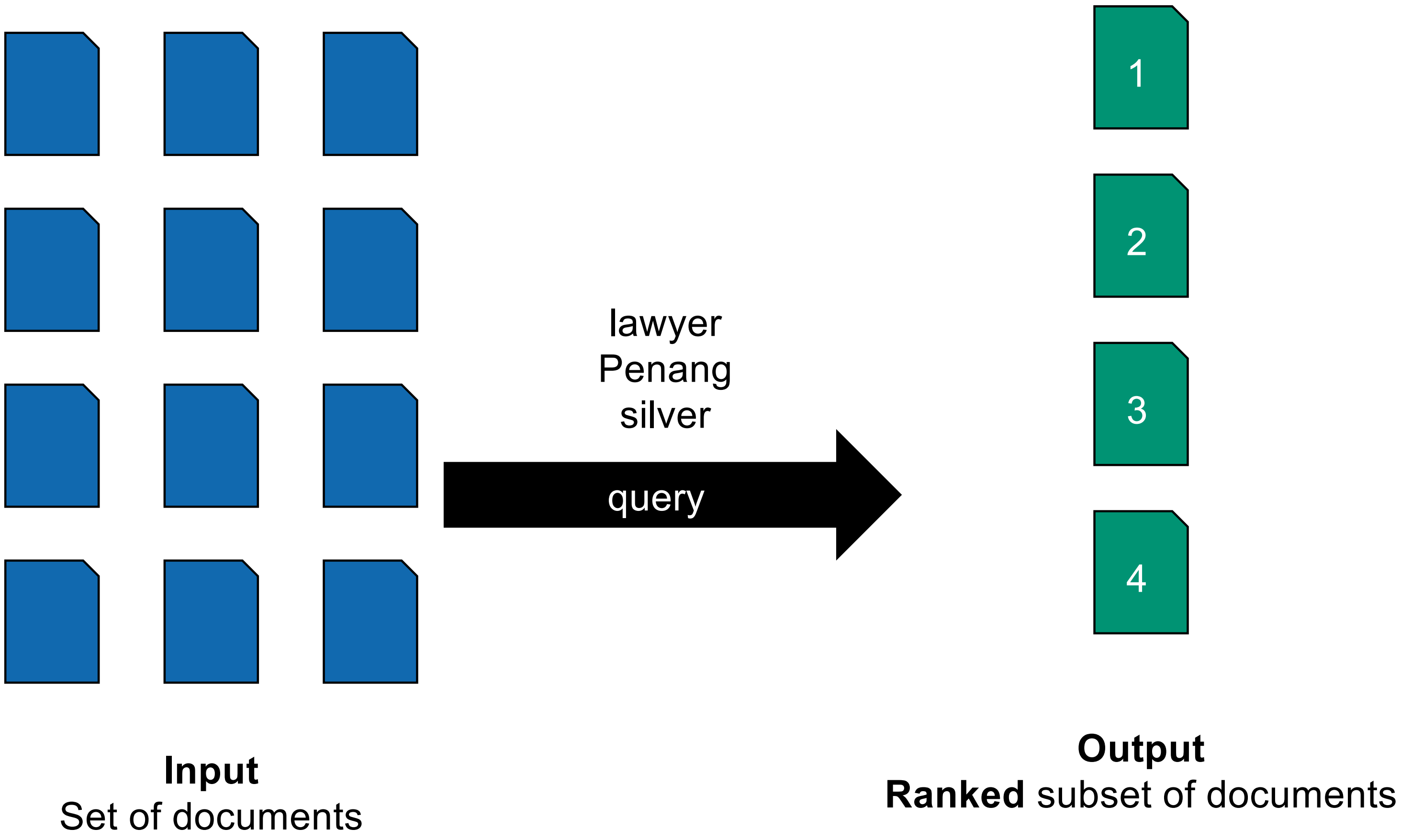


Hash tables

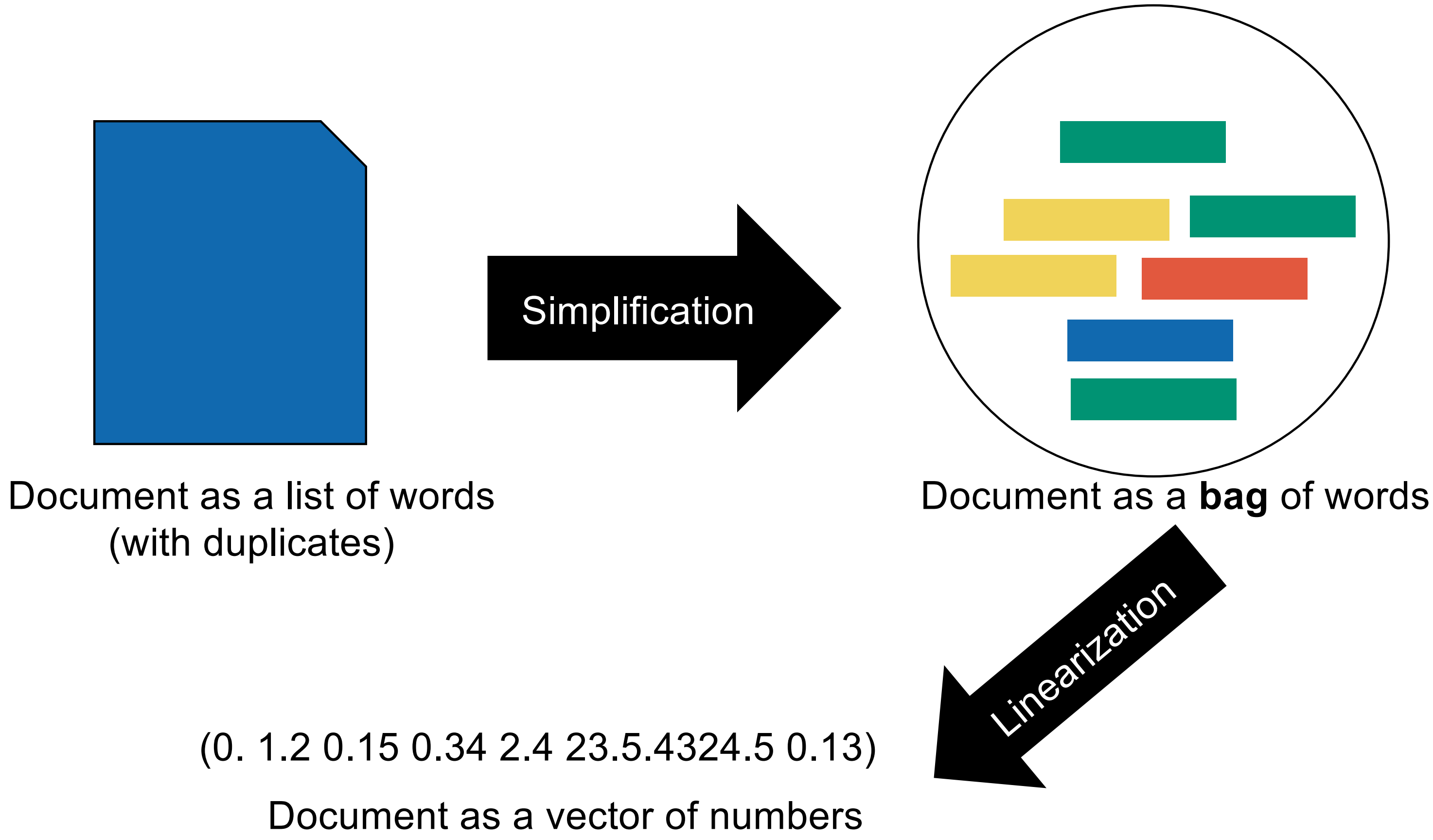


Trees (B, B+)

# Ranked retrieval

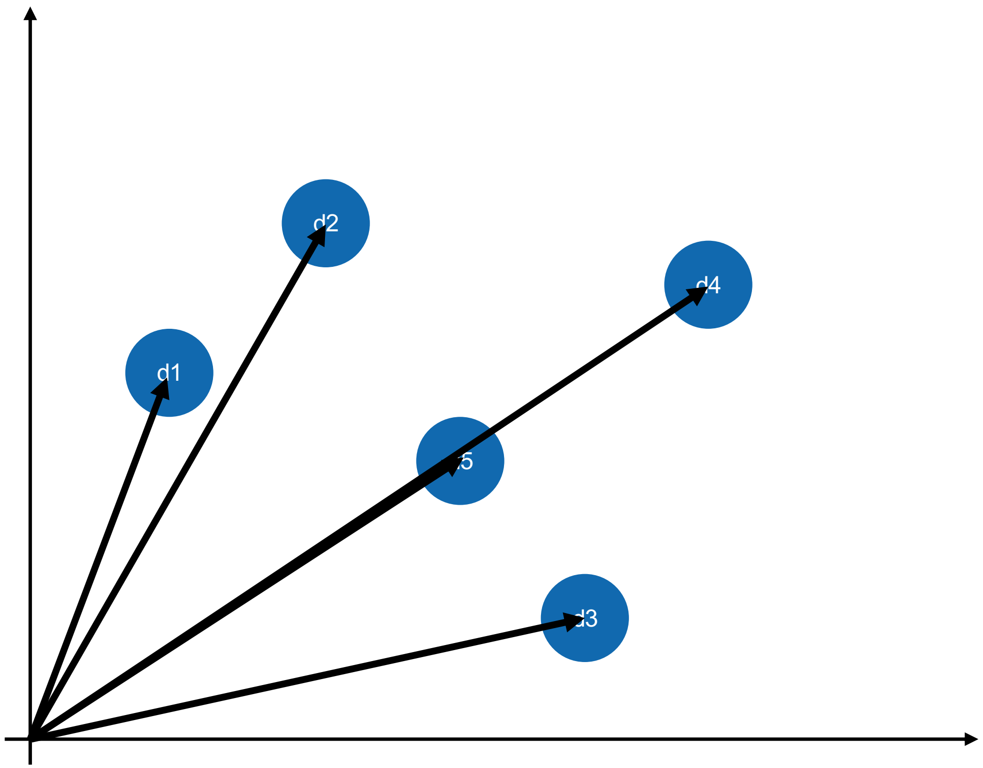


# Model and abstraction

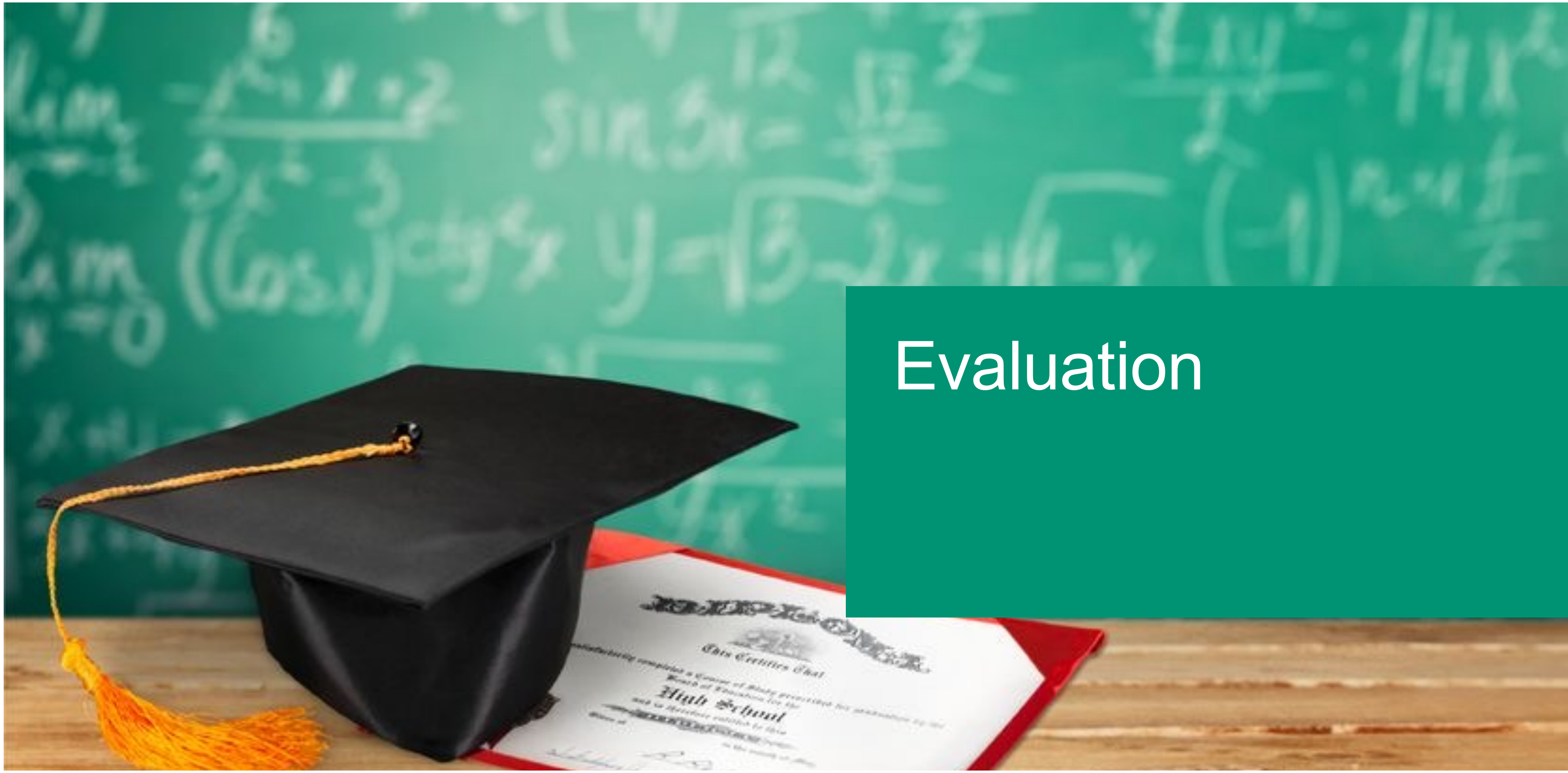


# Vector-Space Model

Documents
= vectors in the
first quadrant
**of**
$\mathbb{R}^M$



ETH zürich



# Evaluation

Design choices

Do we use stop words

?

Design choices

Should we exclude numbers?

?

## Design choices

What is the best value of r for champion lists?

?

## Design choices

Do we use stemming
or lemmatization?

?

Design choices

Should I weight query terms?

?

Design choices

Should I renormalize scores with $L_2$?

?

# Design choices

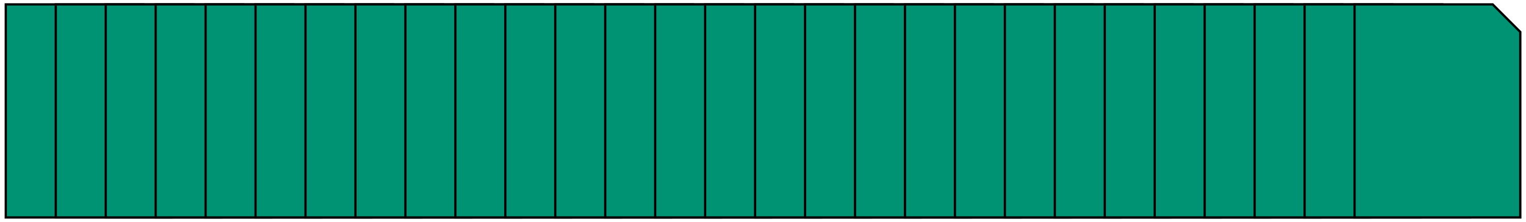
Which document weights
should I use?

?

Our goal

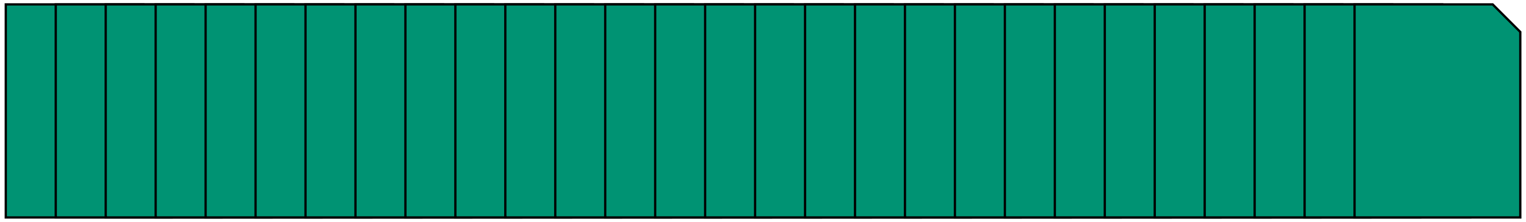
Optimize **relevance**

# Evaluation setup

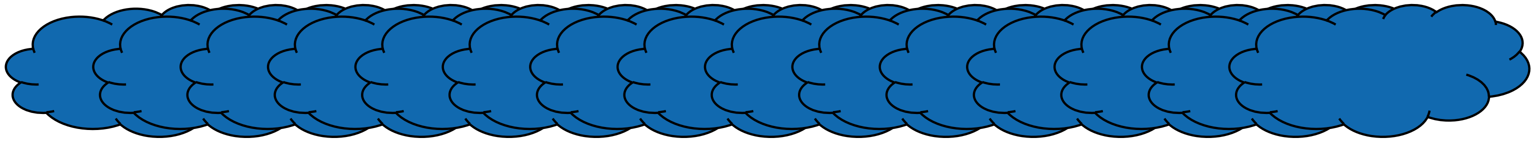


Collection of documents

# Evaluation setup

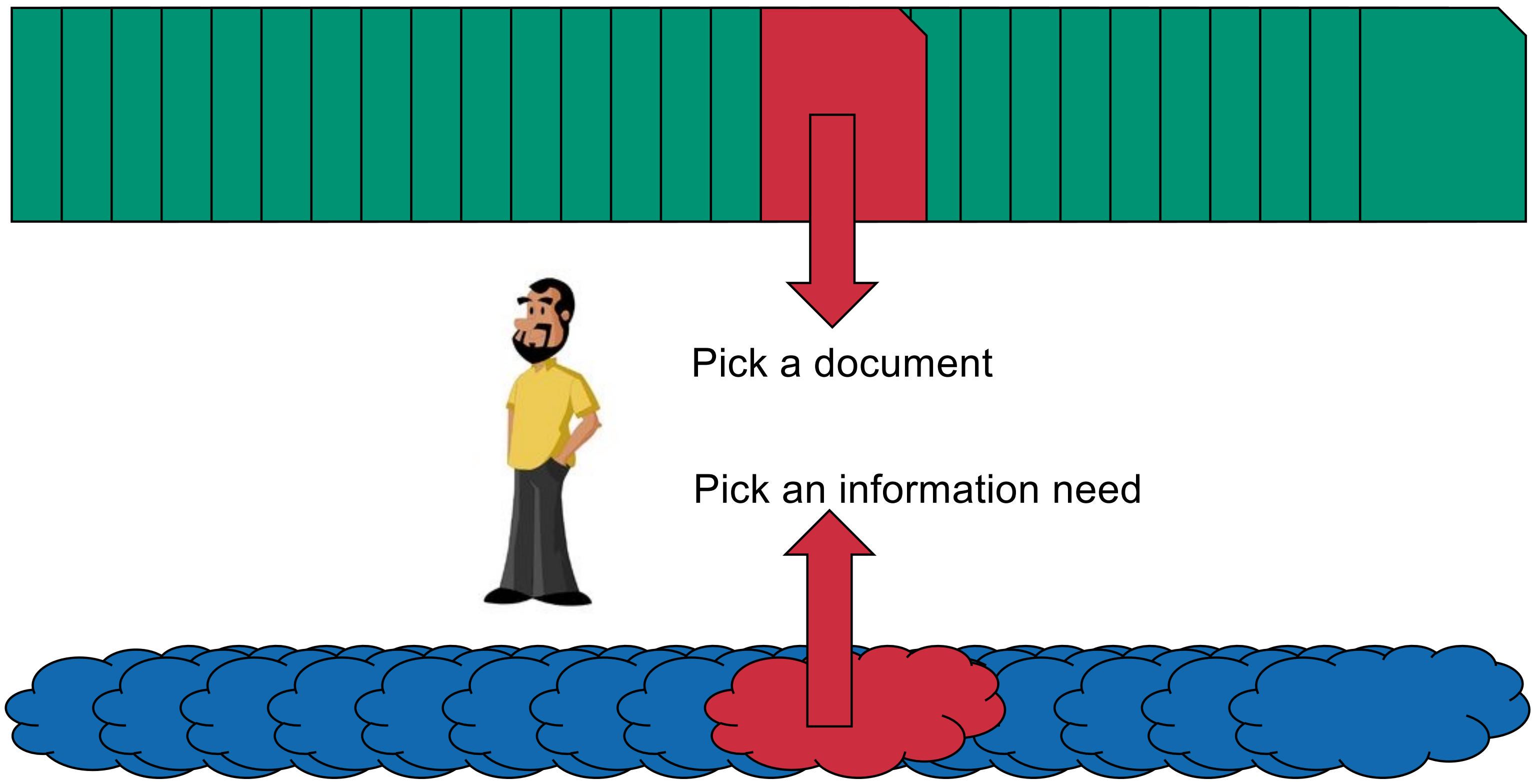


Collection of documents

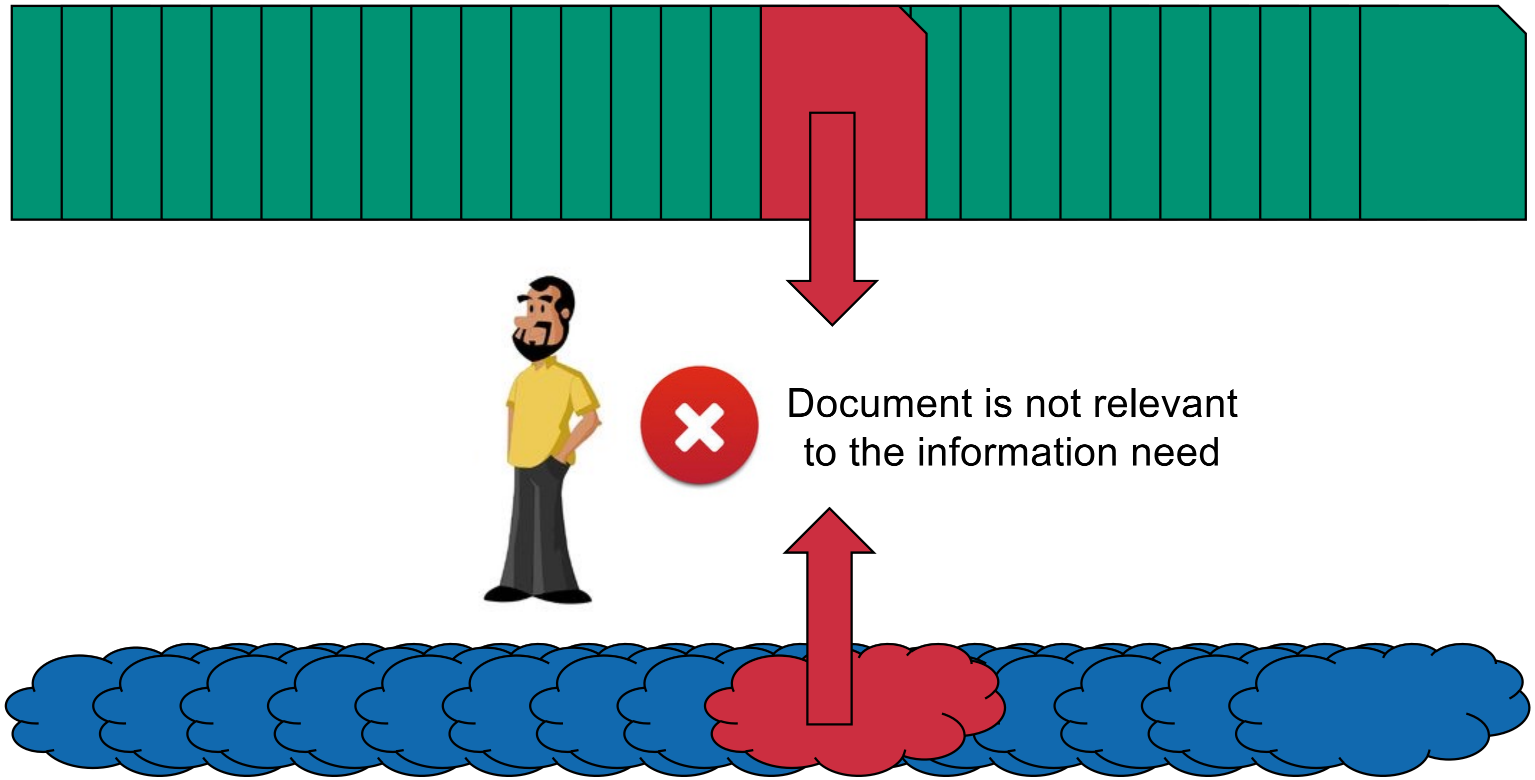


Set of information needs

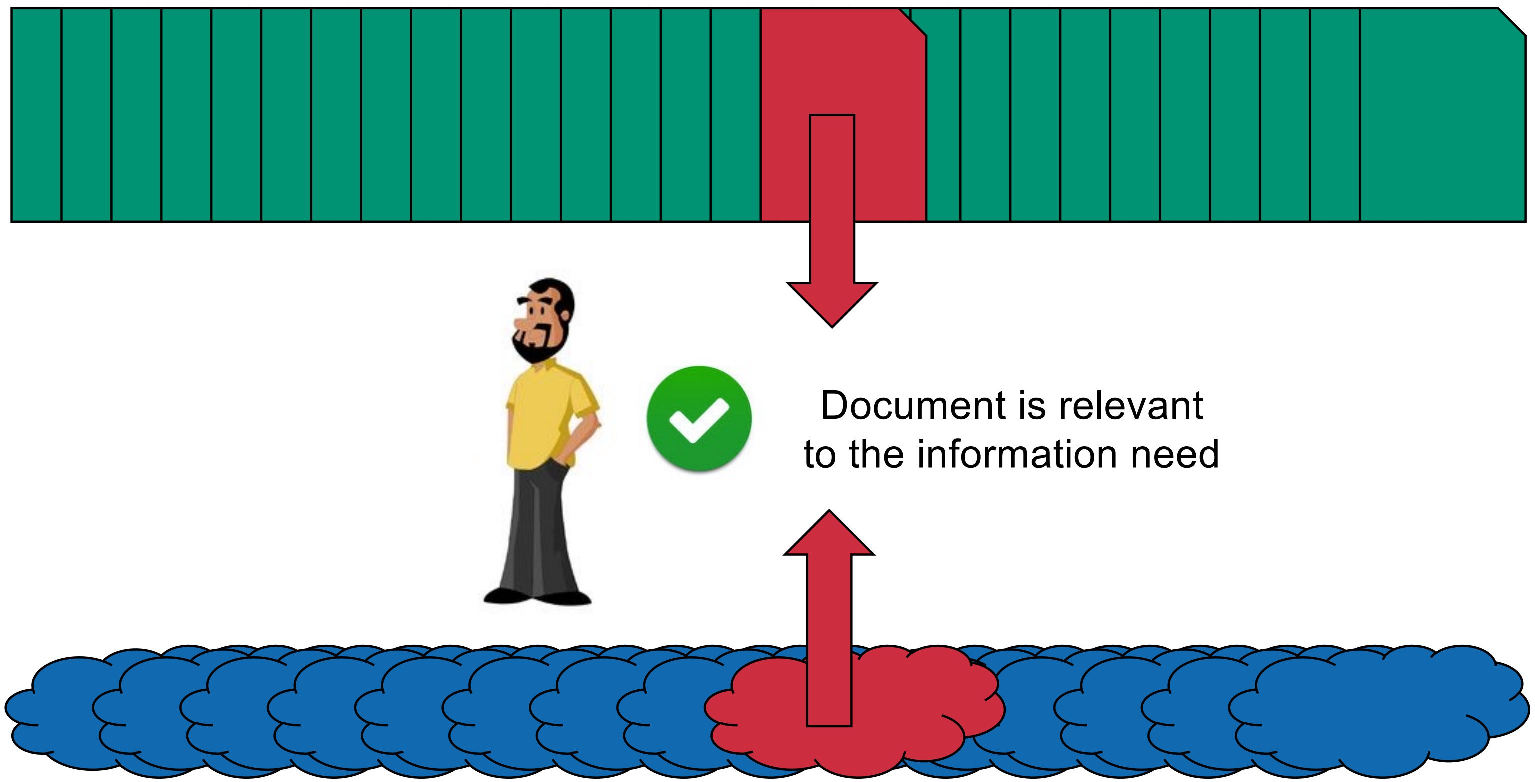
# Evaluation setup



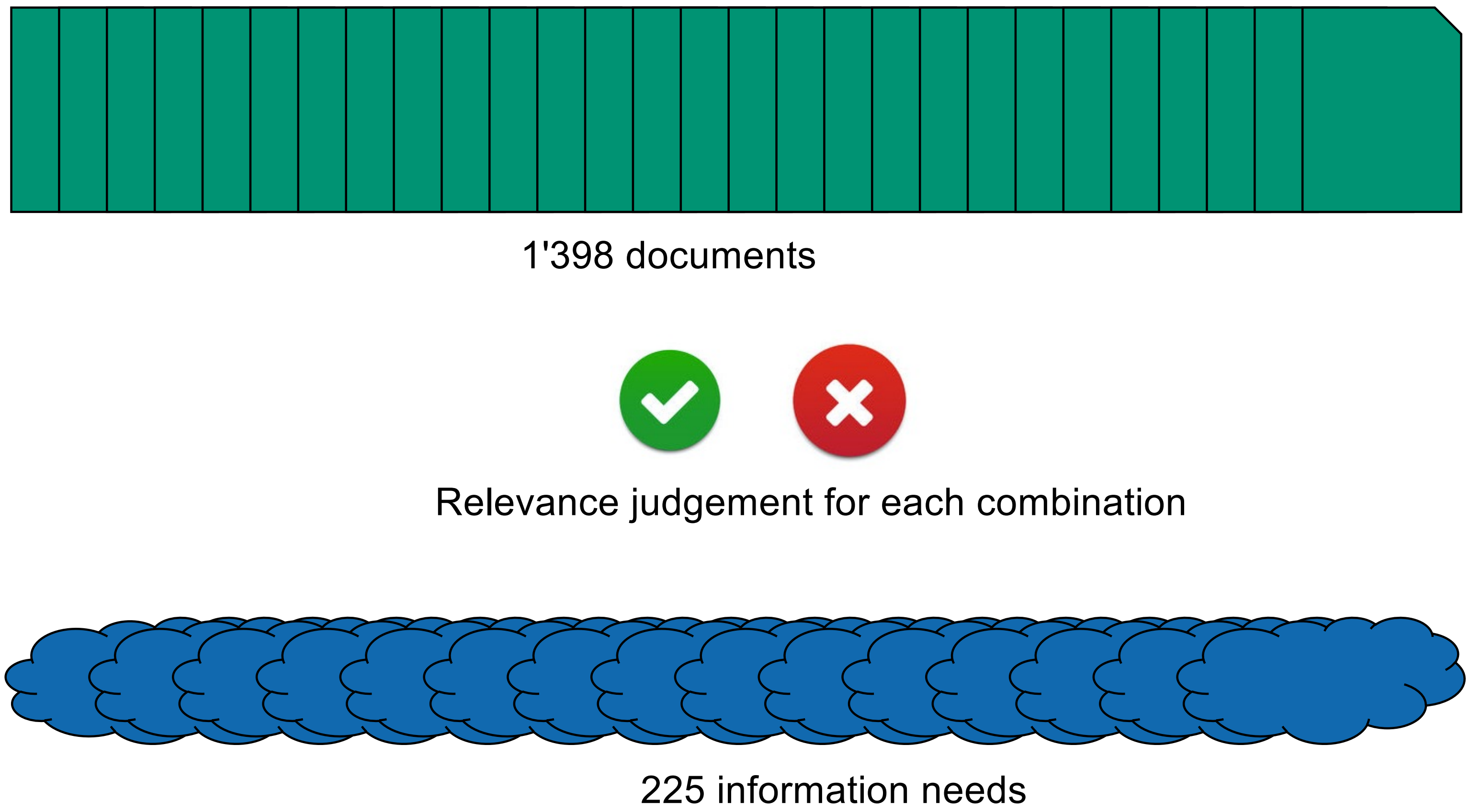
# Evaluation setup



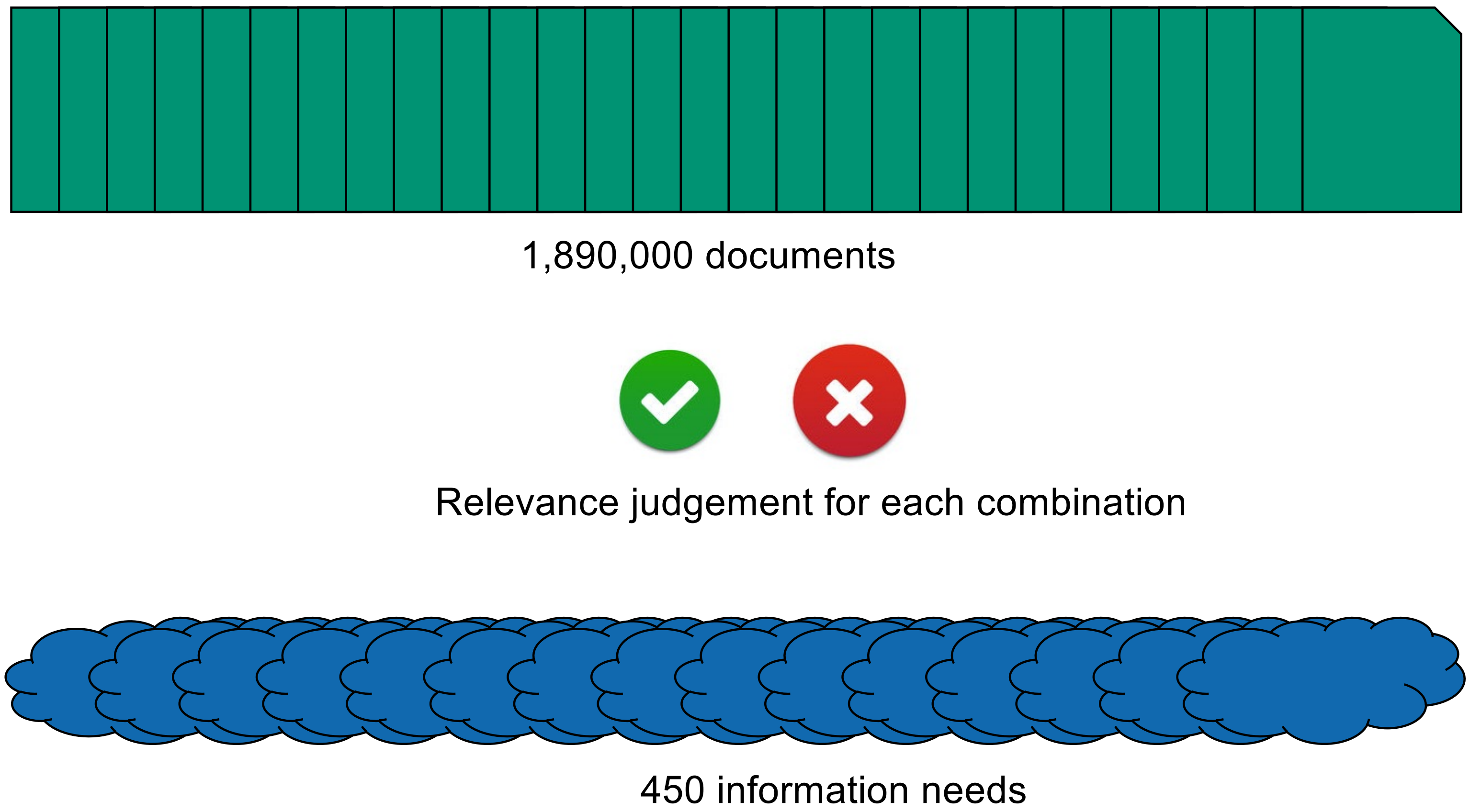
# Evaluation setup



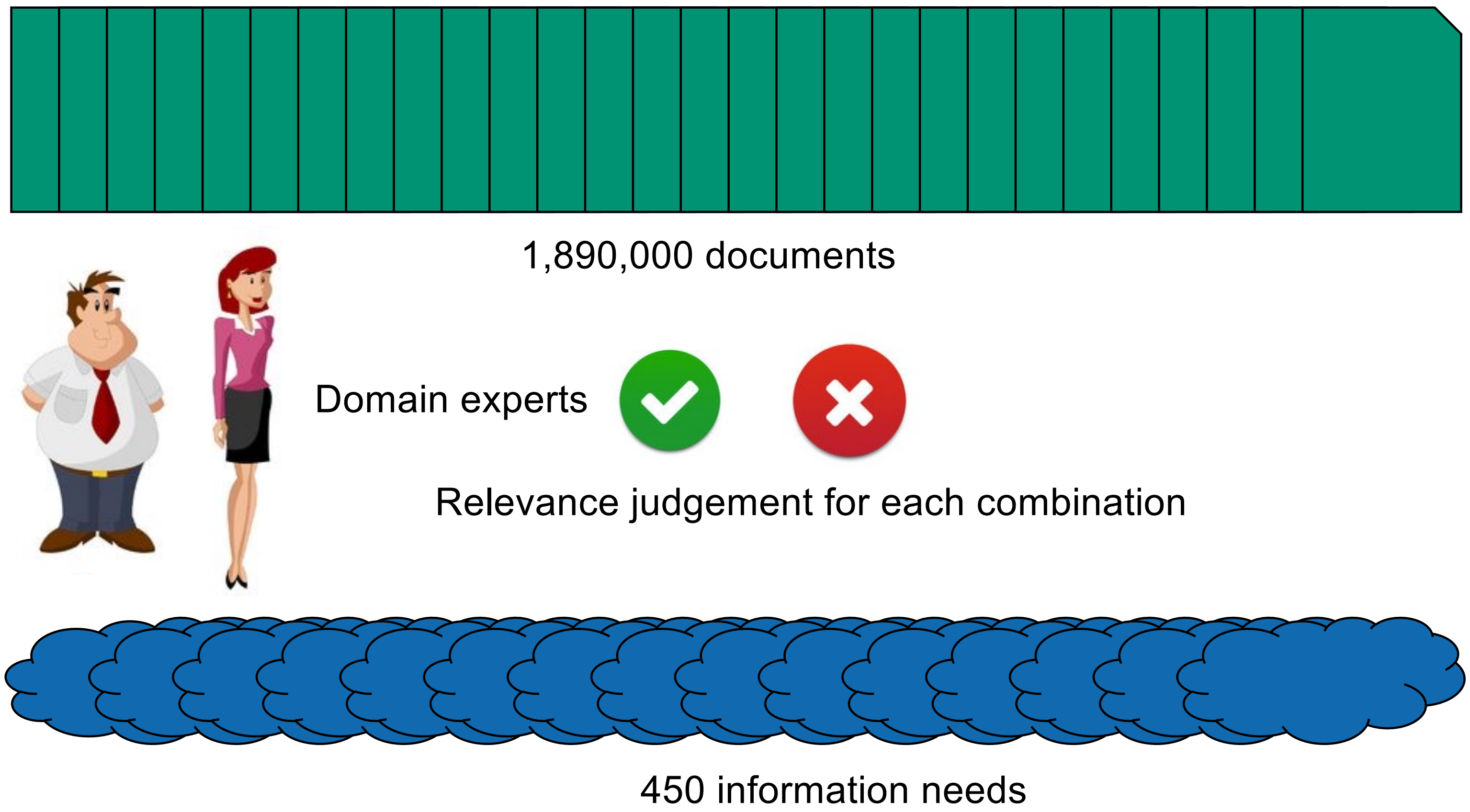
# Cranfield dataset



# TREC dataset



# TREC dataset



ETH zürich

# Information need

# Information need vs. query

# Information need vs. query

Alice is searching for a higher-education institution

ETH Zurich

# Information need vs. query

Alice is searching for a higher-education institution

ETH Zurich

Bob is searching for Ethereum cryptocurrency in Zurich, Kansas

# Information need vs. query

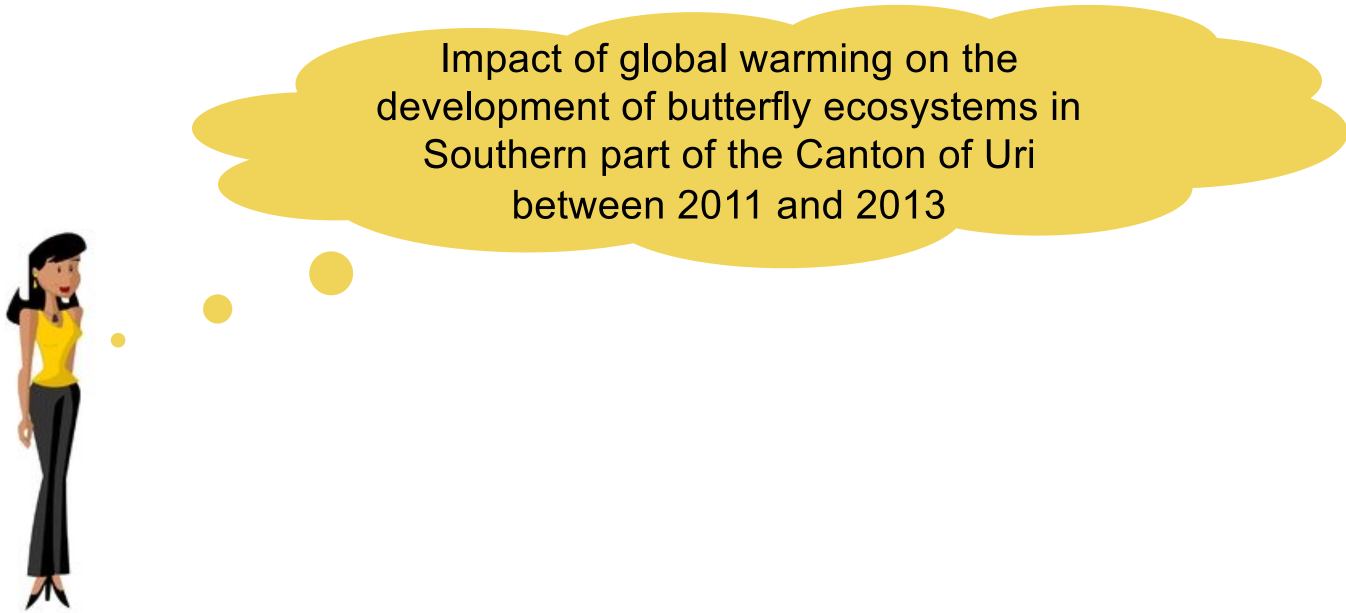
Alice is searching for a higher-education institution

ETH Zurich

Bob is searching for Ethereum cryptocurrency in Zurich, Kansas

Carlos is searching for the extended trading hours on insurance stocks.

# Information need



# Information need

Impact of global warming on the development of butterfly ecosystems in Southern part of the Canton of Uri between 2011 and 2013

```
butterfly AND climate AND uri AND south
```

# Information need

Impact of global warming on the development of butterfly ecosystems in Southern part of the Canton of Uri between 2011 and 2013

```
butterfly AND climate AND uri AND south

lepidoptera AND ur AND NOT winter
```

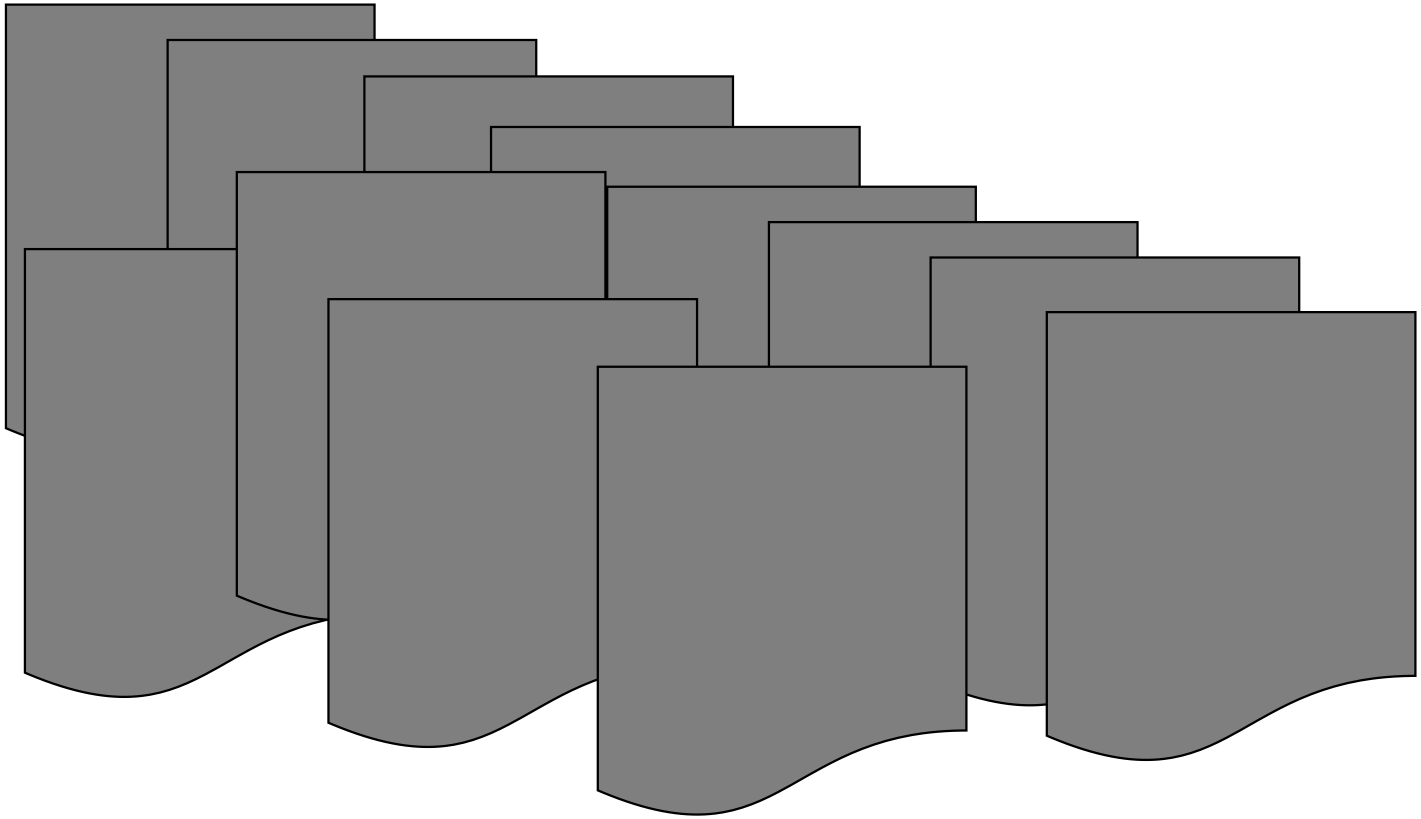
# Information need

Impact of global warming on the development of butterfly ecosystems in Southern part of the Canton of Uri between 2011 and 2013

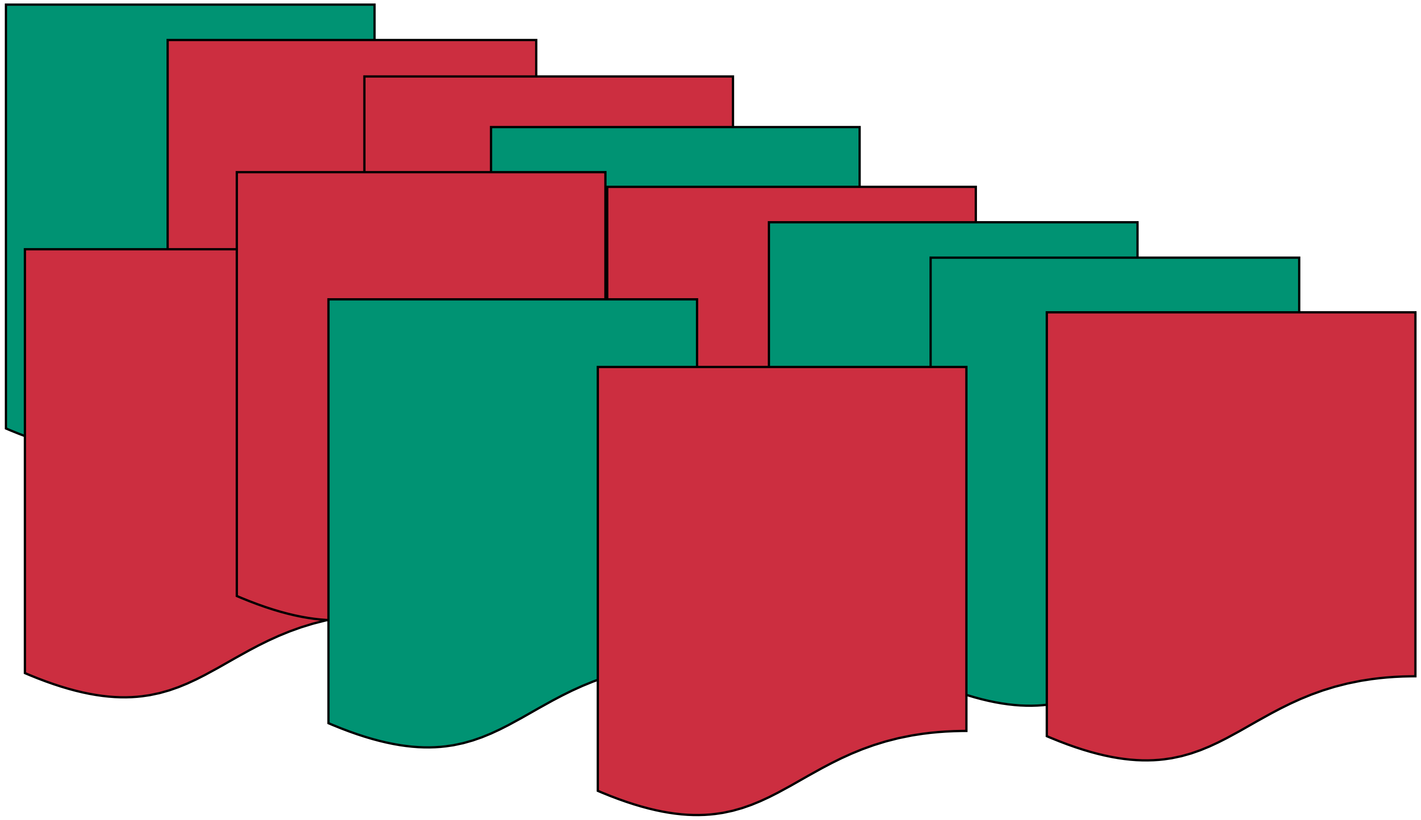
```
butterfly AND climate AND uri AND south

lepidoptera AND ur AND NOT winter

(central NEAR (switzerland OR sweden))
AND NOT (lu OR nw OR ow OR stockholm)
AND warmth
```

# Precision and Recall

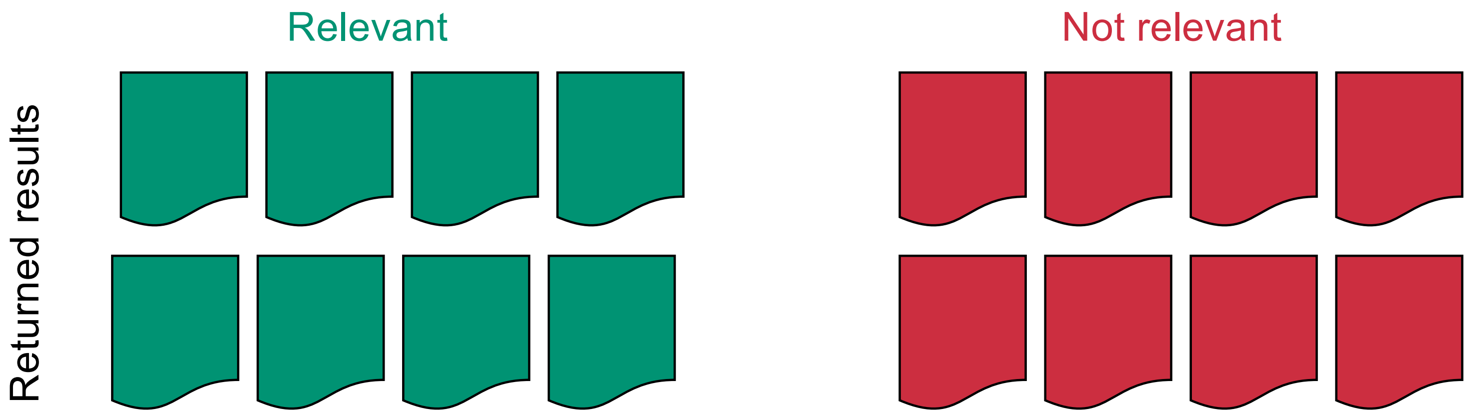
# Results



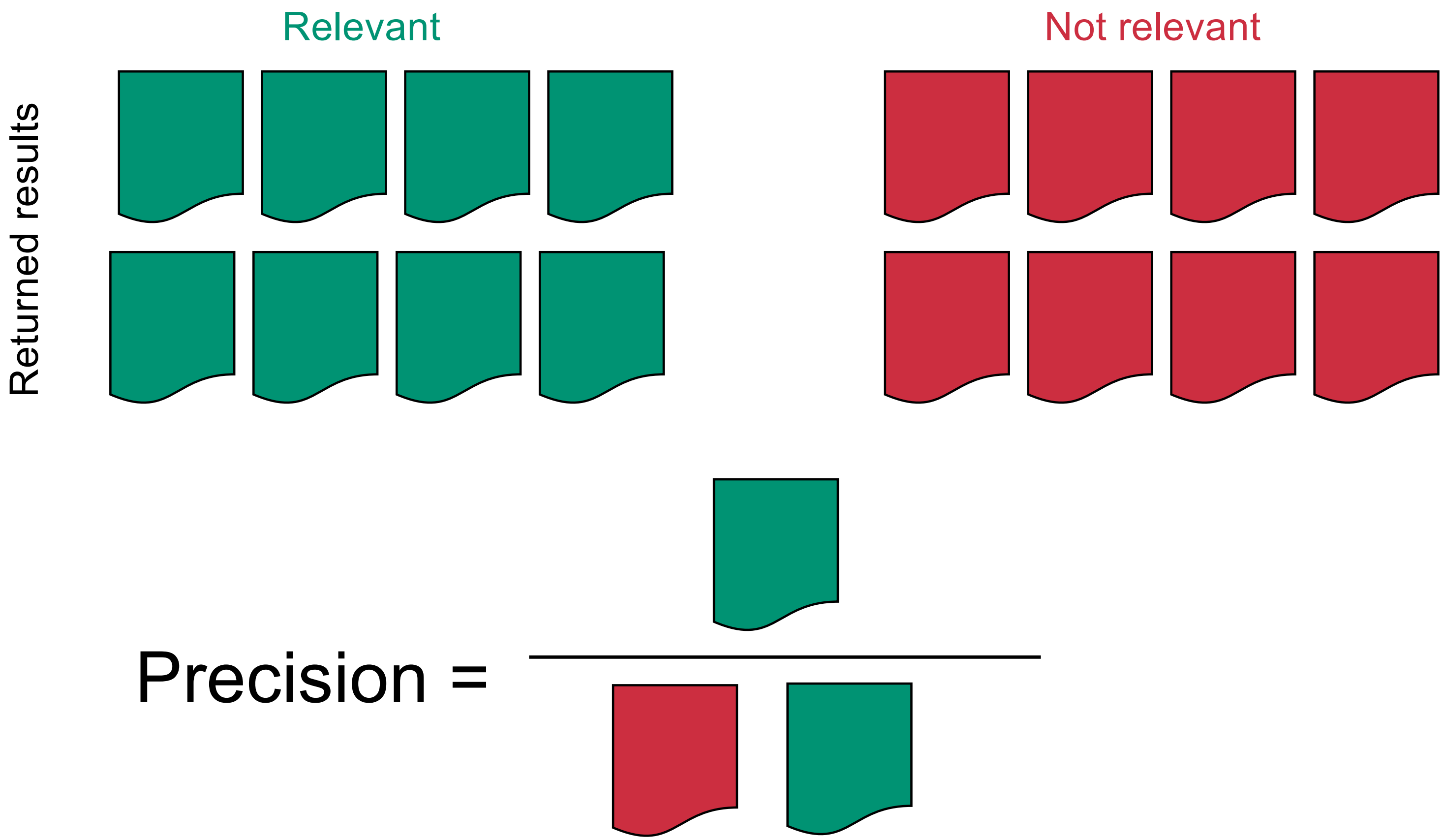
# Results



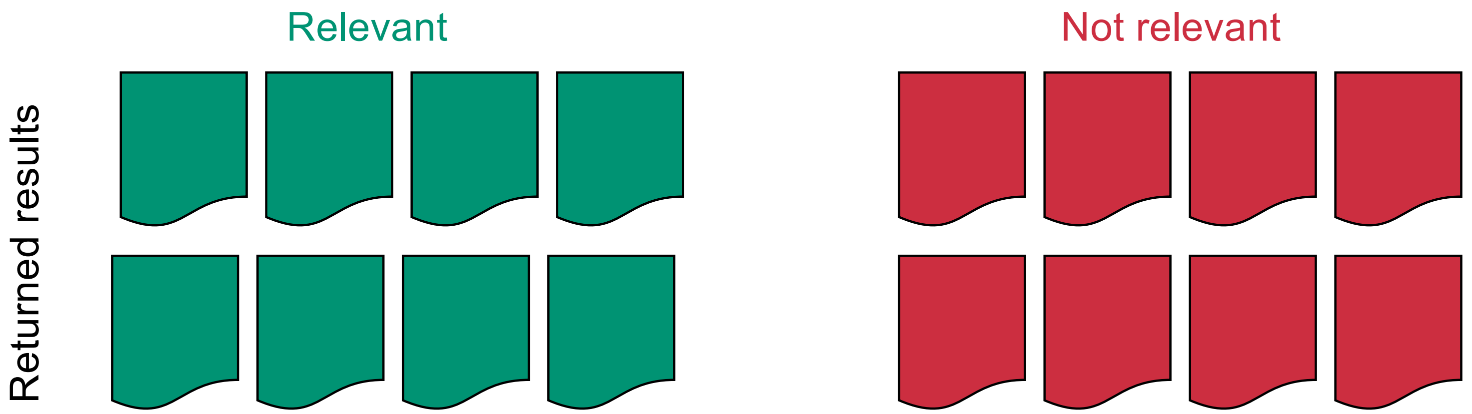
# Precision



# Precision

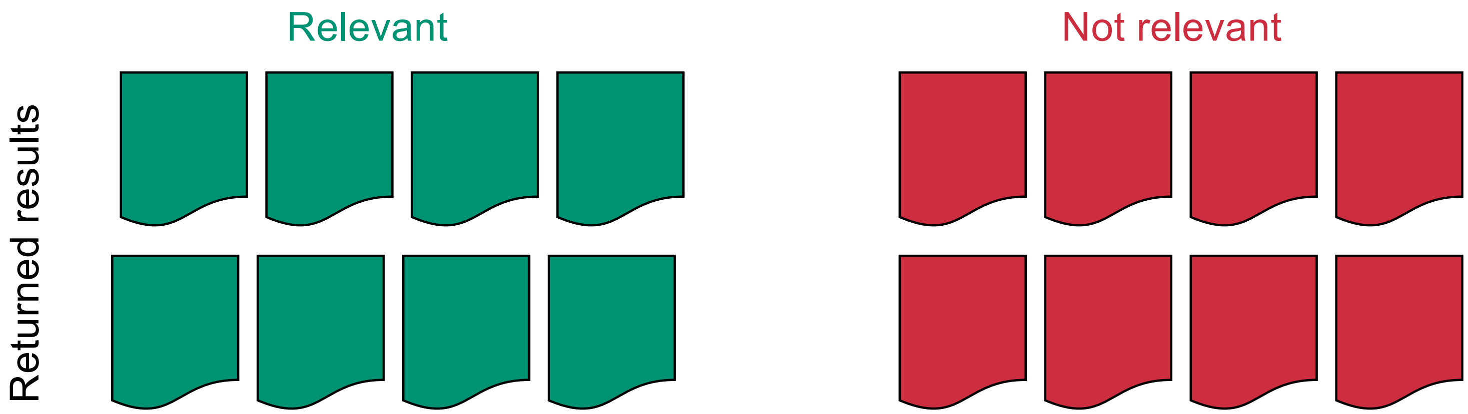


# Precision



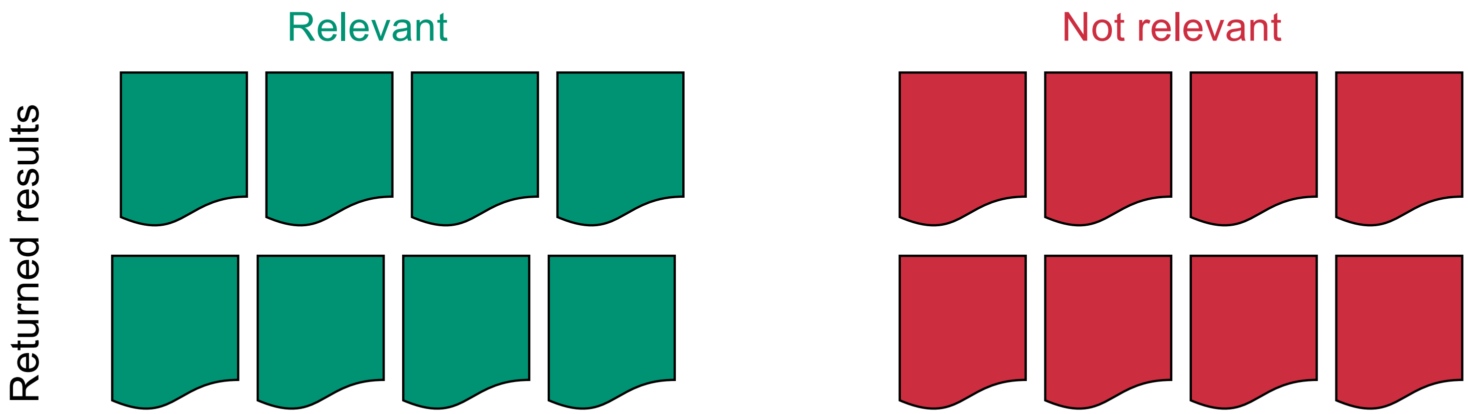

$$\text{Precision} = \frac{\text{true positives}}{\text{returned}}$$

# Precision



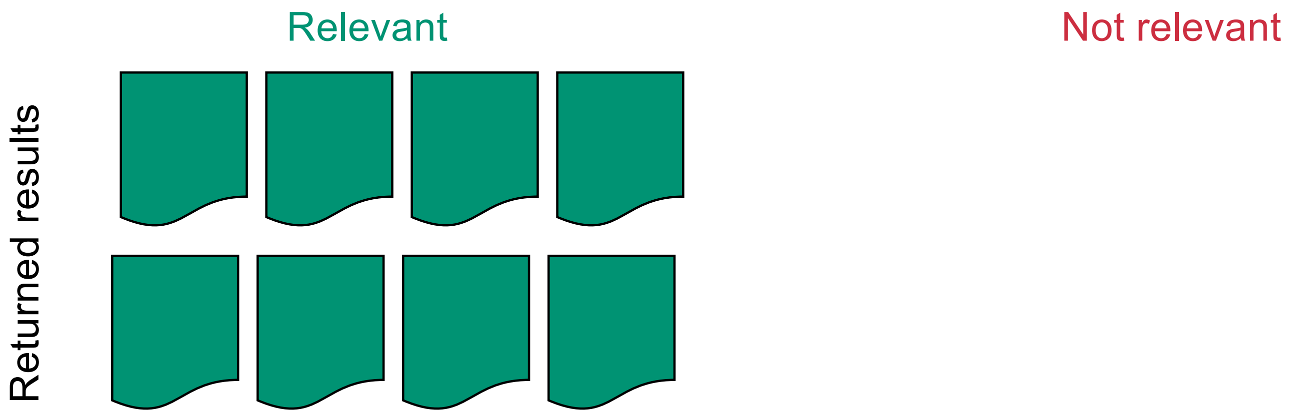

Precision =

# Precision



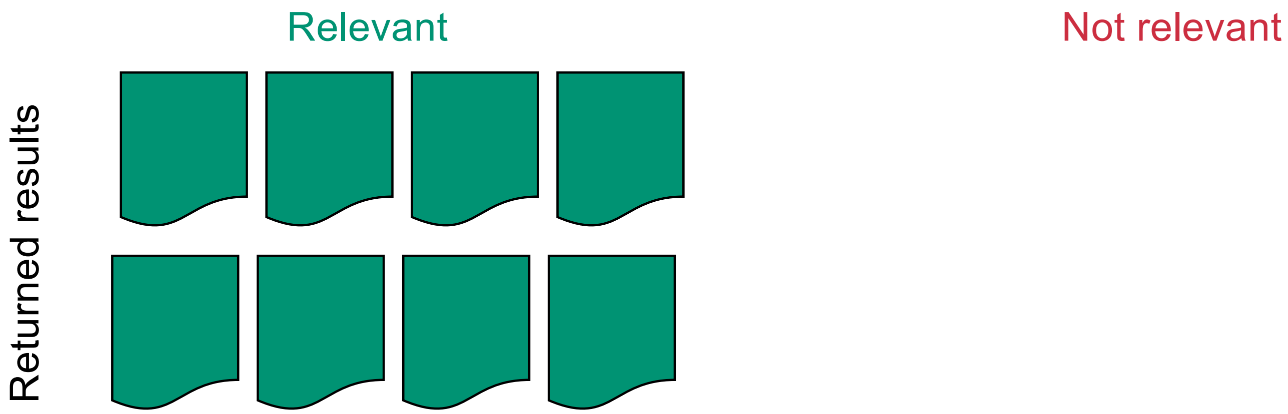

Precision = 50%

# Precision



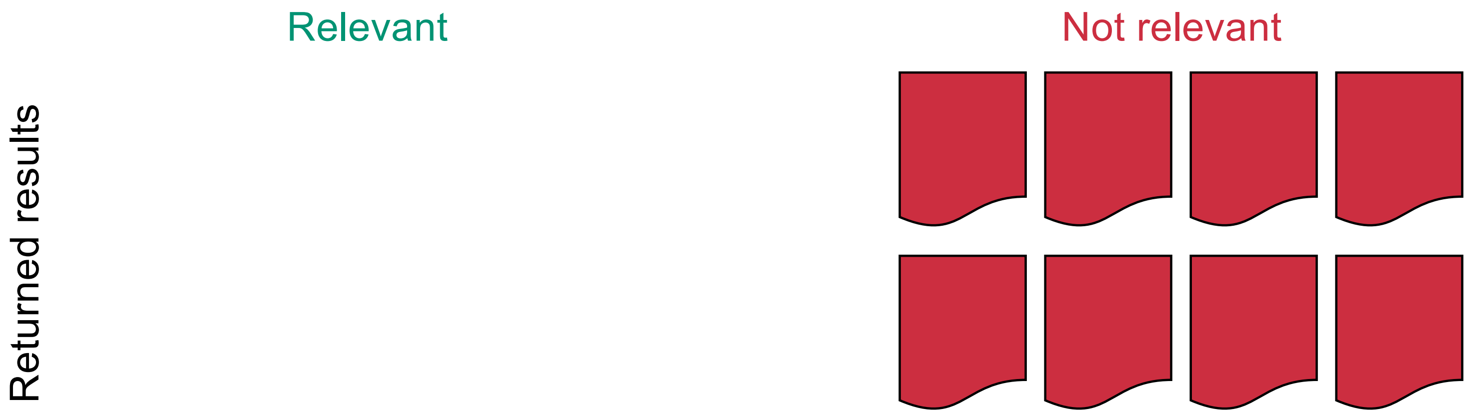

Precision =

# Precision



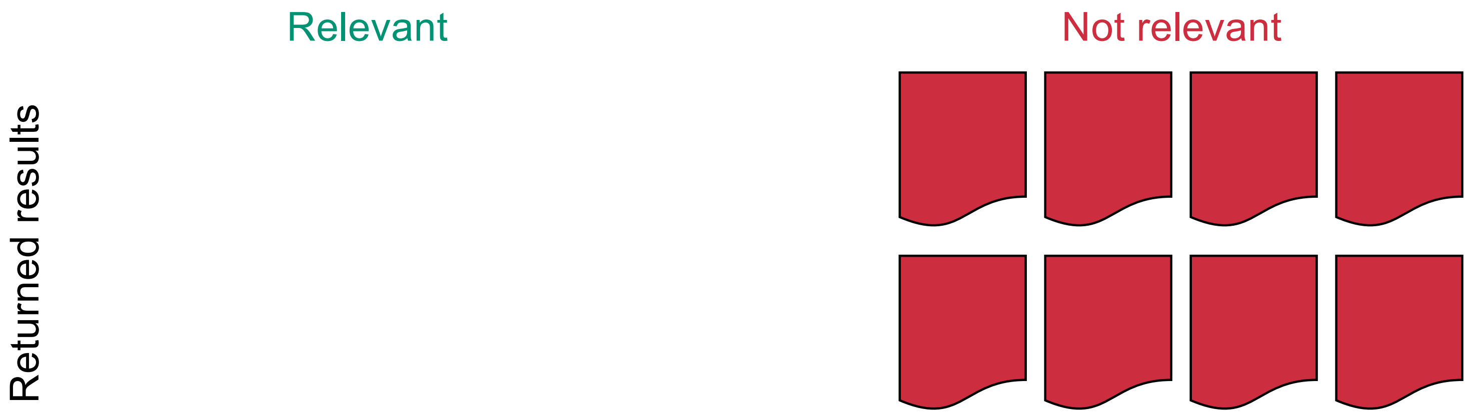

Precision = 100%

# Precision



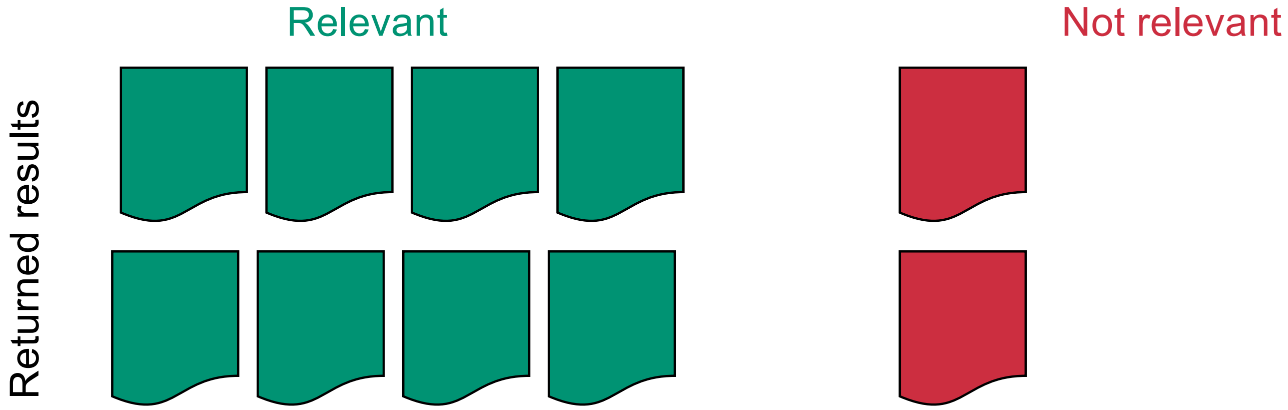

Precision =

# Precision



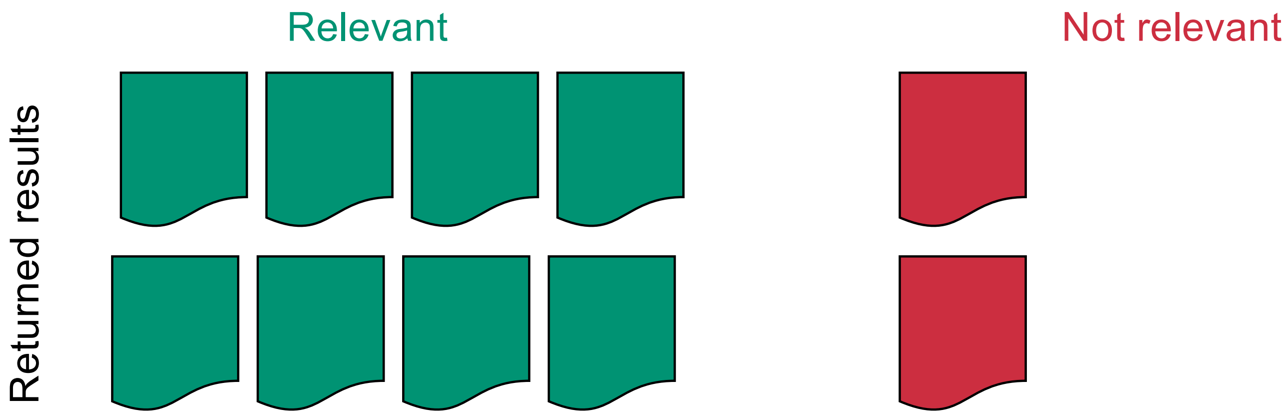

Precision = 0%

# Precision



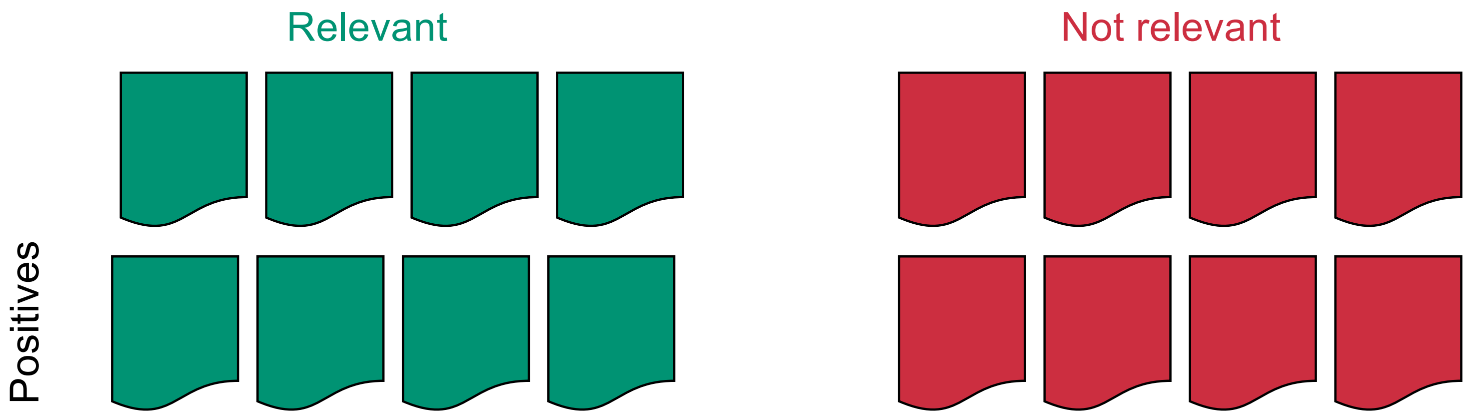

Precision =

# Precision

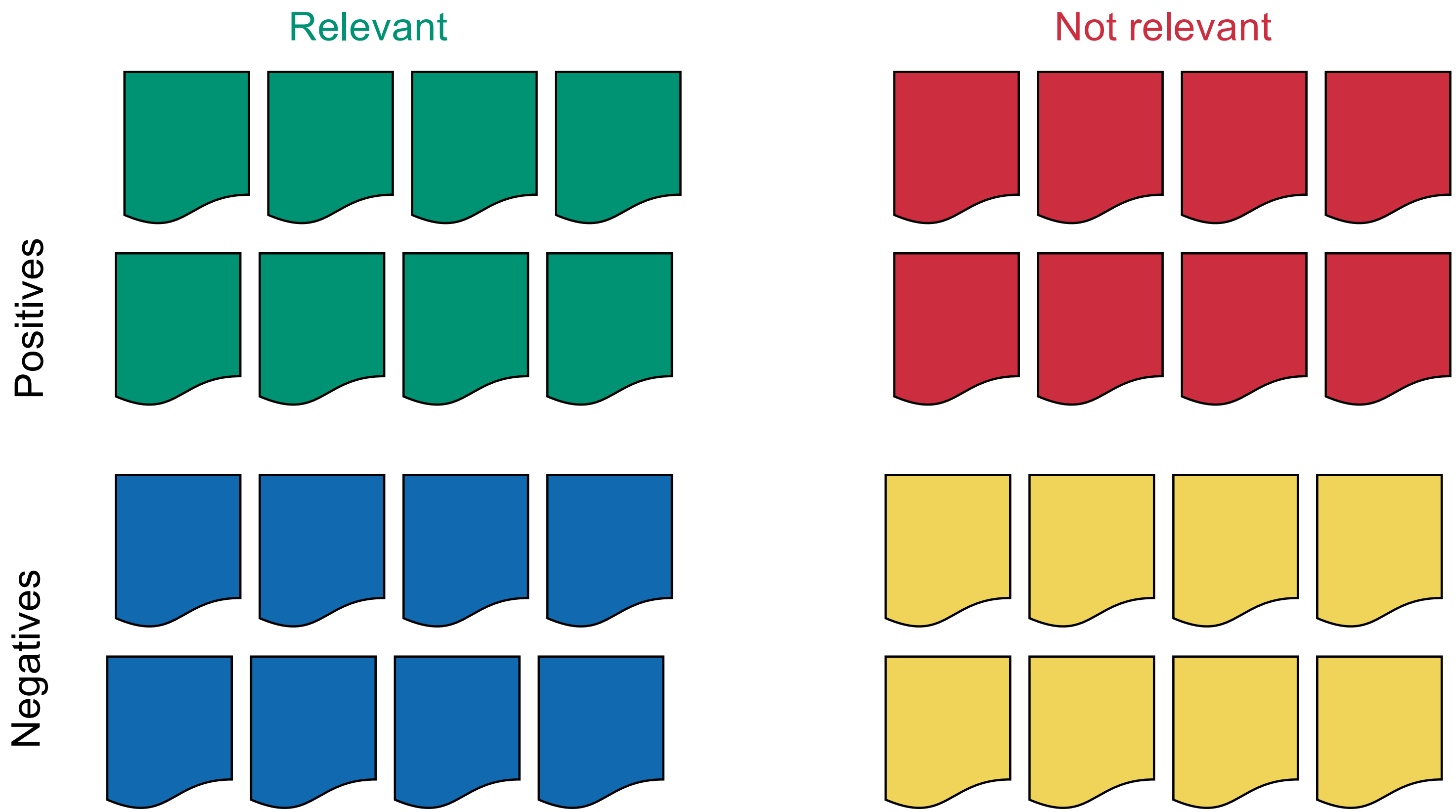



Precision = 80%

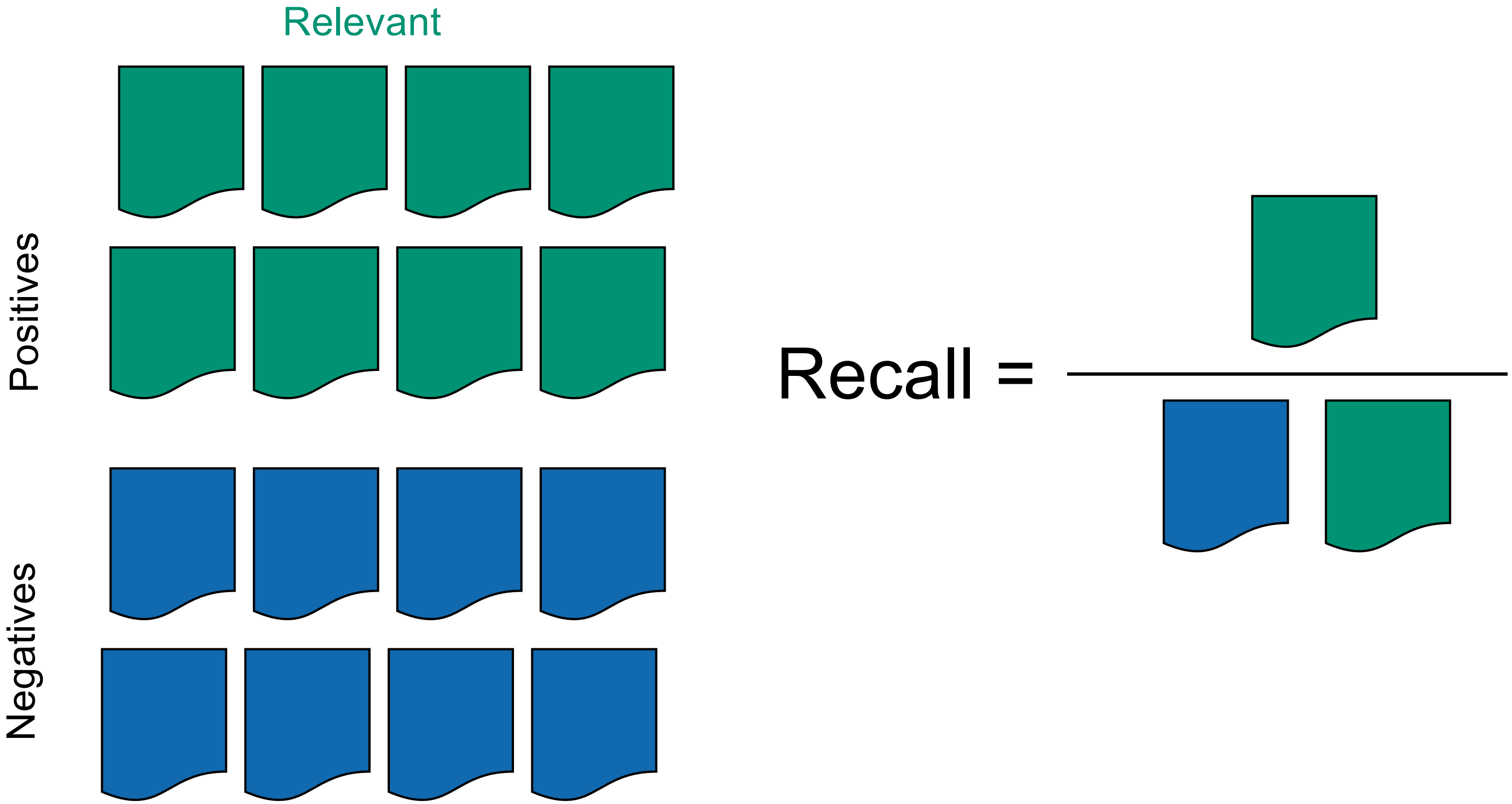
# Precision



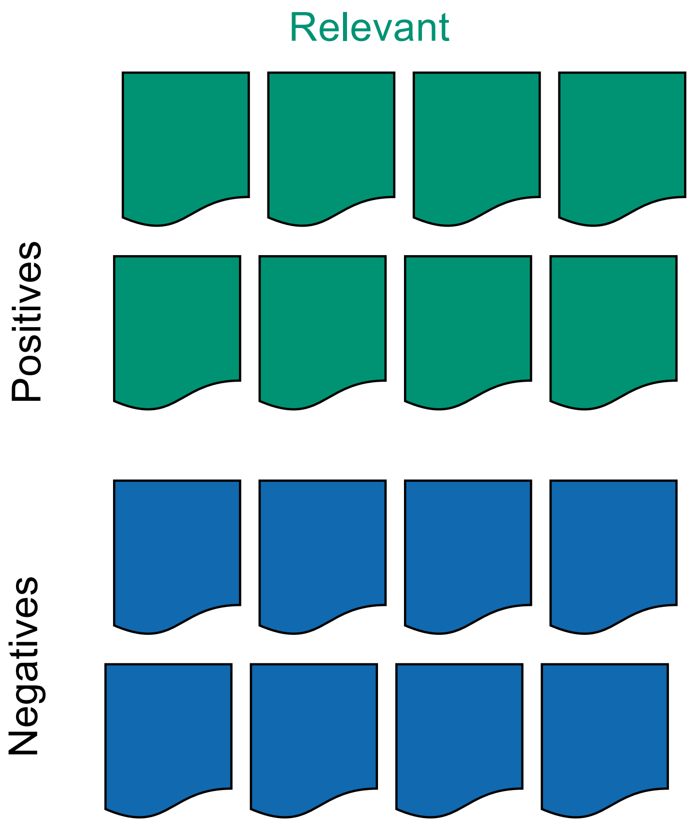
# Results



# Recall

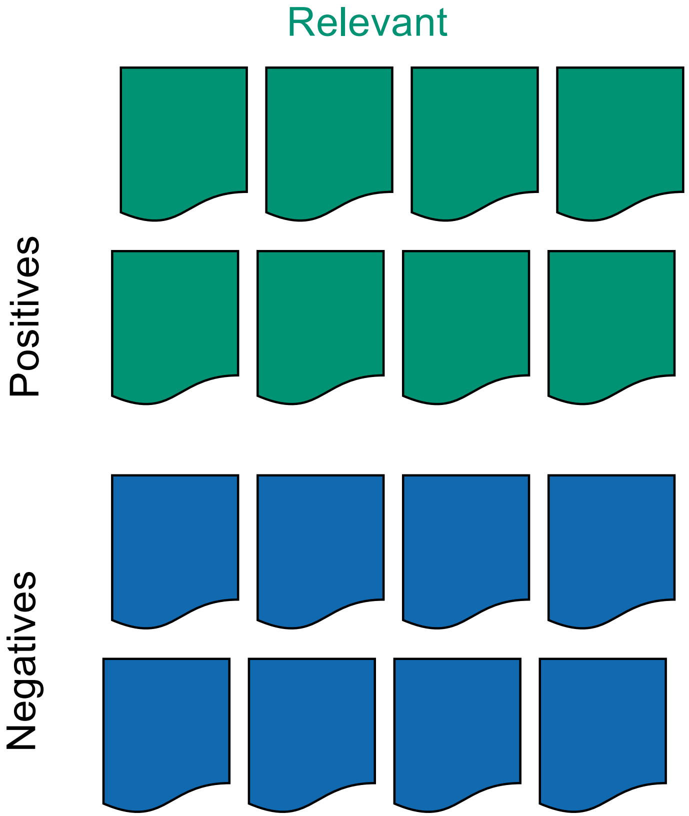


# Recall



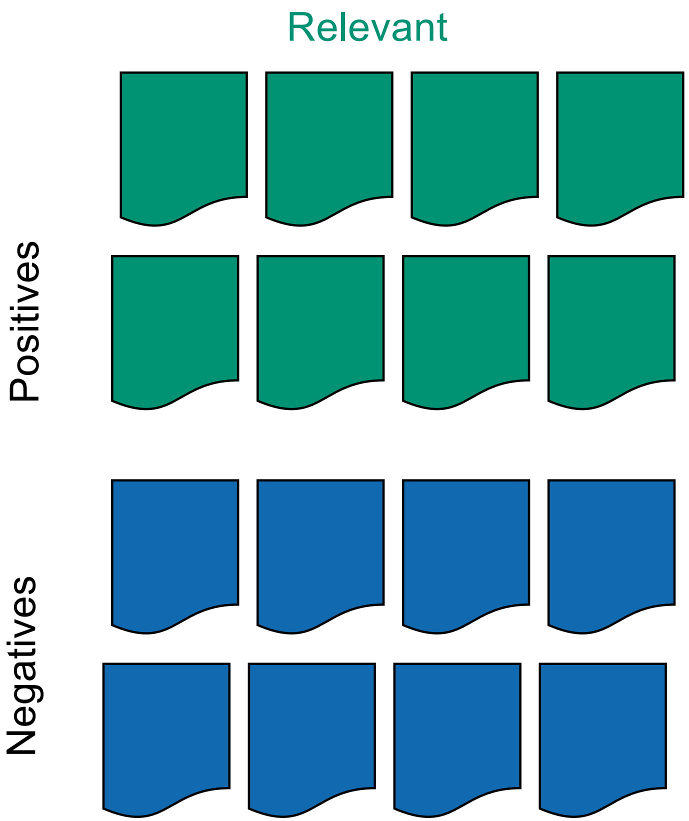

$$\text{Recall} = \frac{\text{true positives}}{\text{relevant}}$$

# Recall



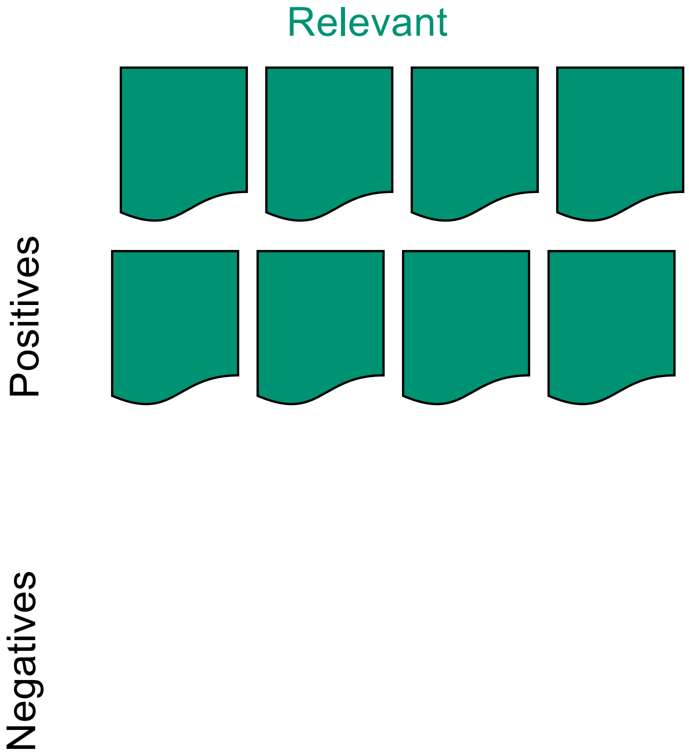

Recall =

# Recall



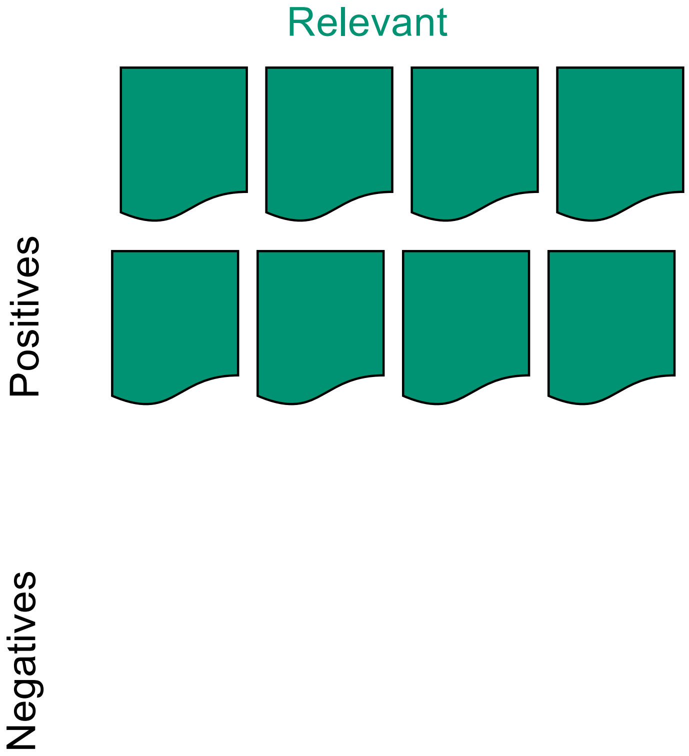

Recall = 50%

# Recall



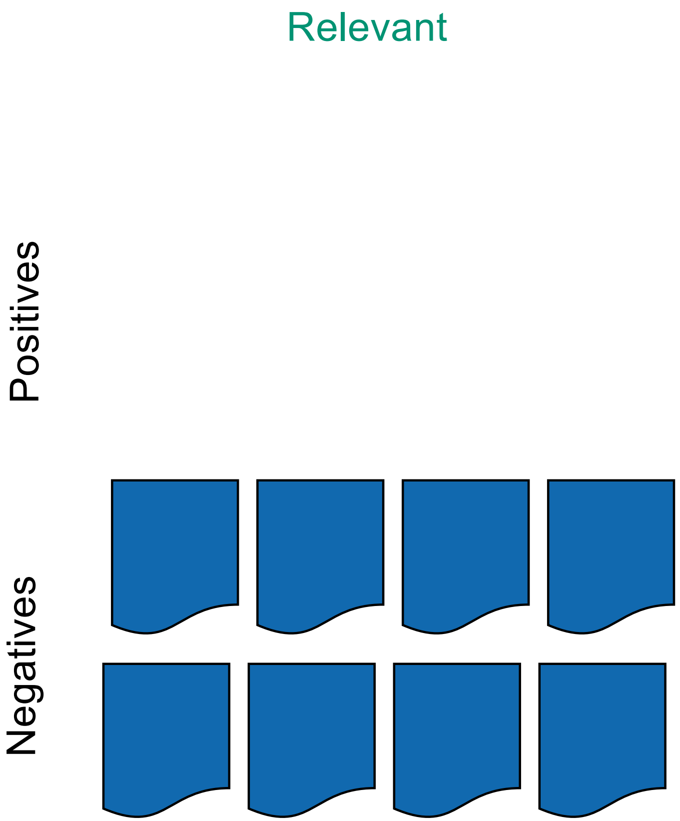

Recall =

# Recall



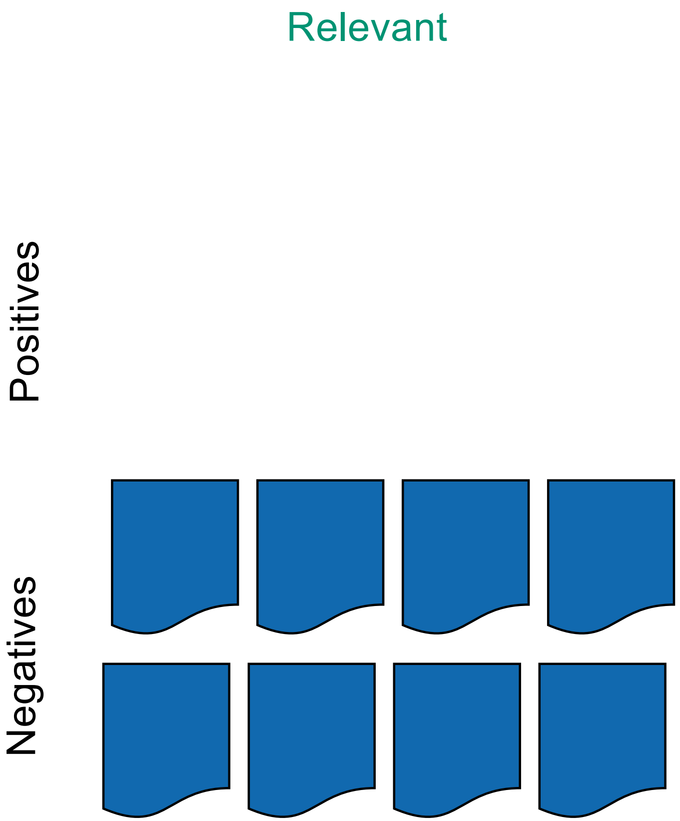

Recall = 100%

# Recall



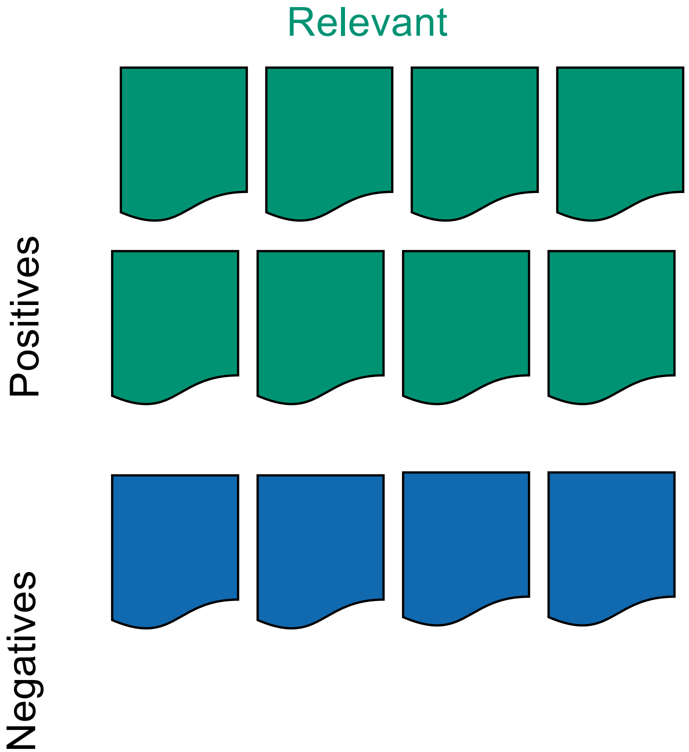

Recall = 0%

# Recall



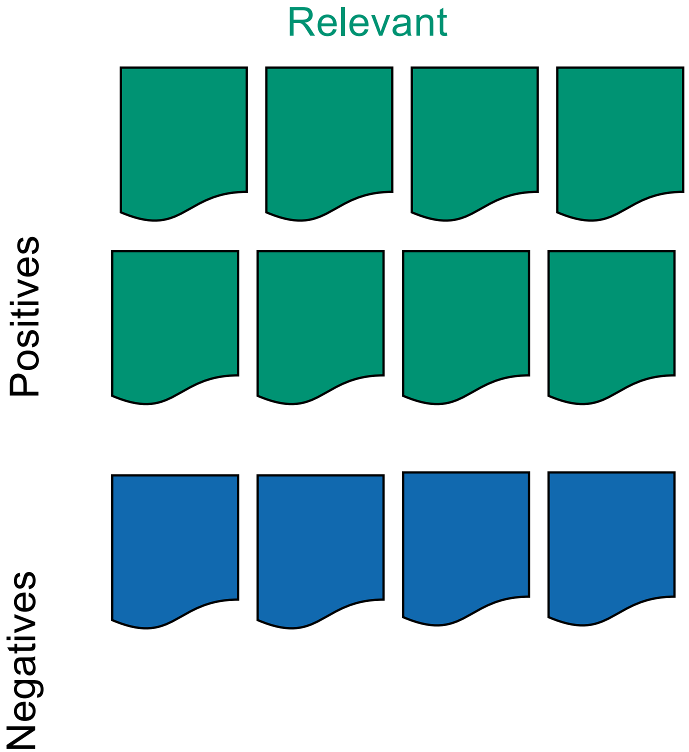

Recall =

# Recall



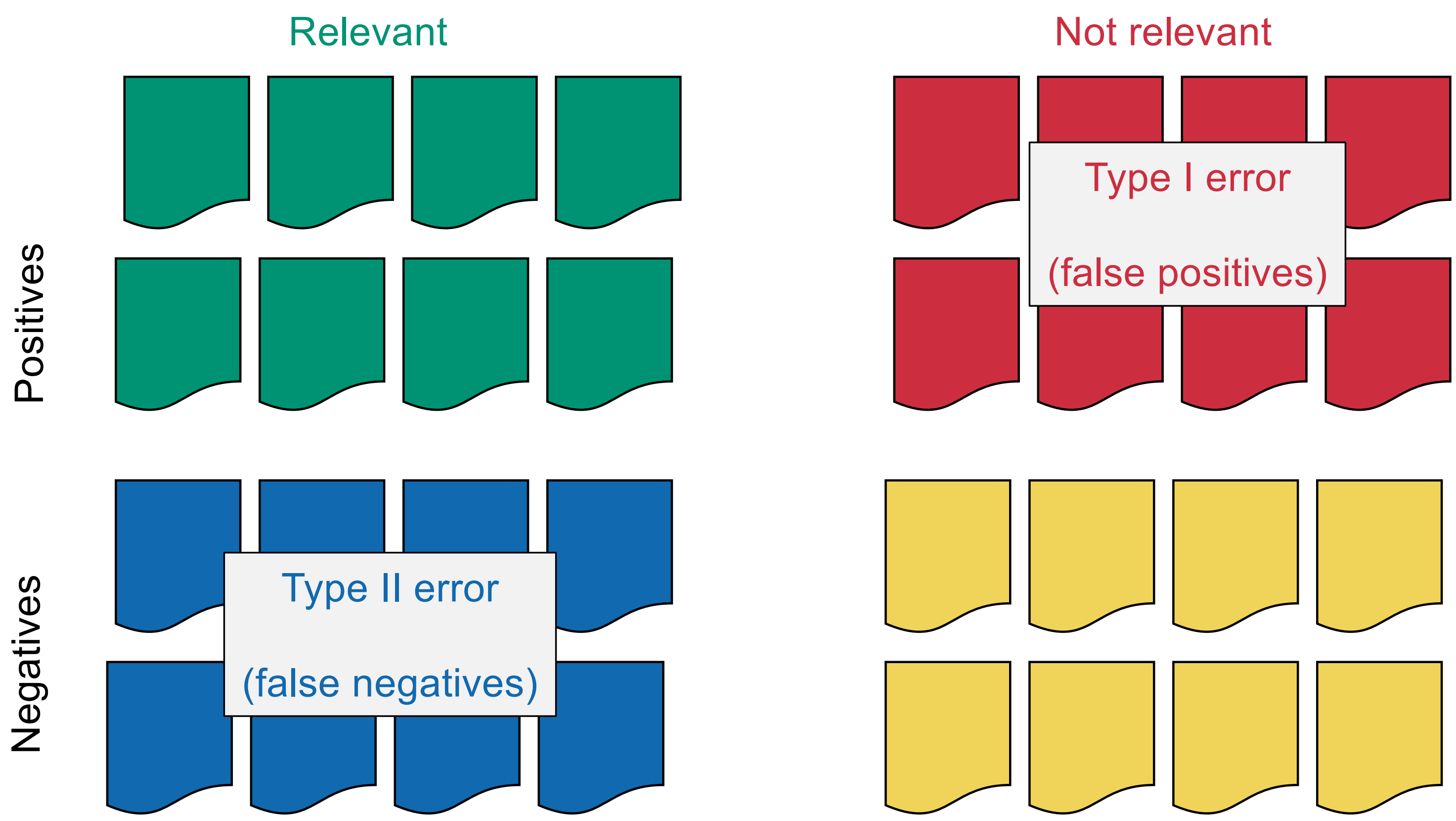

Recall =

# Recall

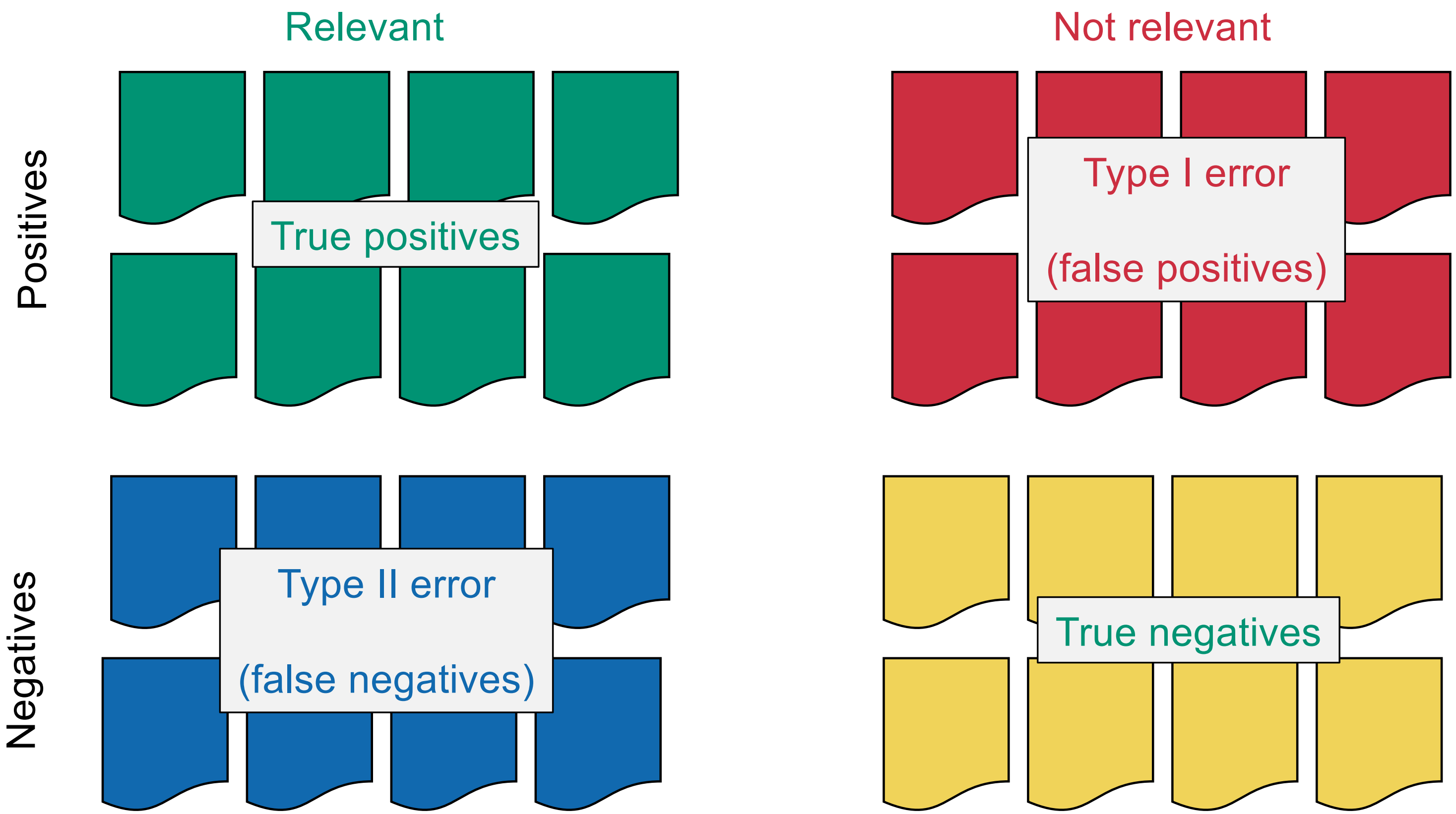



Recall = 67%

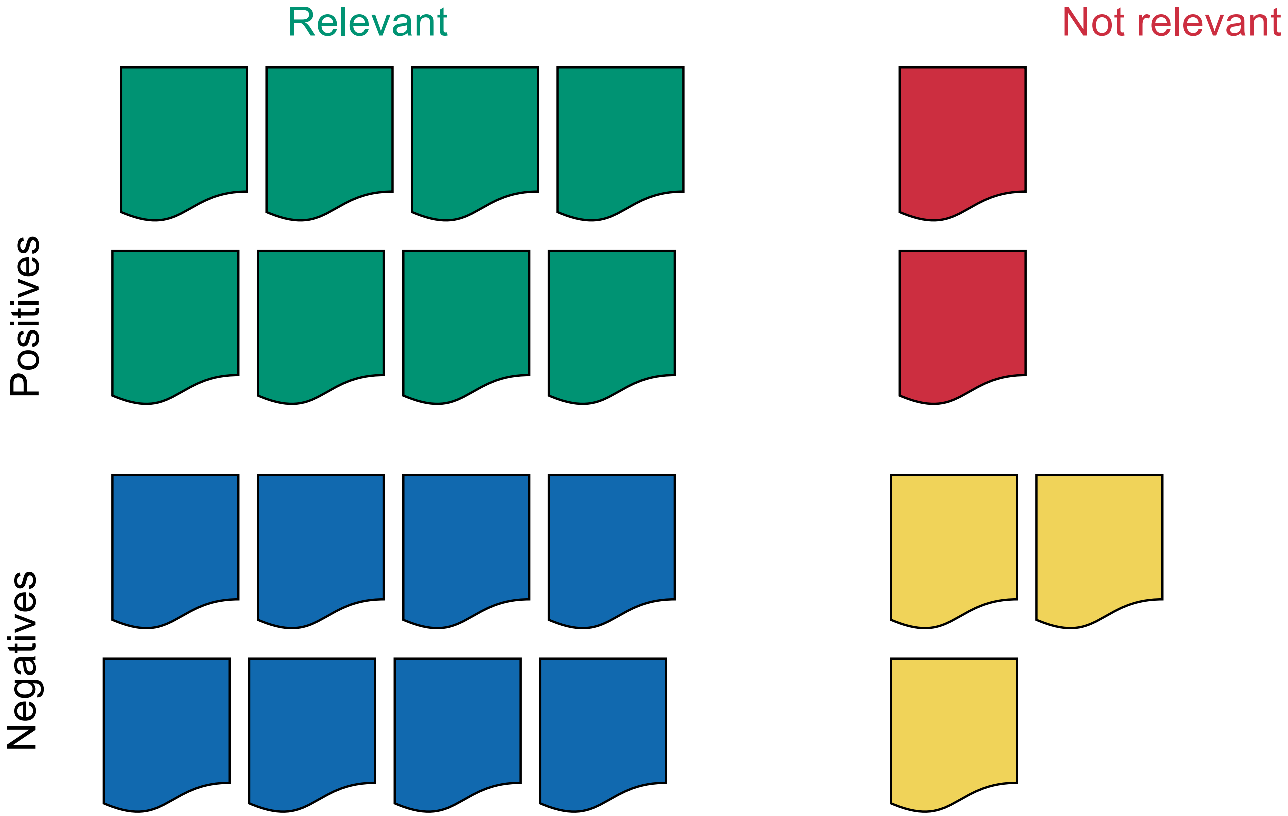
# Type I vs. Type II error



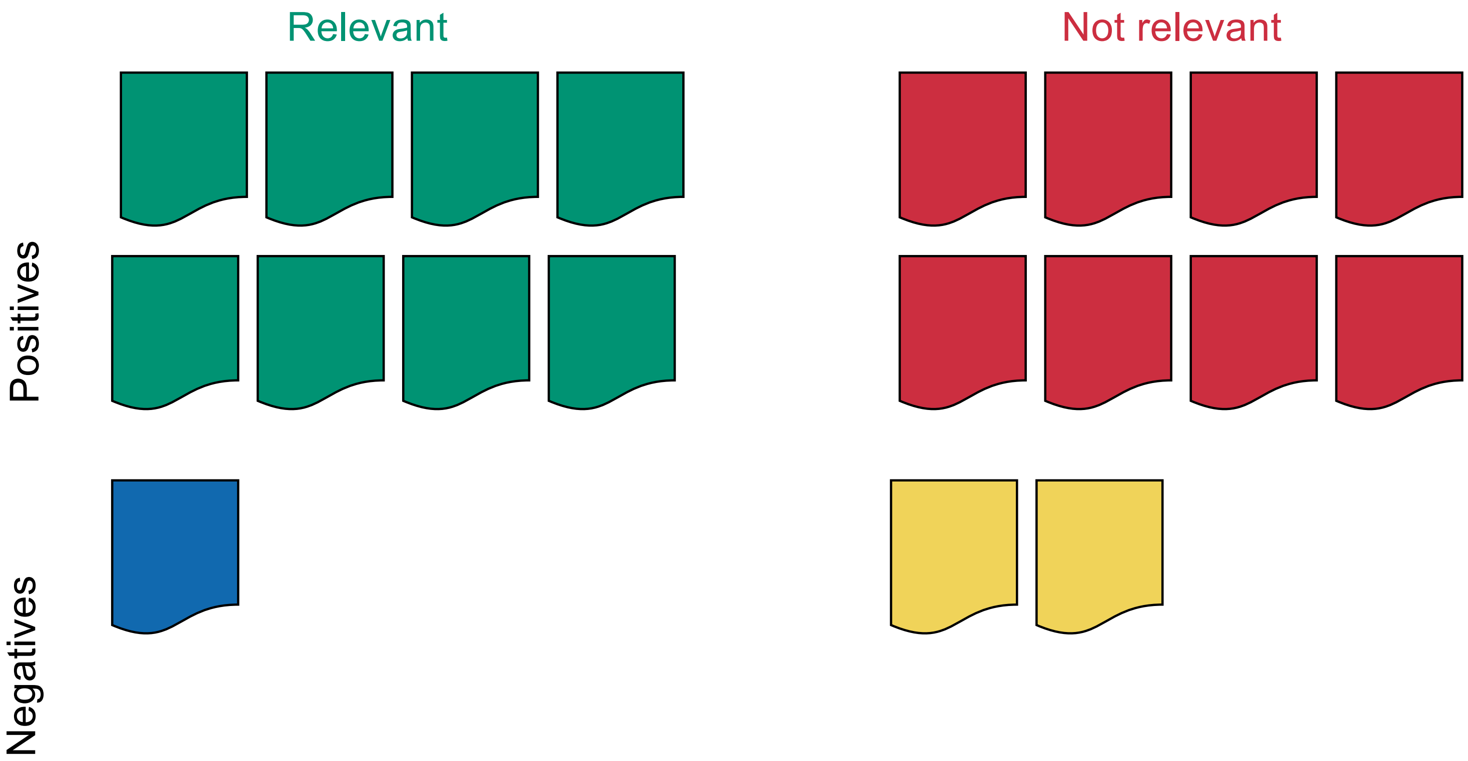
# Type I vs. Type II error



# High precision

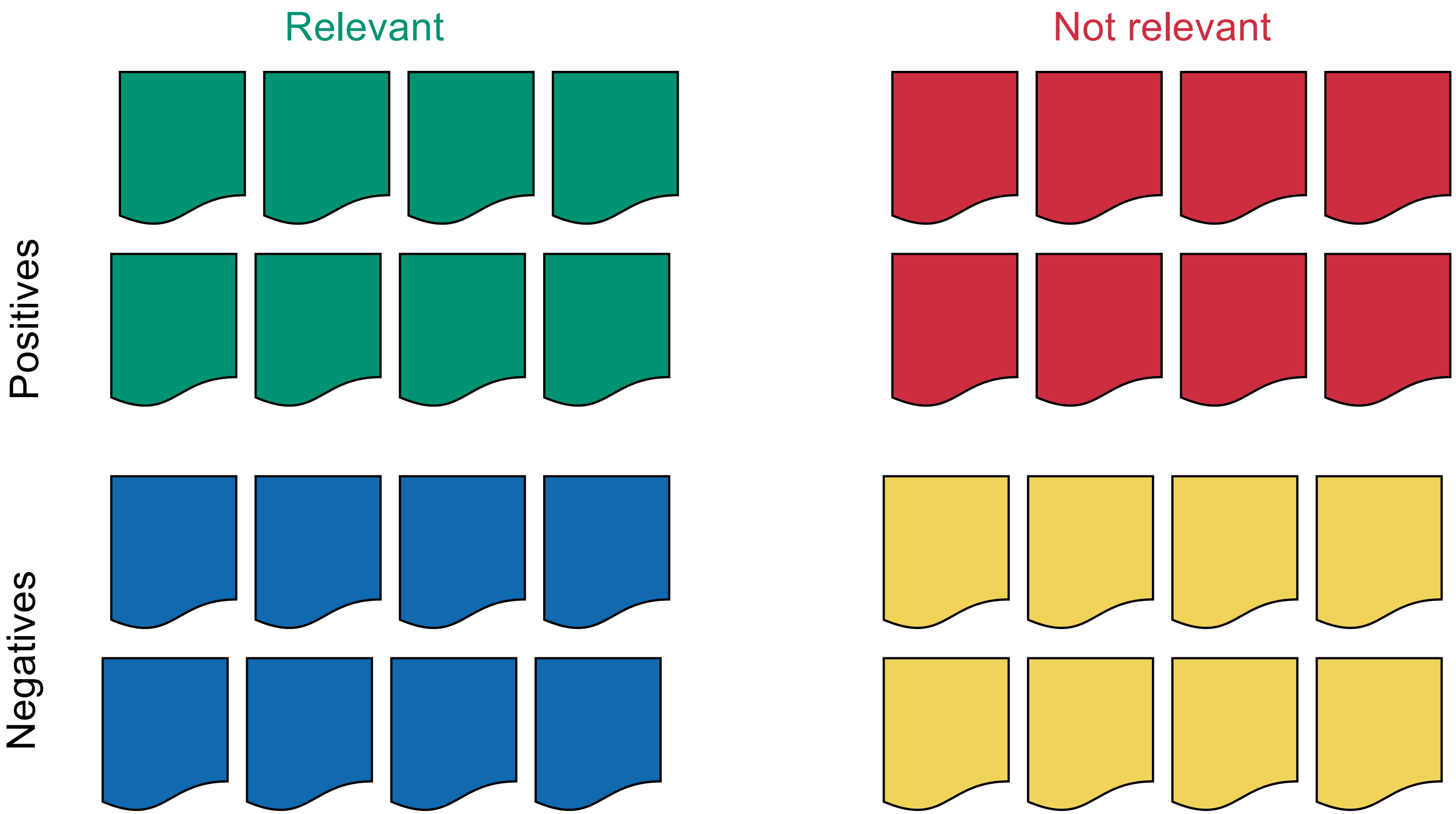


# High recall



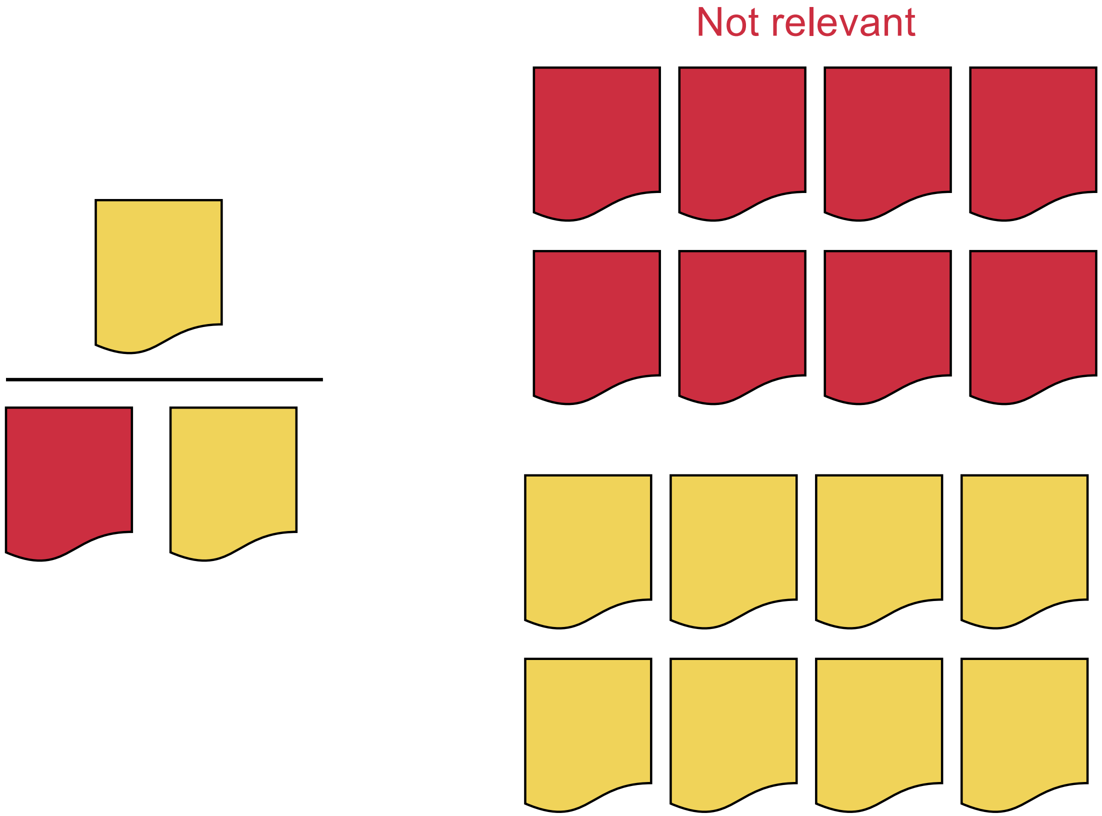
# Compromise

# Specificity



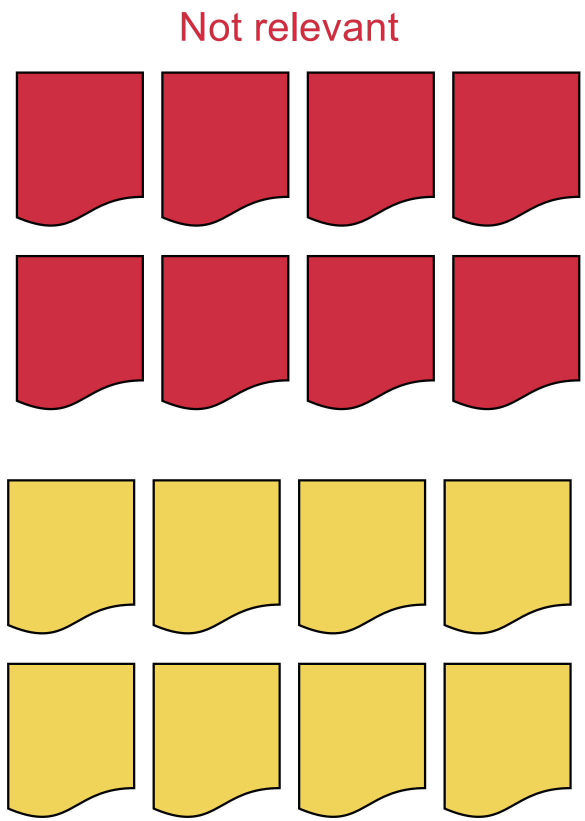
# Specificity

Specificity =



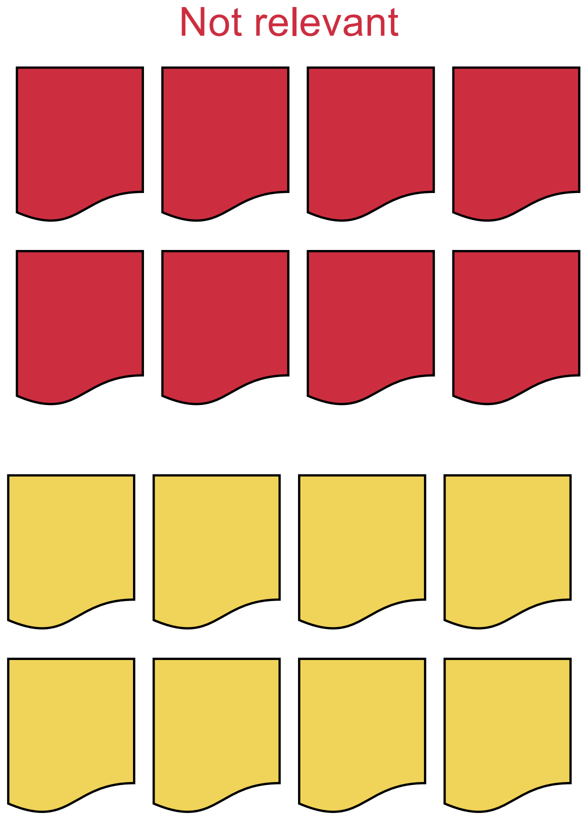
# Specificity

$$\text{Specificity} = \frac{\text{true negatives}}{\text{all negatives}}$$



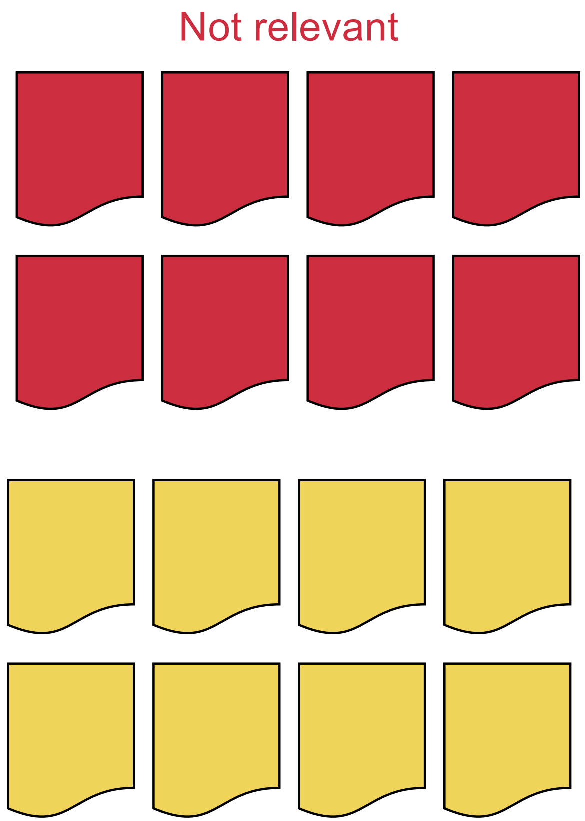
Specificity

Specificity =



## Specificity

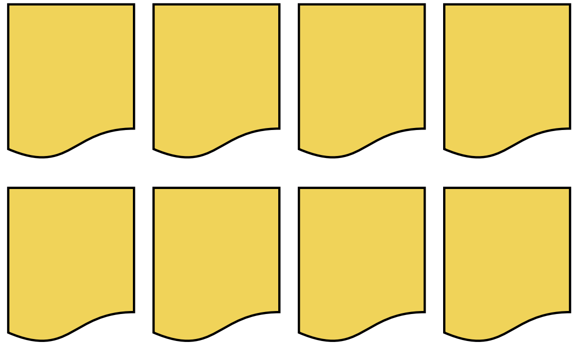
Specificity = 50%



Specificity

Not relevant

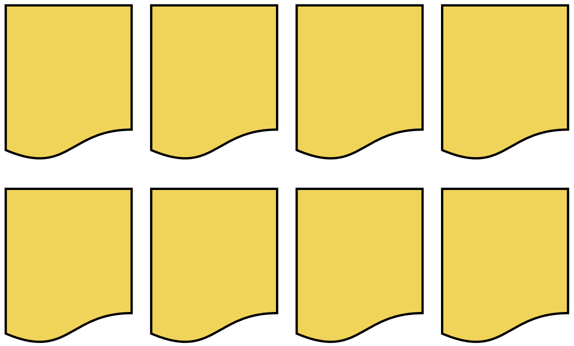
Specificity =



Specificity

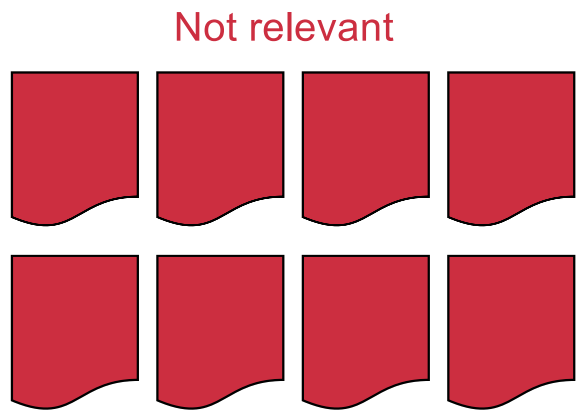
Not relevant

Specificity = 100%



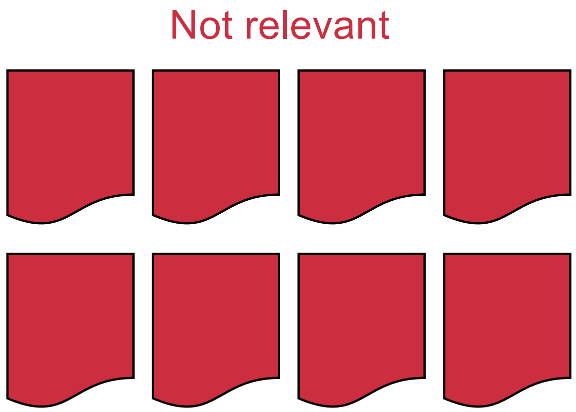
# Specificity

Specificity =



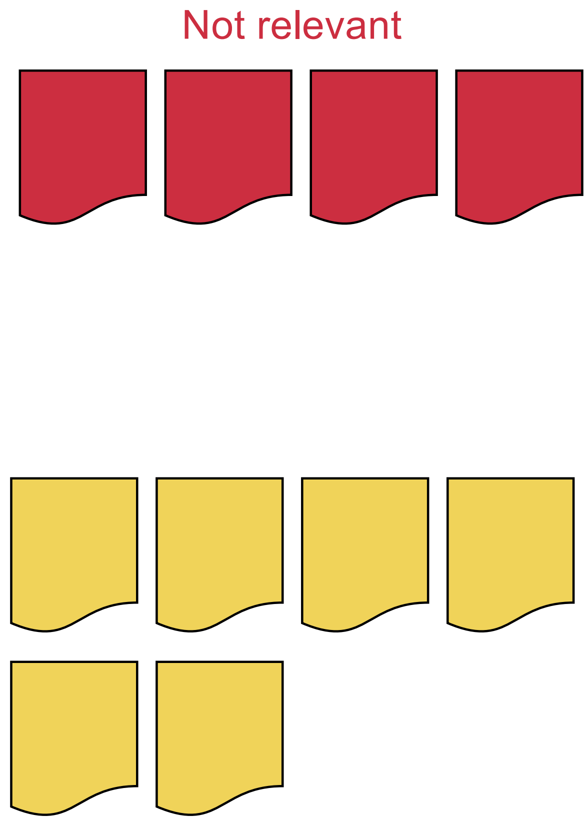
Specificity

Specificity = 0%



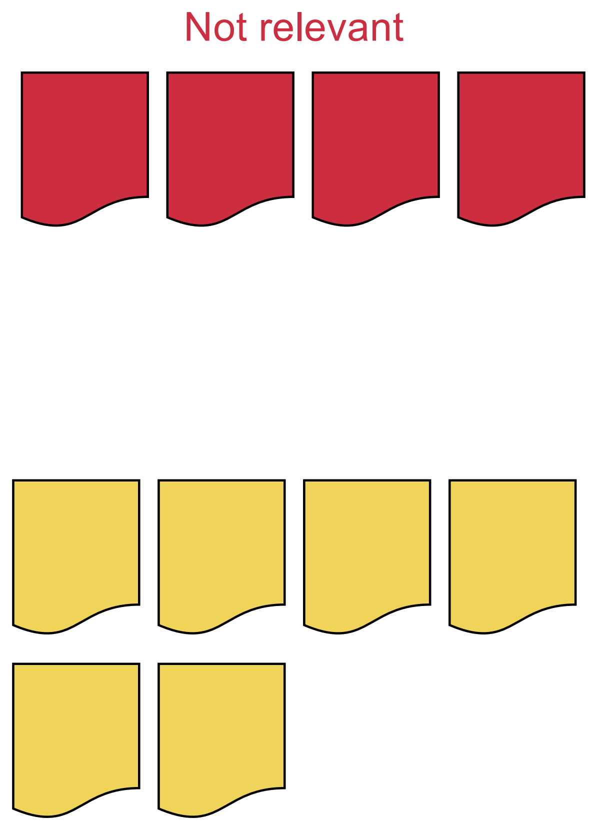
# Specificity

Specificity =

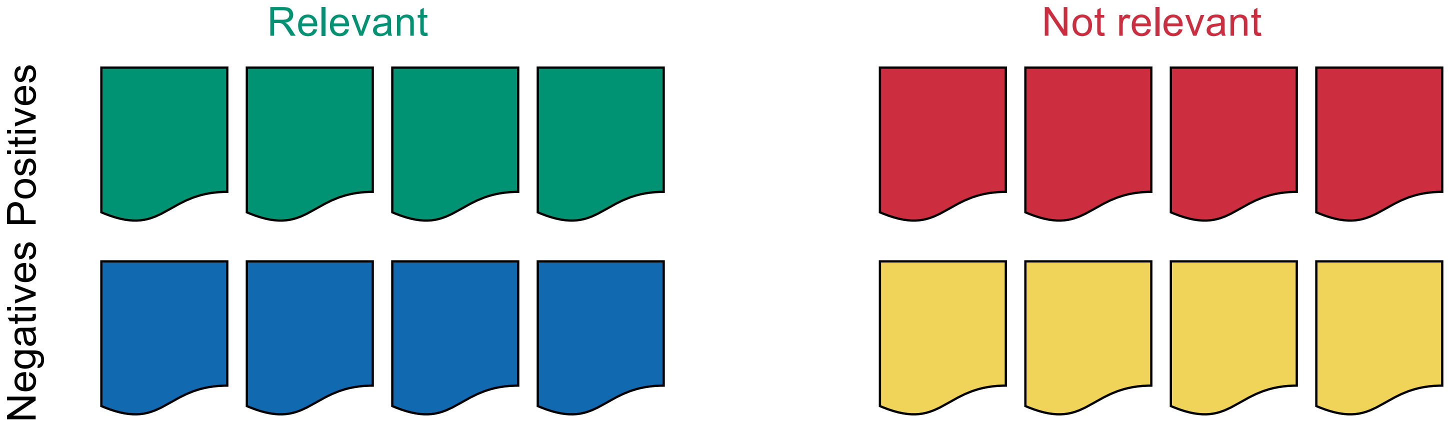


Specificity

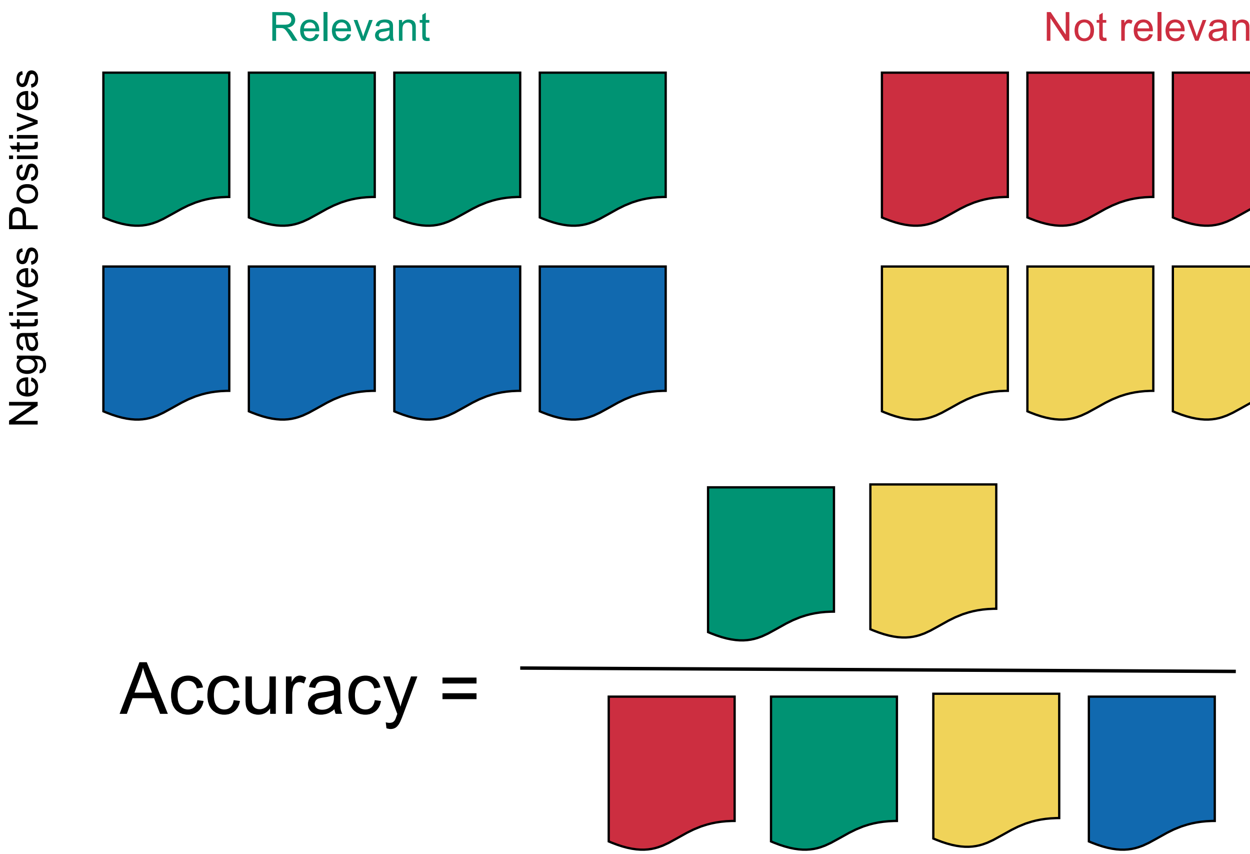
Specificity = 60%



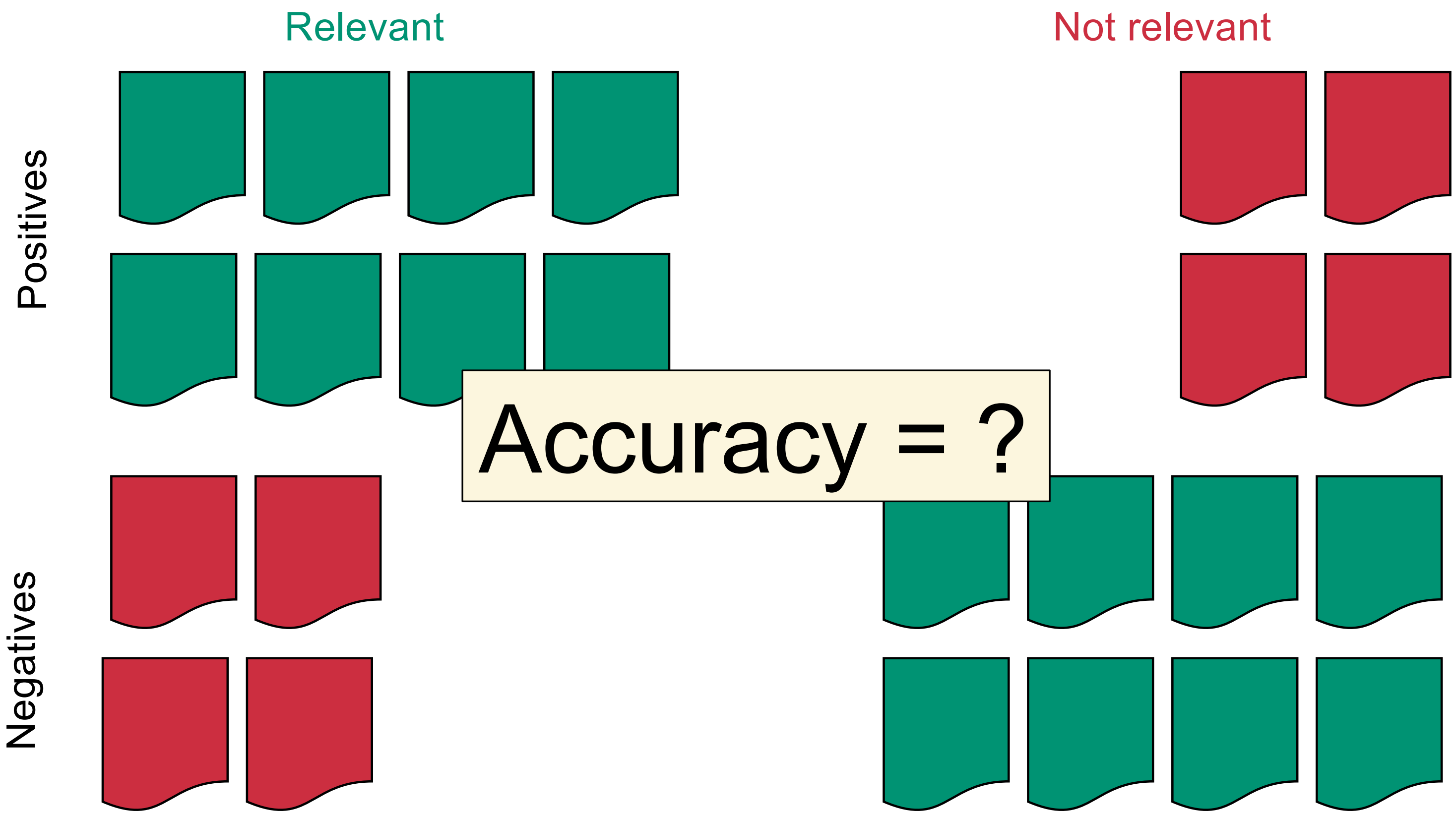
# Accuracy



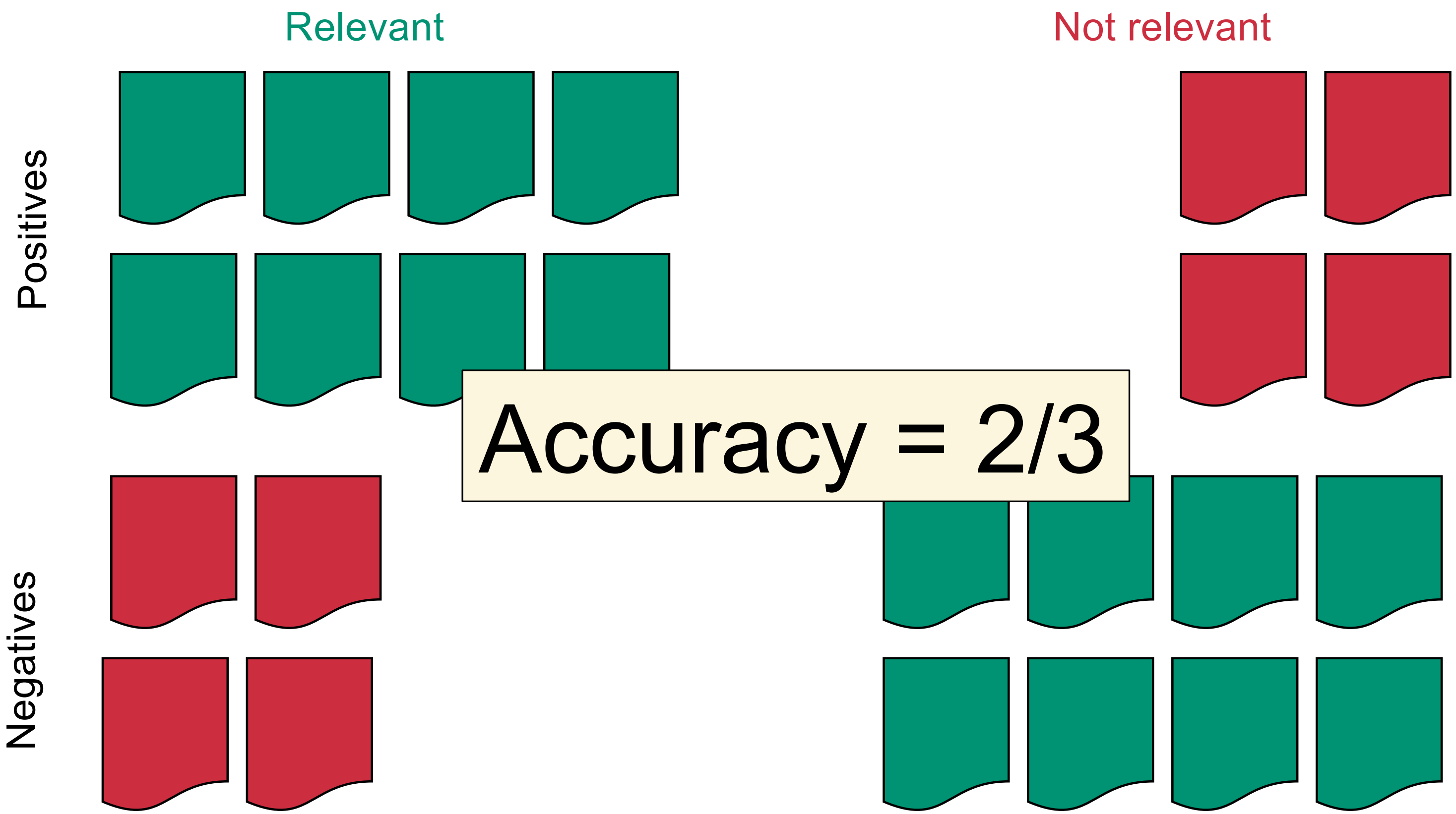
# Accuracy



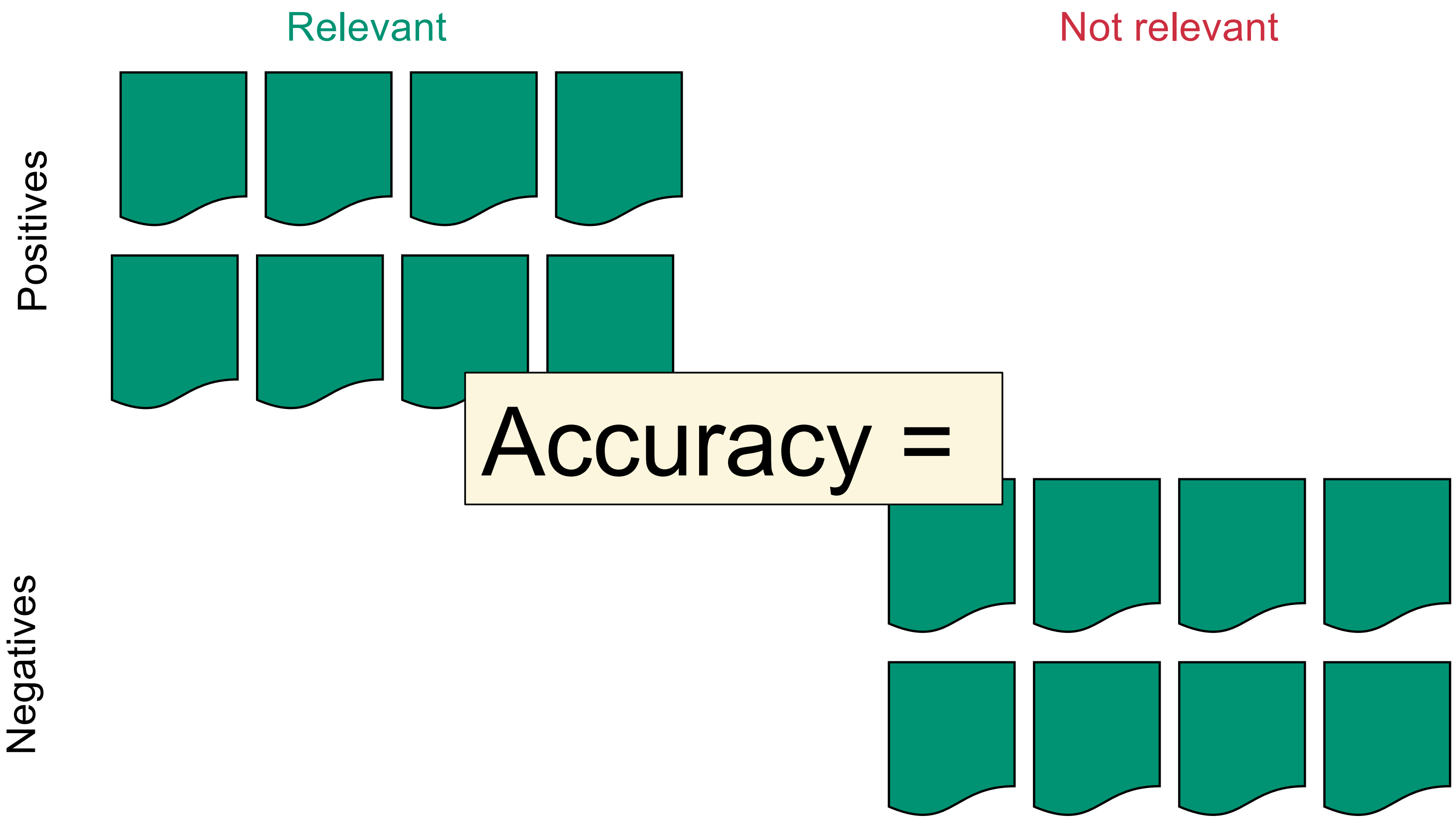
# Alternate quantification: accuracy



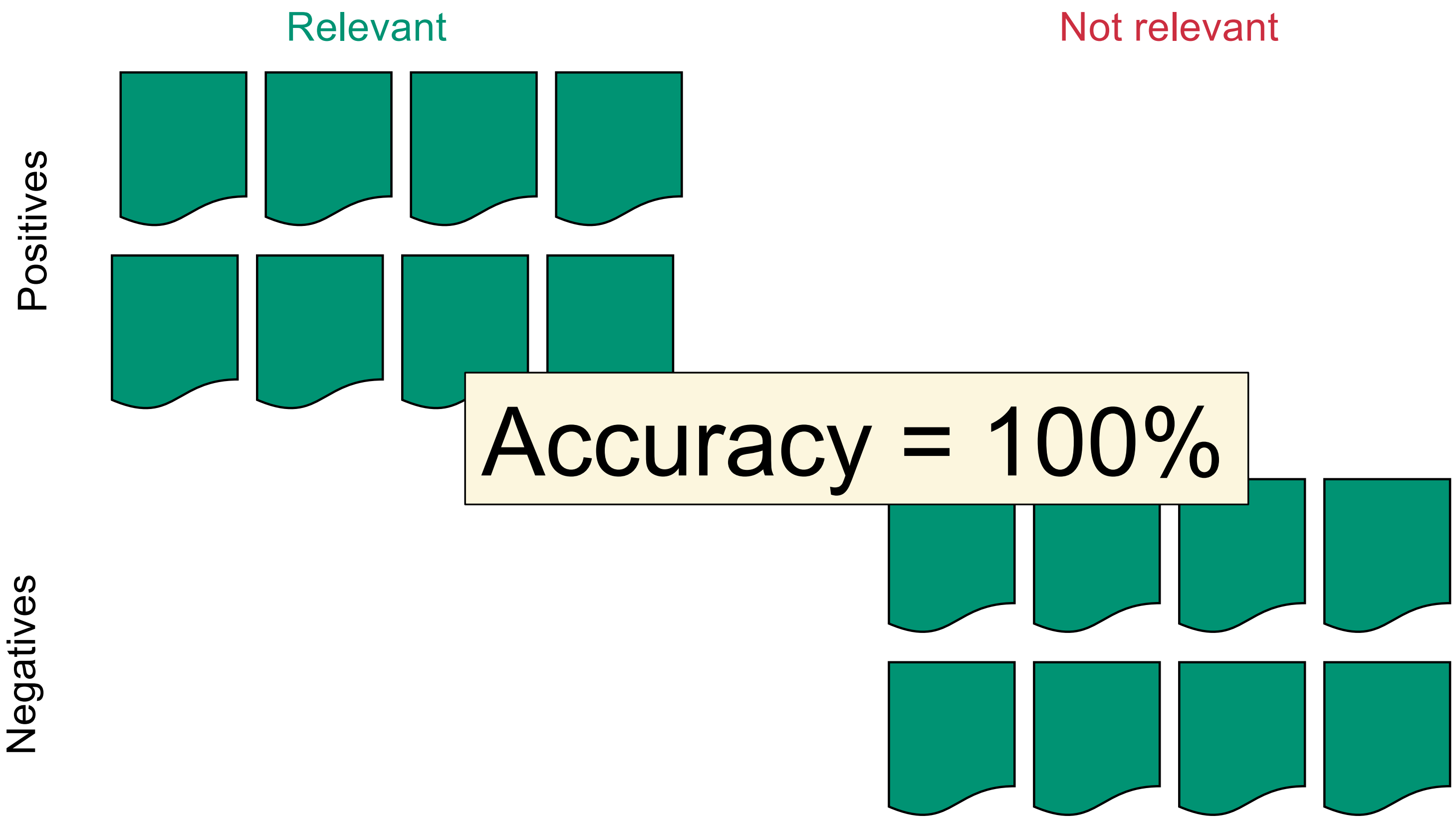
# Alternate quantification: accuracy



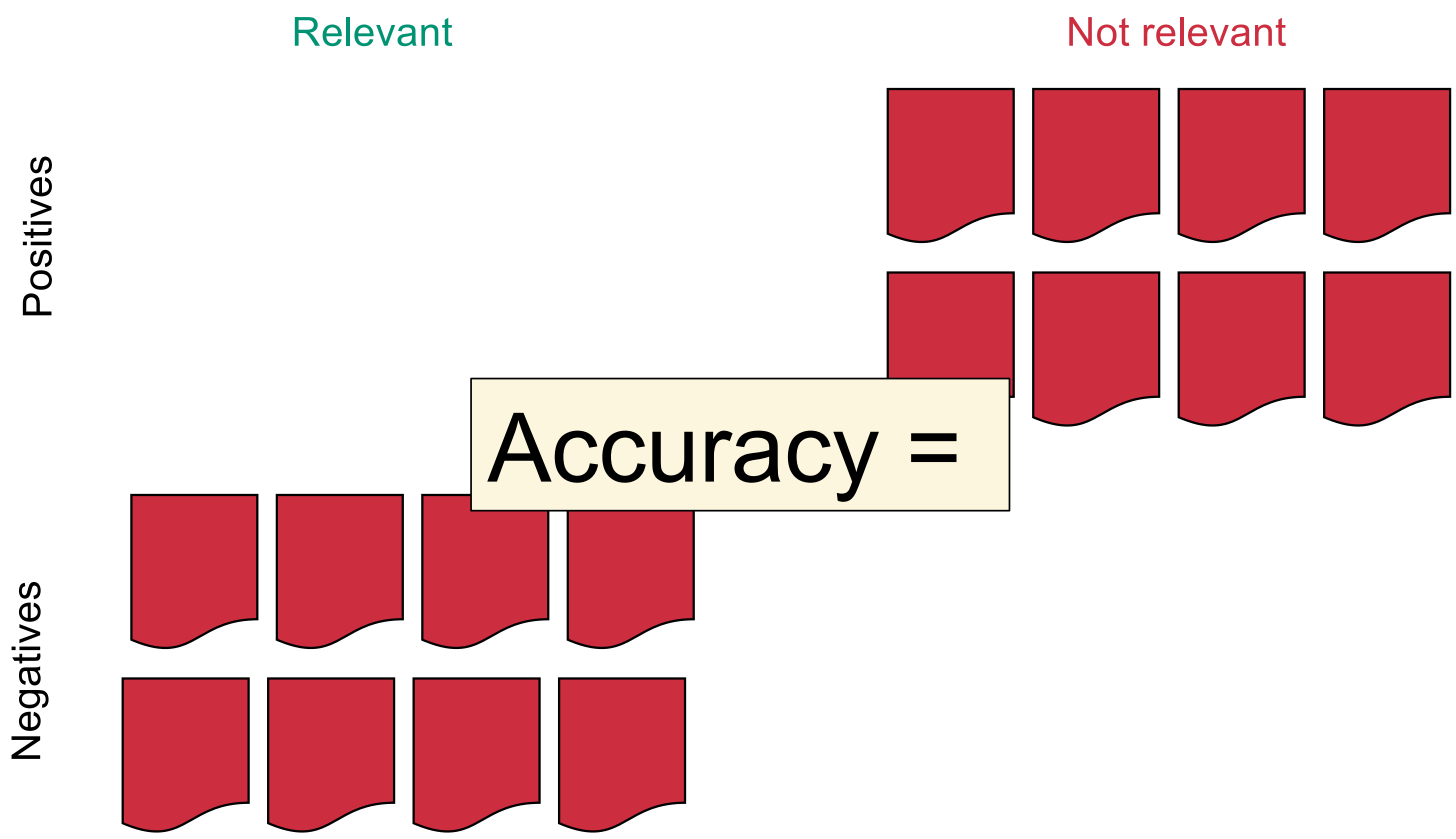
# Alternate quantification: accuracy



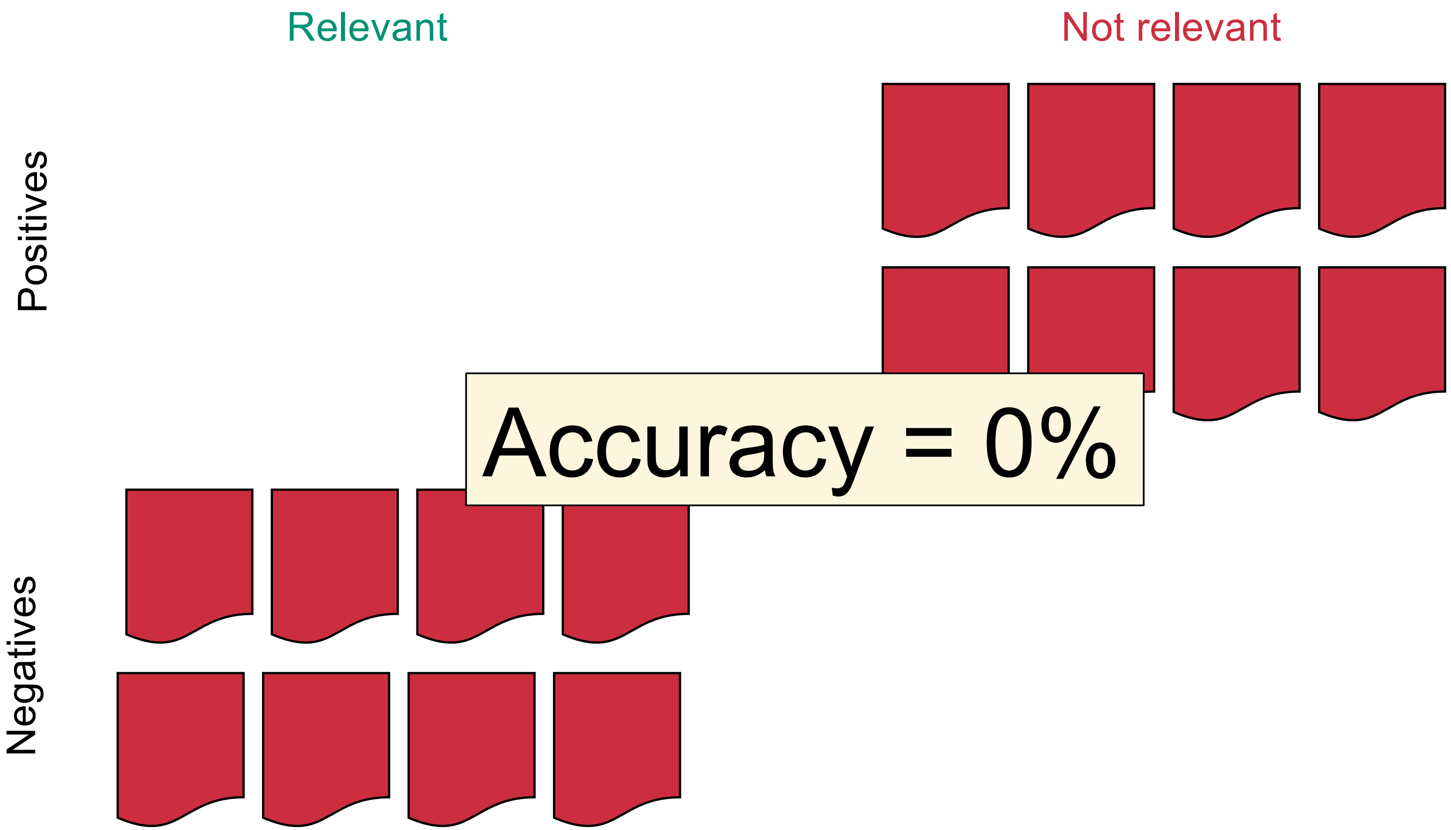
# Alternate quantification: accuracy



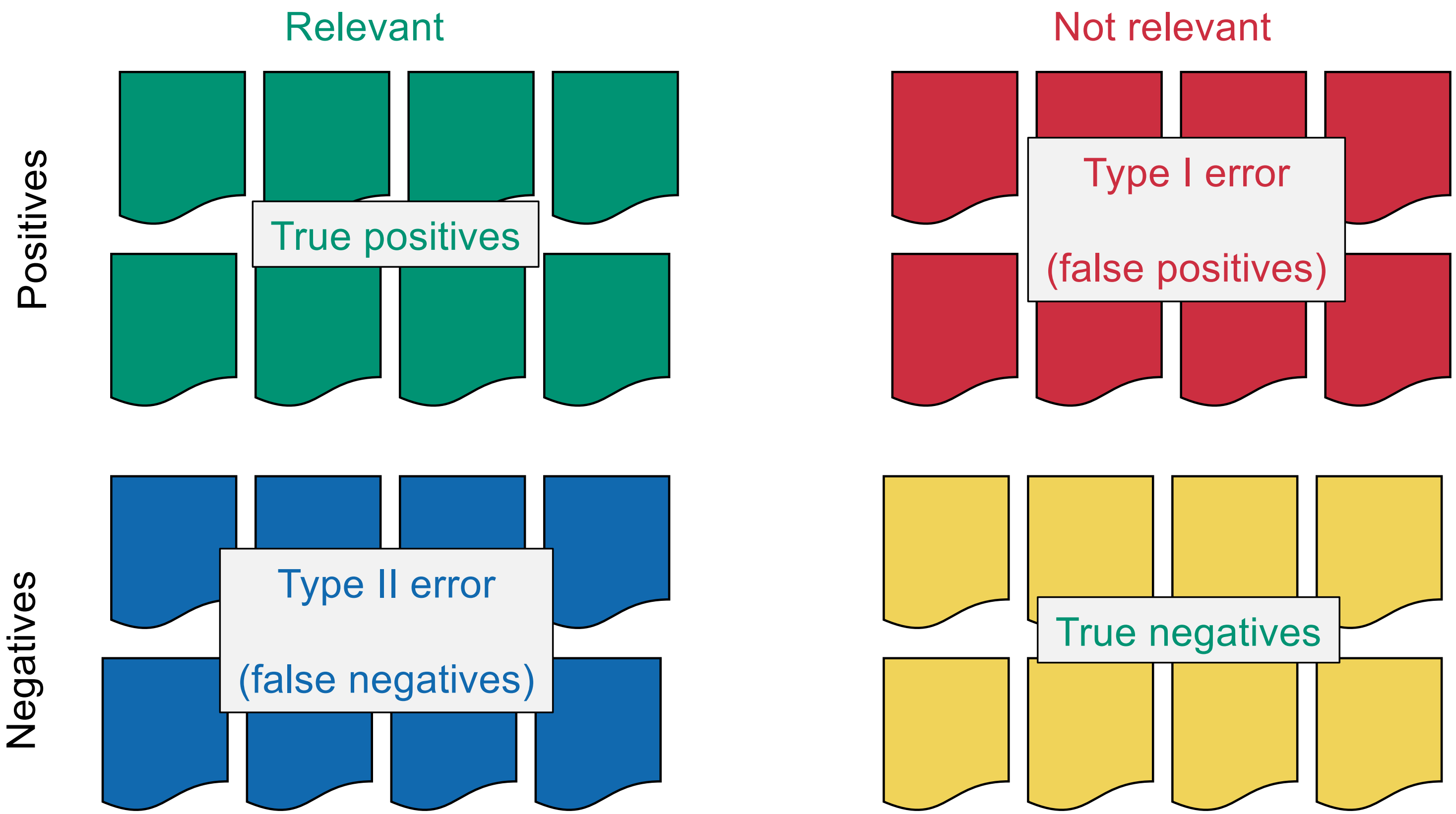
# Alternate quantification: accuracy



# Alternate quantification: accuracy



# Hacking accuracy?

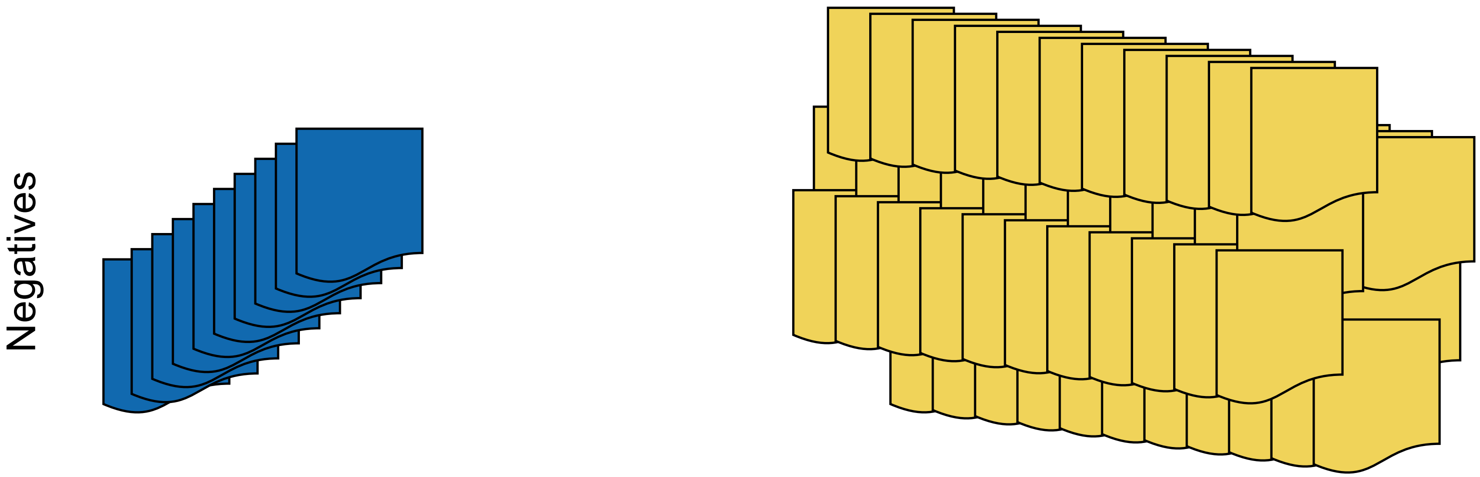


# Hacking accuracy...

Relevant

Not relevant

Never return anything



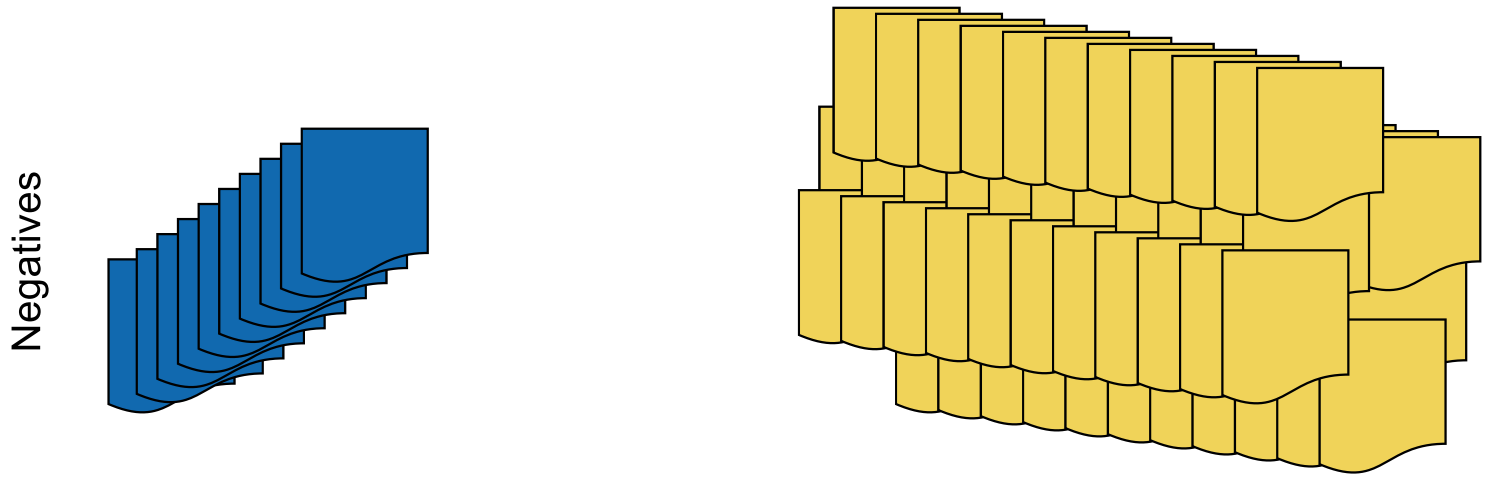
Hacking accuracy...

Relevant

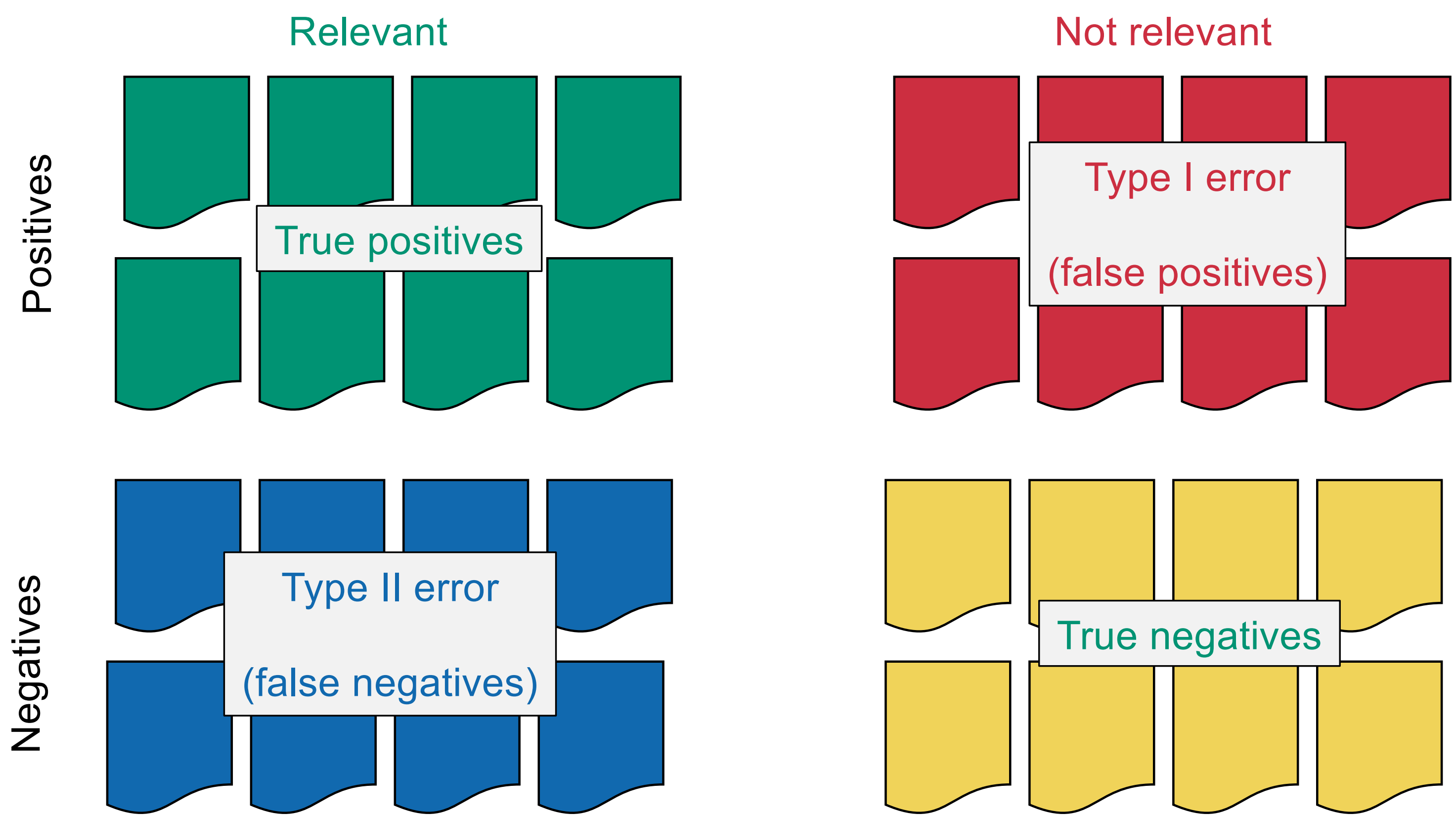
Not relevant

Never return anything

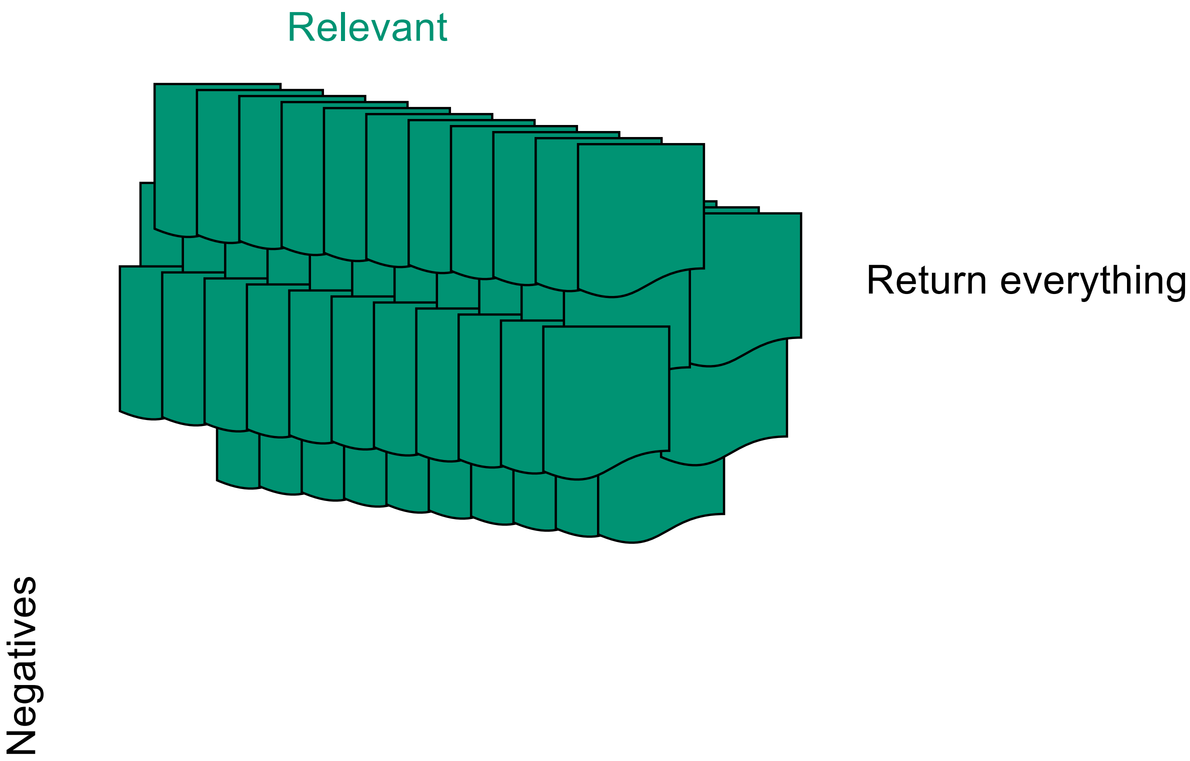
**Accuracy = 99%**



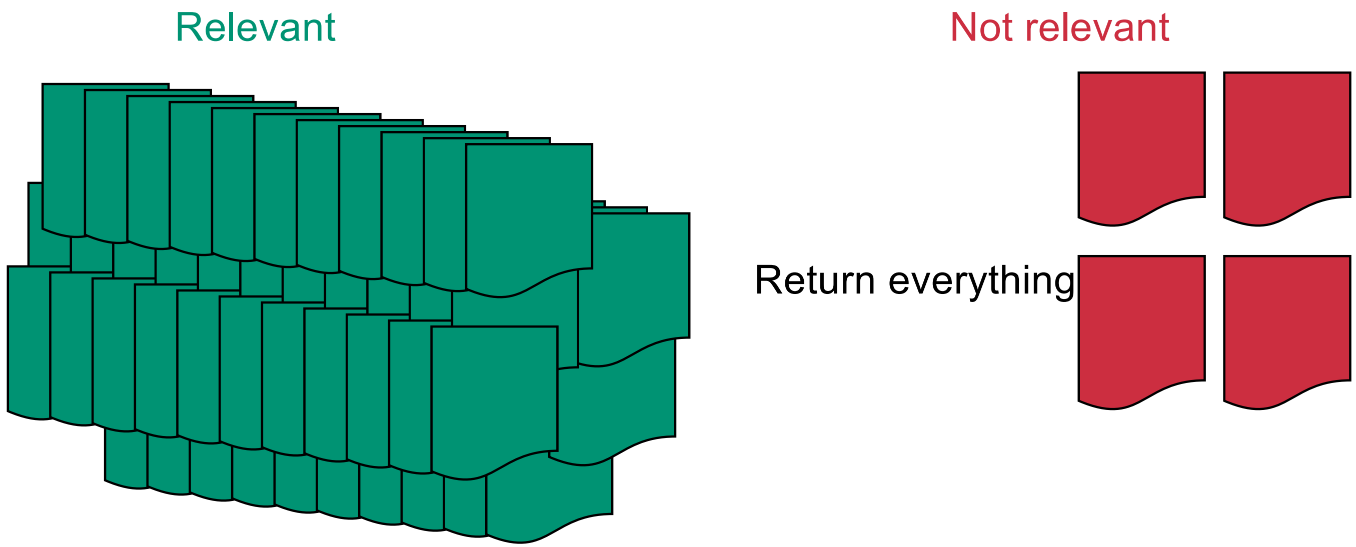
# Hacking recall?



# Hacking recall...



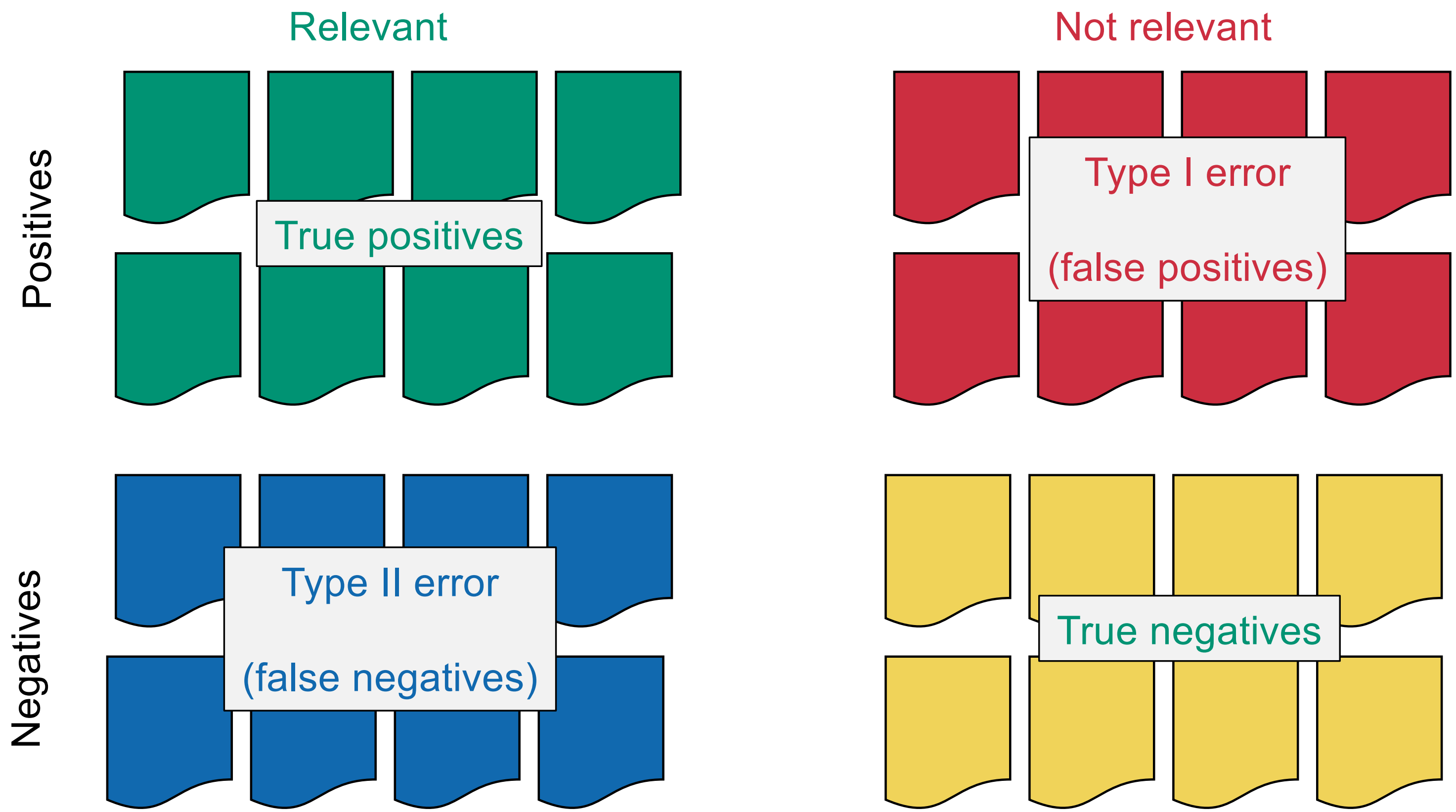
# Hacking recall...




Negatives

**Recall = 100%**

# "Merging" Precision and Recall



First idea:

First idea: arithmetic mean

$$M = \frac{P + R}{2}$$

Poll

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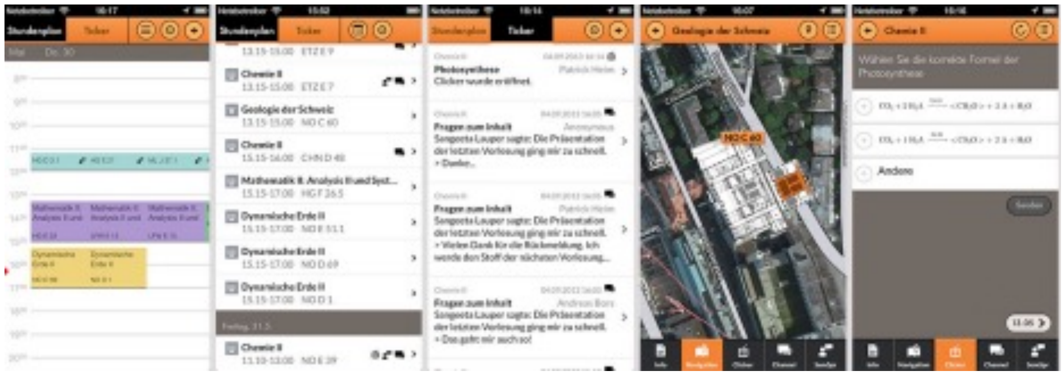


F measure: harmonic mean

$$F_{0.5} = \frac{2}{\frac{1}{P} + \frac{1}{R}}$$

F measure: harmonic mean

$$F_{0.5} = \frac{2PR}{P + R}$$

F measure: harmonic mean

$$F_\alpha = \frac{1}{\frac{\alpha}{P} + \frac{1-\alpha}{R}}$$

F measure: harmonic mean

$$F_\alpha = \frac{1}{\frac{\alpha}{P} + \frac{1-\alpha}{R}}$$

$$\alpha = 0$$

$$\alpha = 1$$

# F measure: harmonic mean

$$F_\alpha = \frac{1}{\frac{\alpha}{P} + \frac{1-\alpha}{R}}$$

$\alpha = 0$

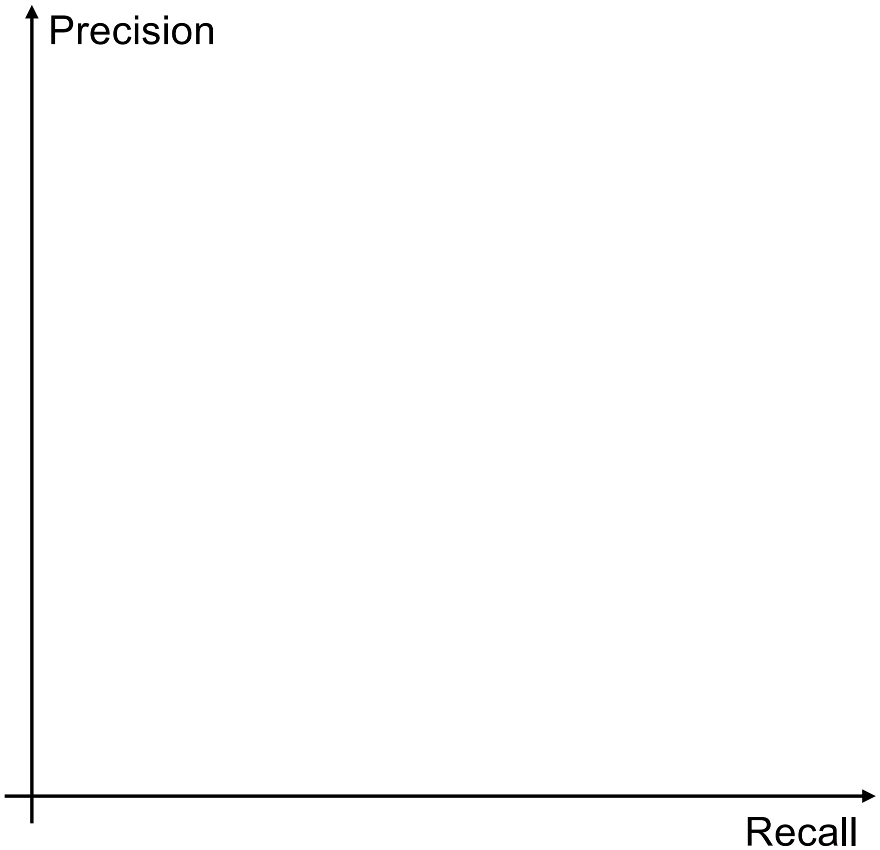
$\alpha = 1$

Recall-heavy

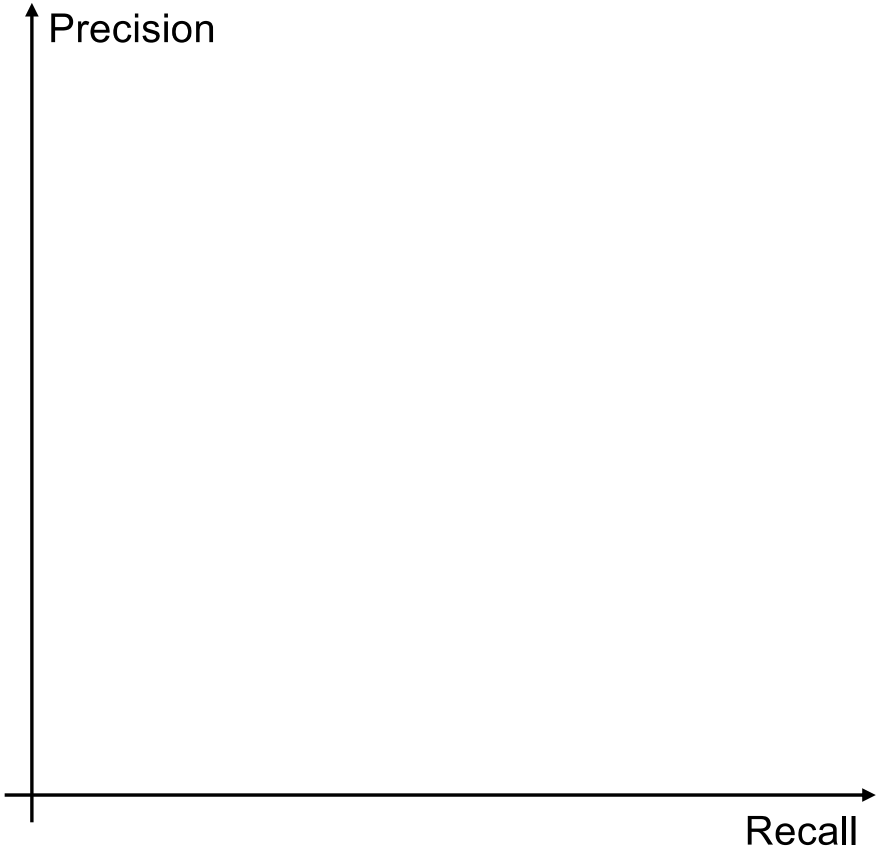

Precision-heavy

# Precision and Recall in Ranked Retrieval

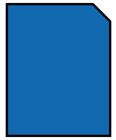
# Precision vs. Recall



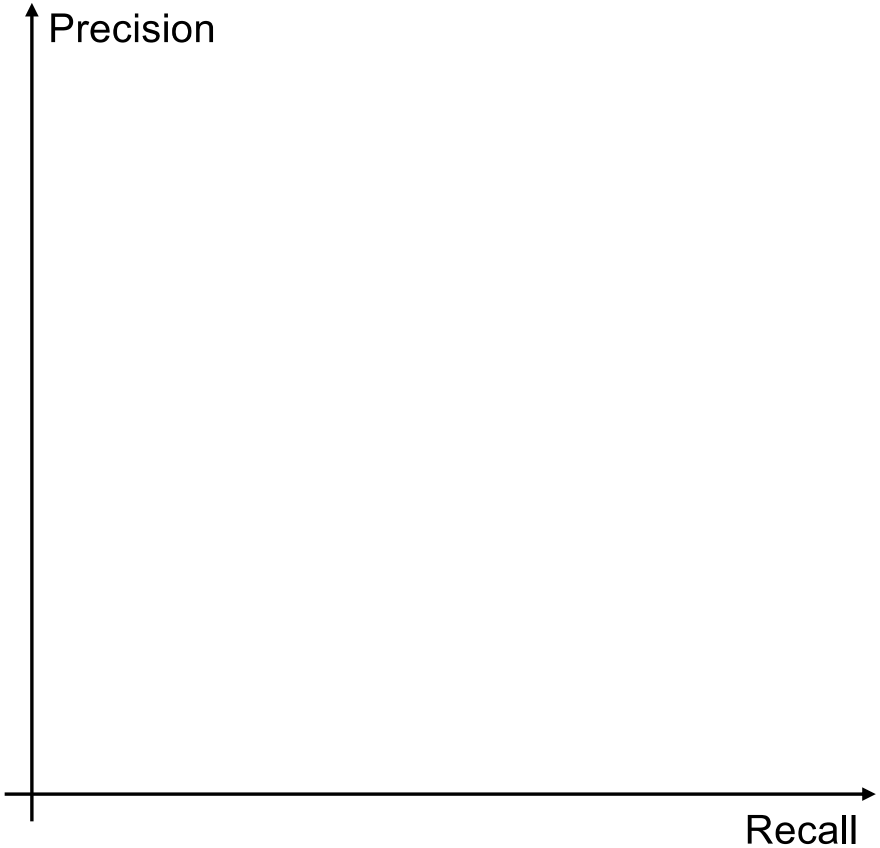
# Precision vs. Recall



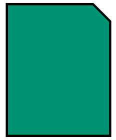

Top 1



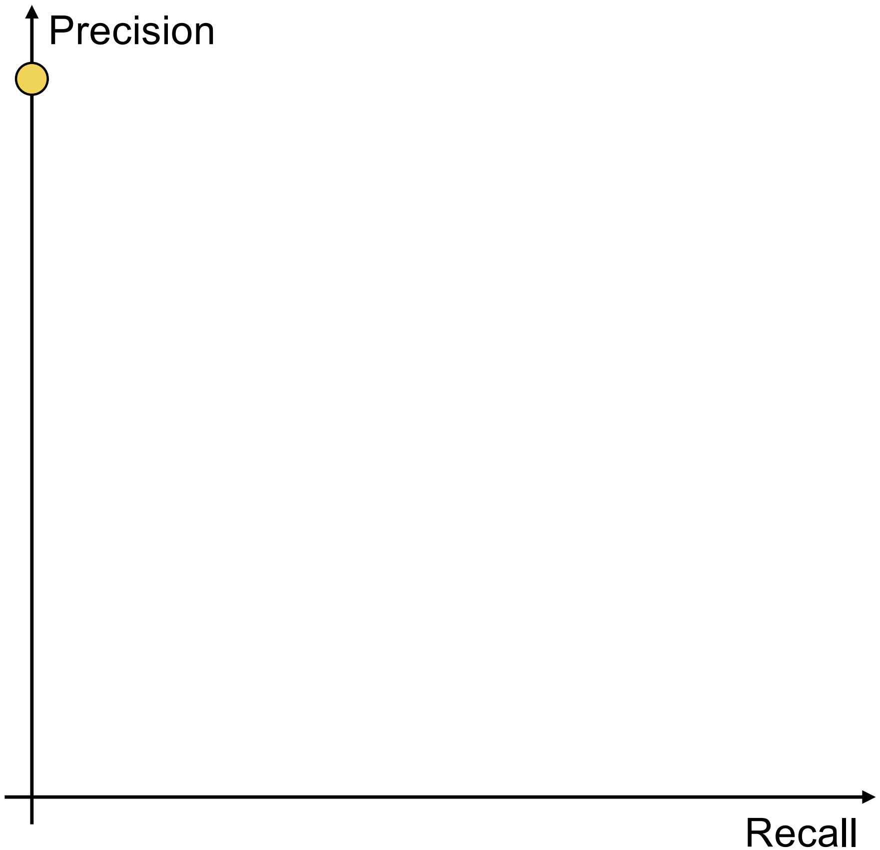
# Precision vs. Recall



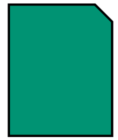

Top 1



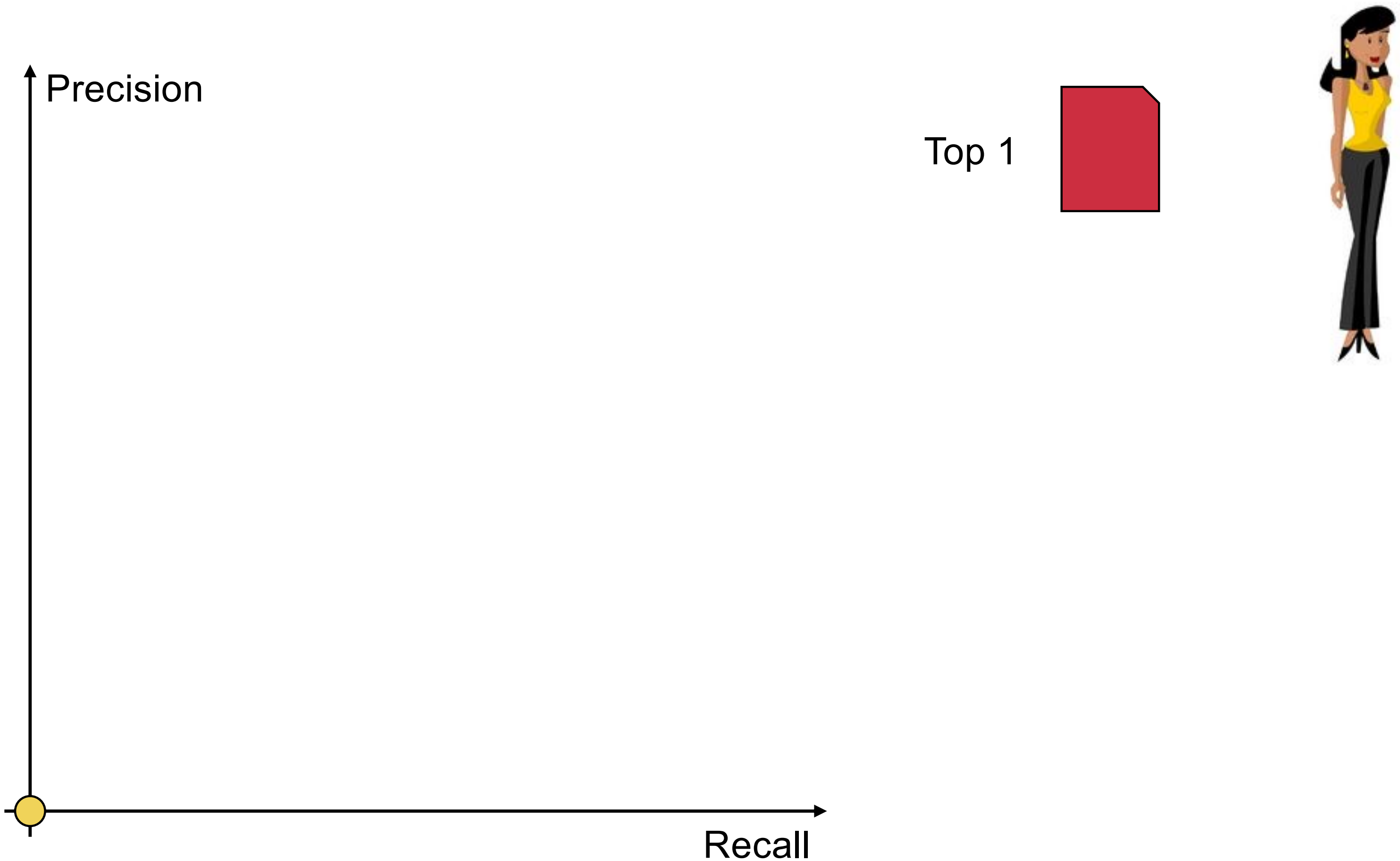
# Precision vs. Recall



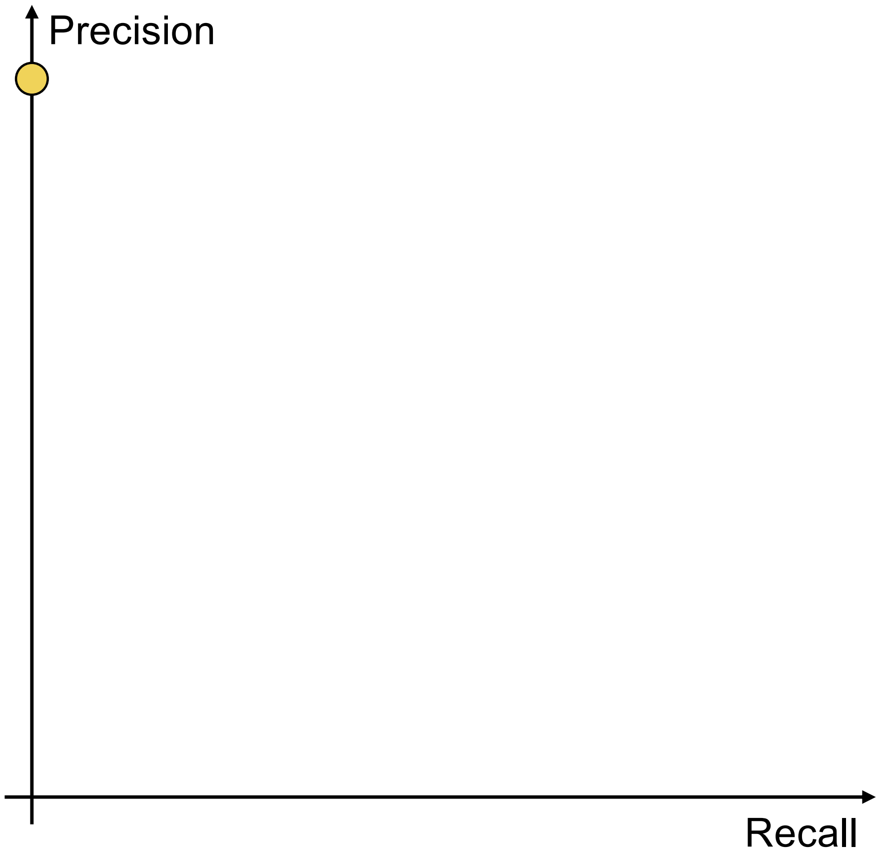

Top 1

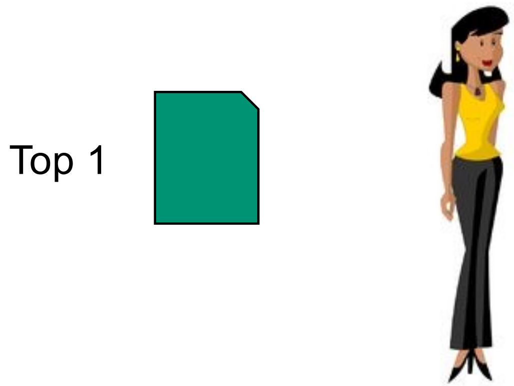


# Precision vs. Recall

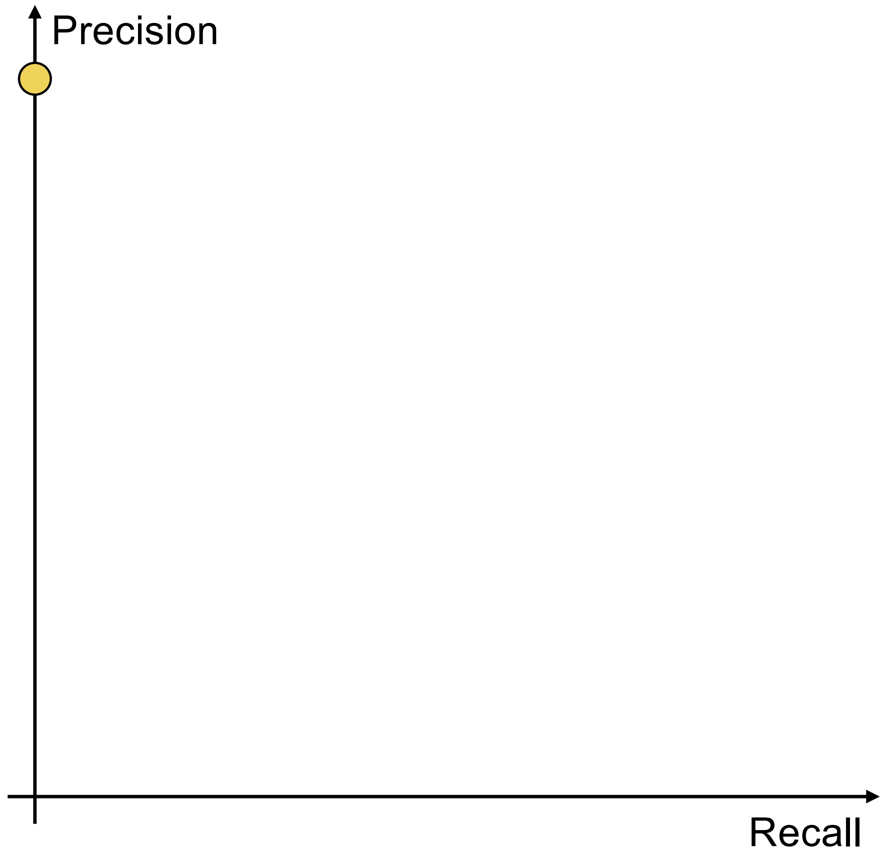


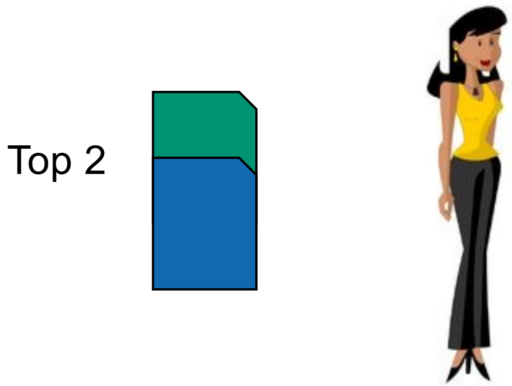
# Precision vs. Recall



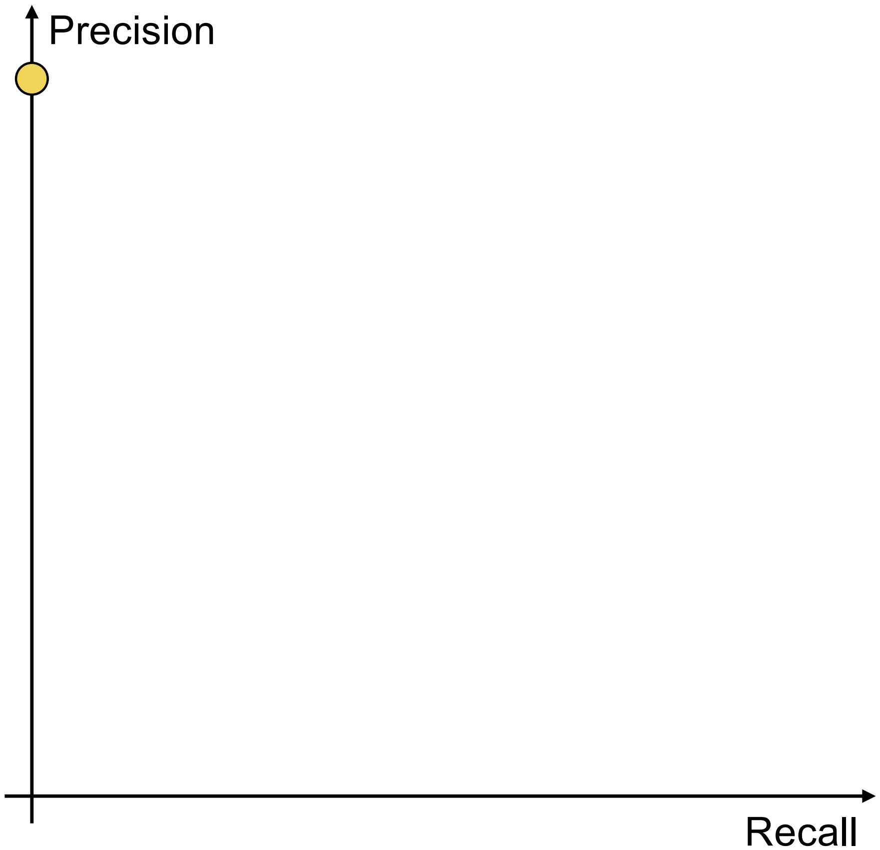



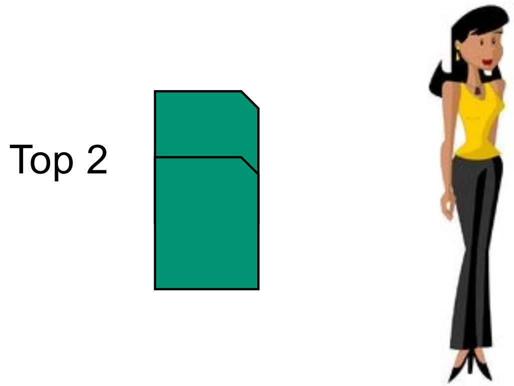
# Precision vs. Recall



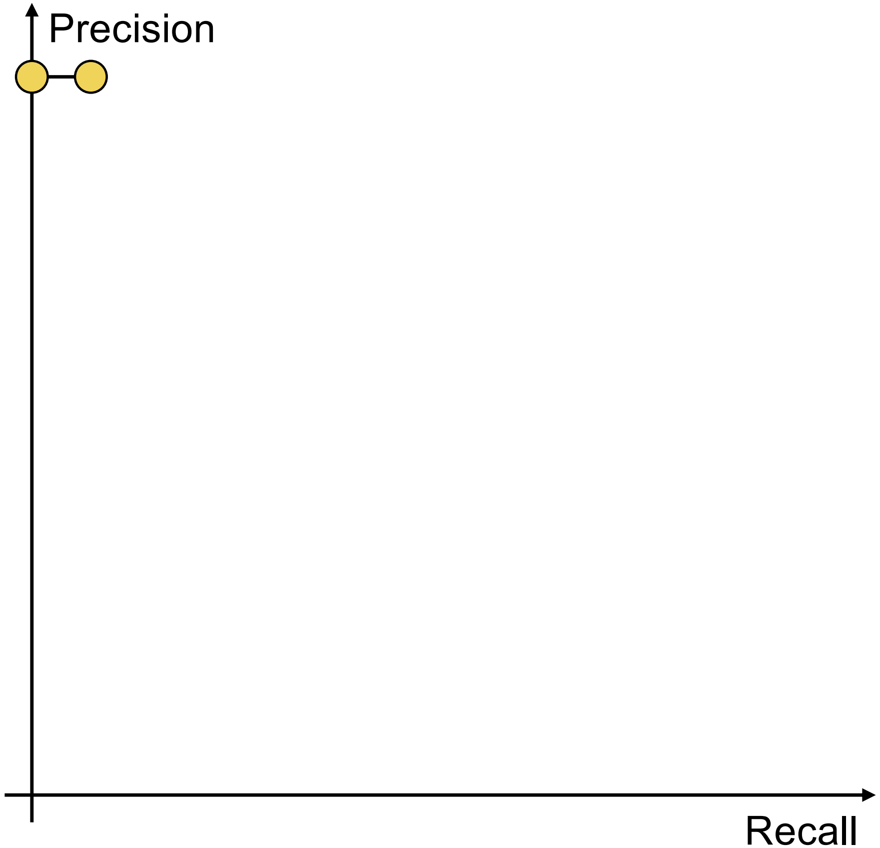



# Precision vs. Recall

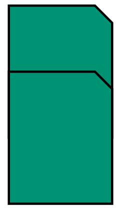


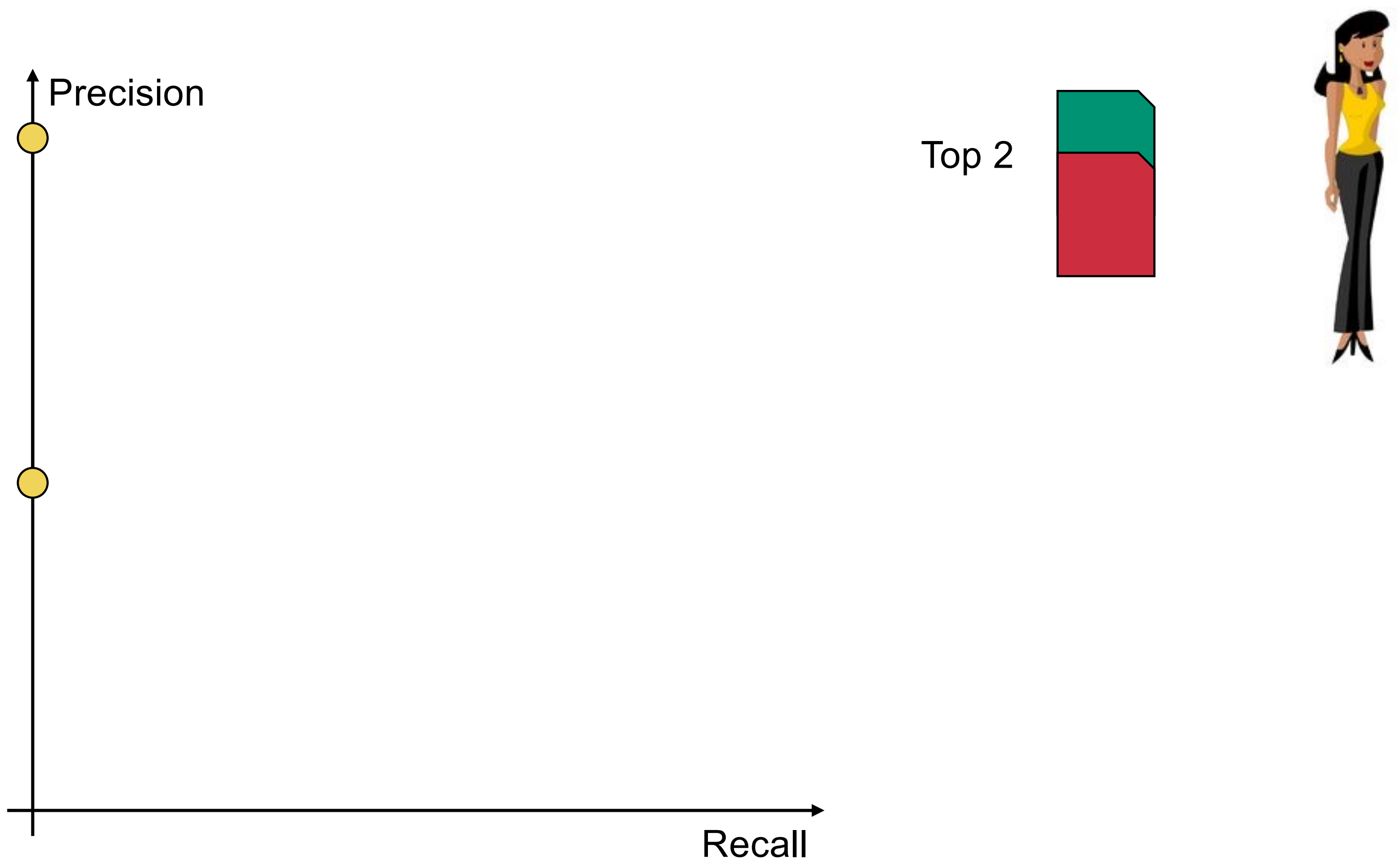
# Precision vs. Recall



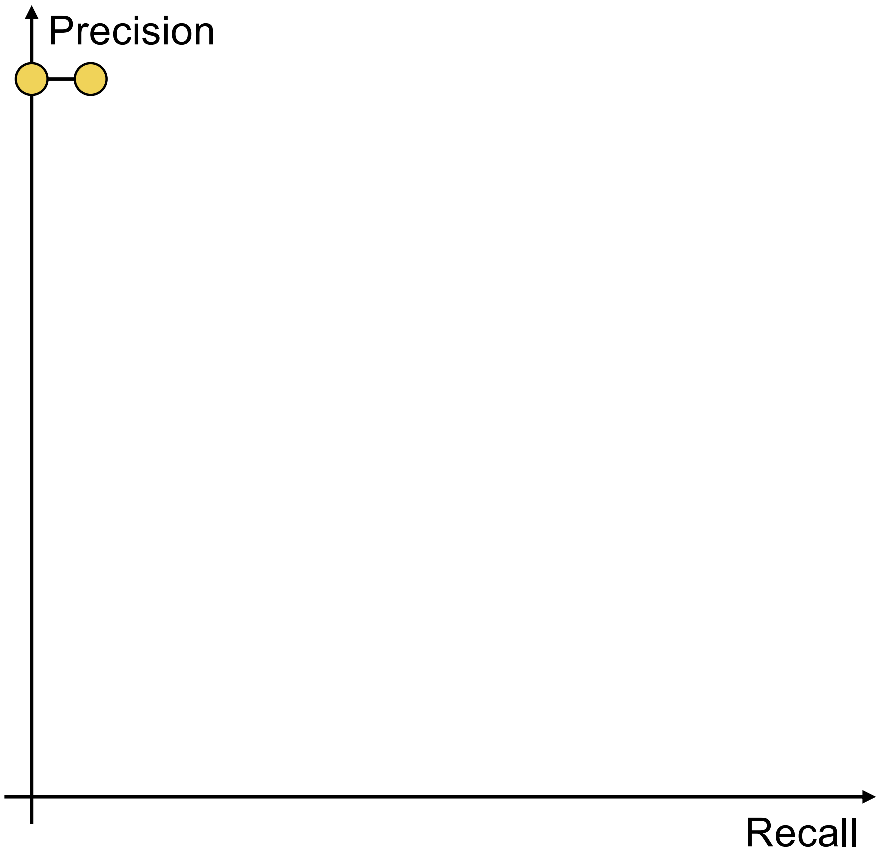

Top 2



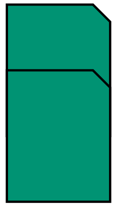
# Precision vs. Recall



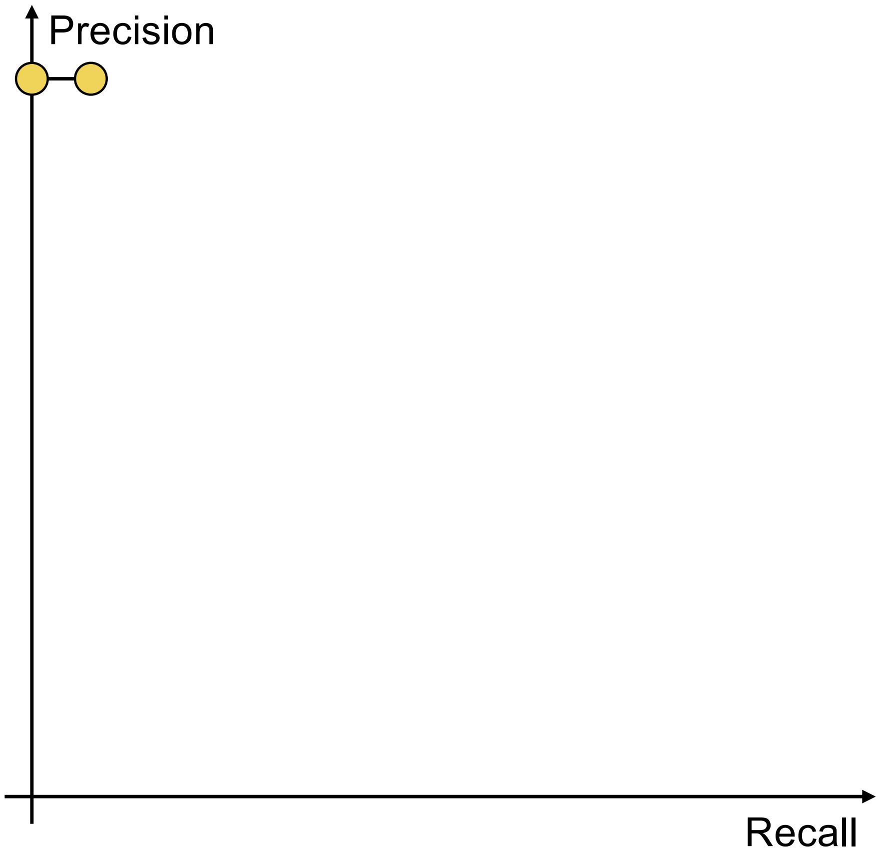
# Precision vs. Recall

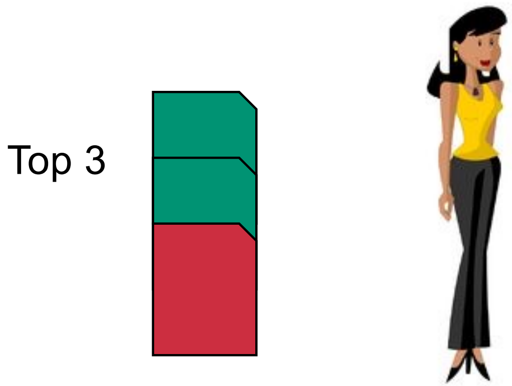



Top 2

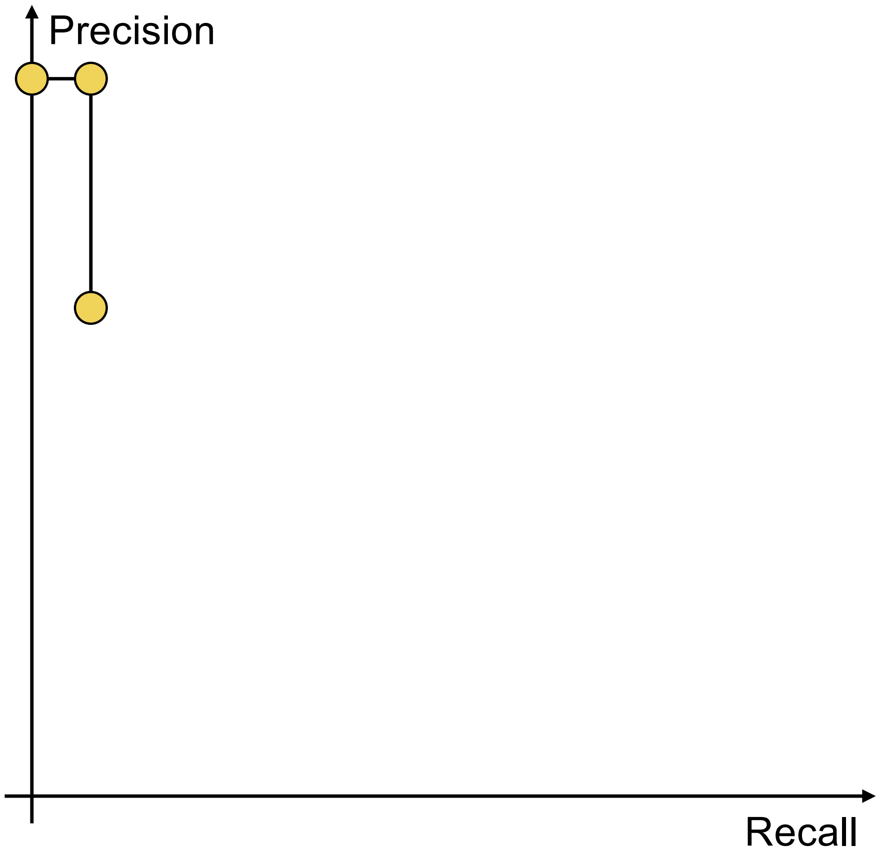


# Precision vs. Recall



# Precision vs. Recall

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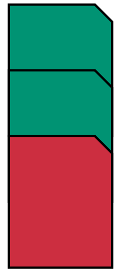

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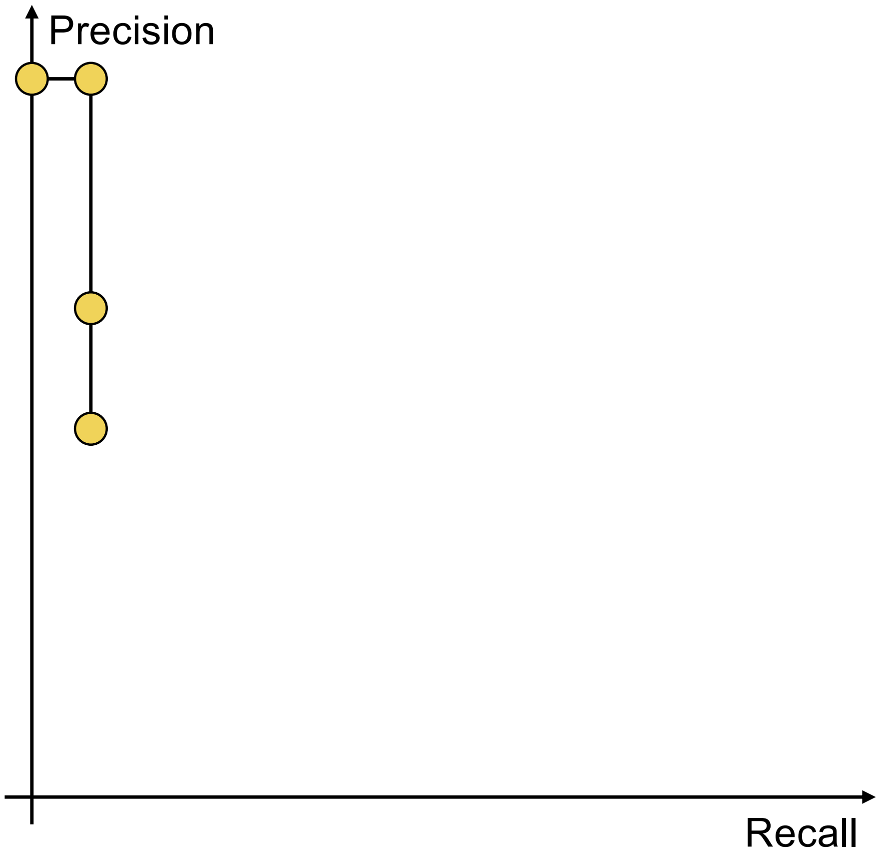


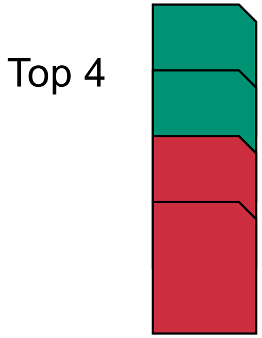


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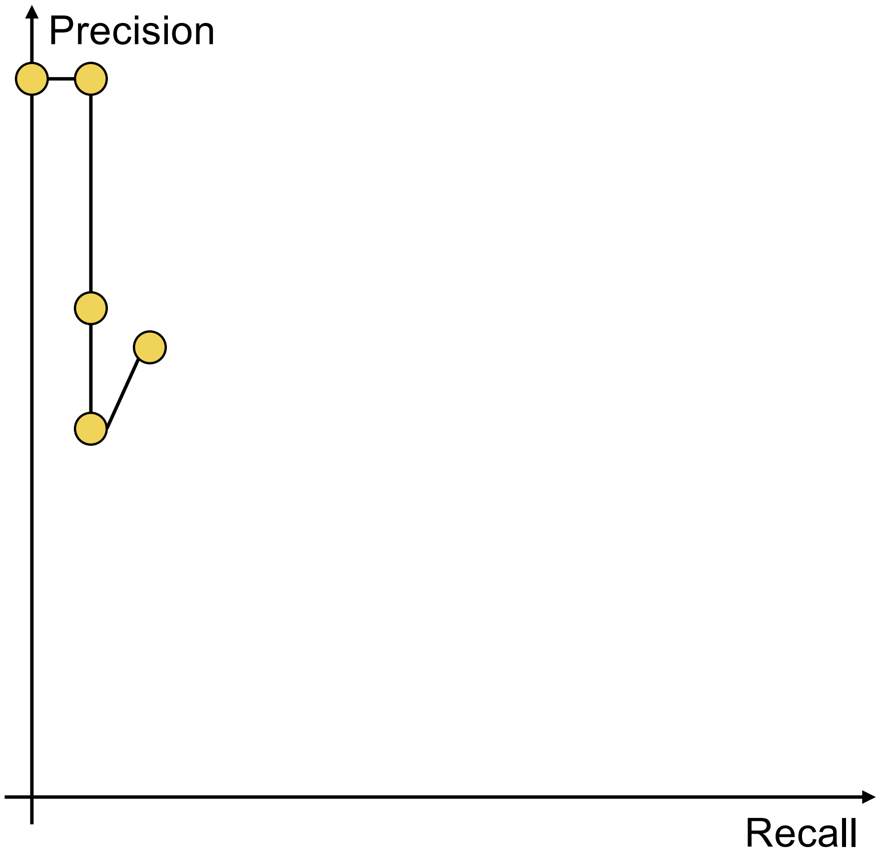


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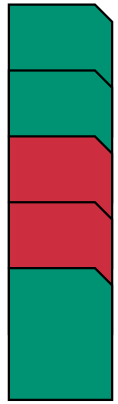
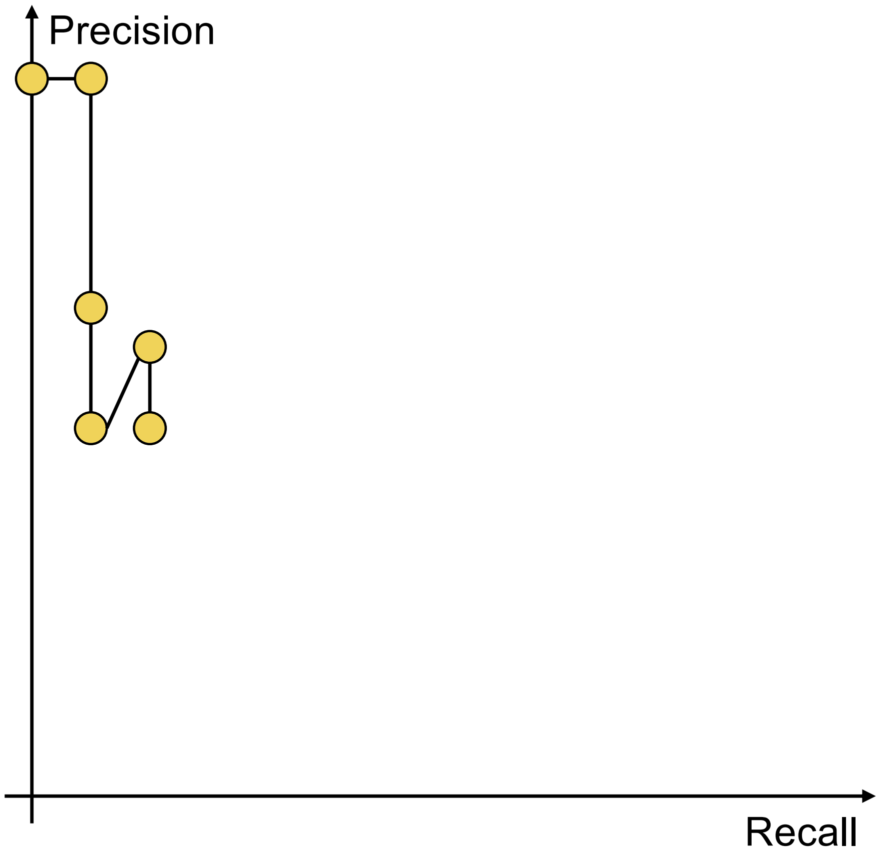


# Precision vs. Recall

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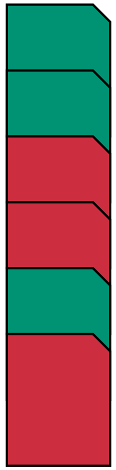
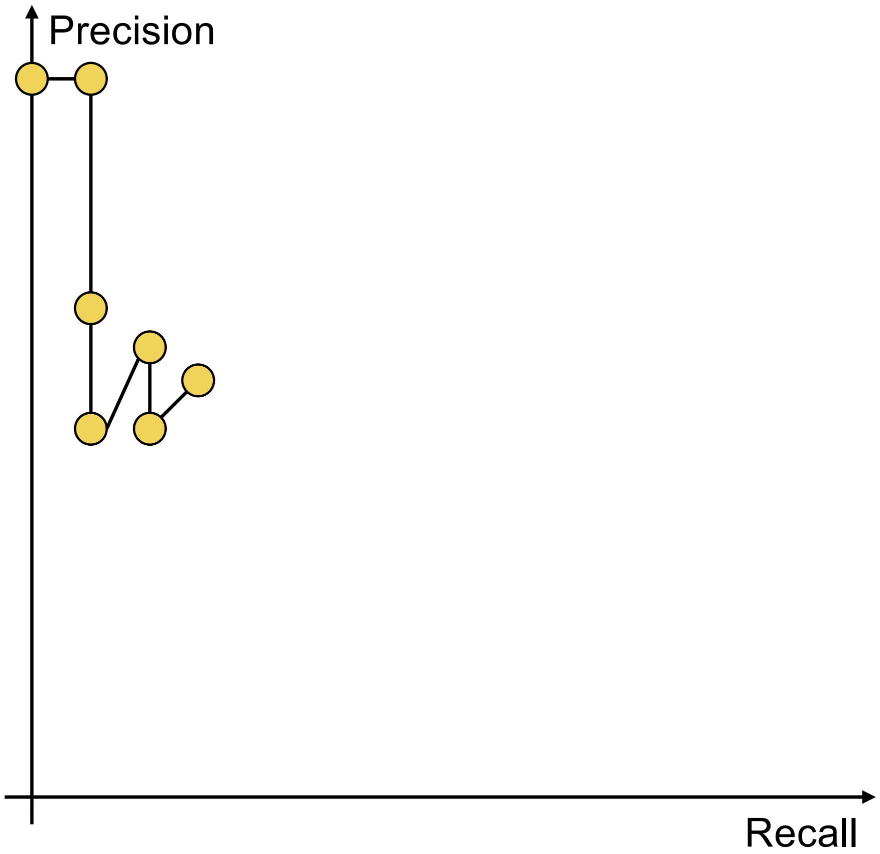
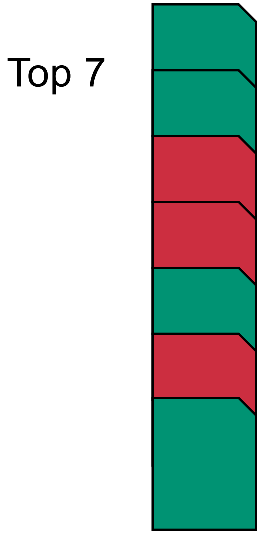


# Precision vs. Recall

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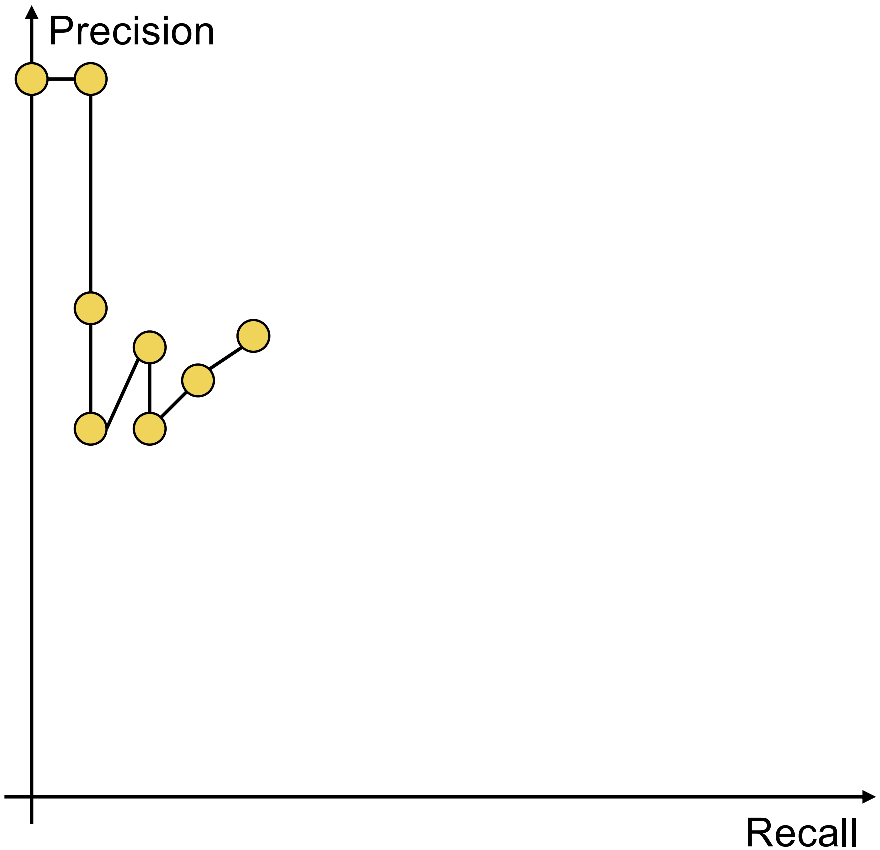


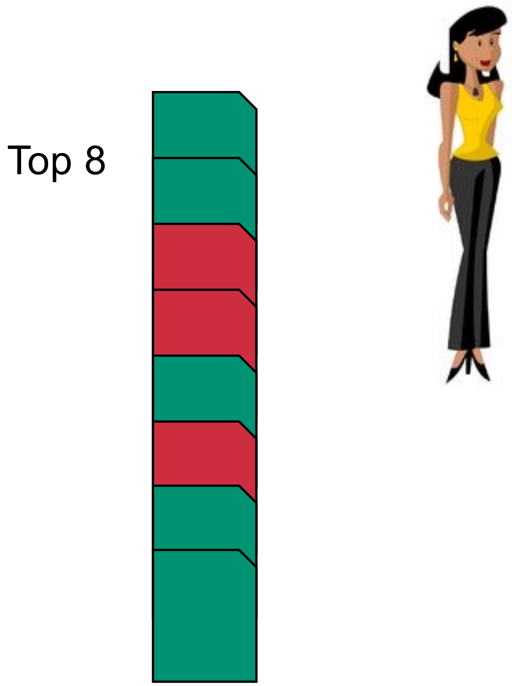

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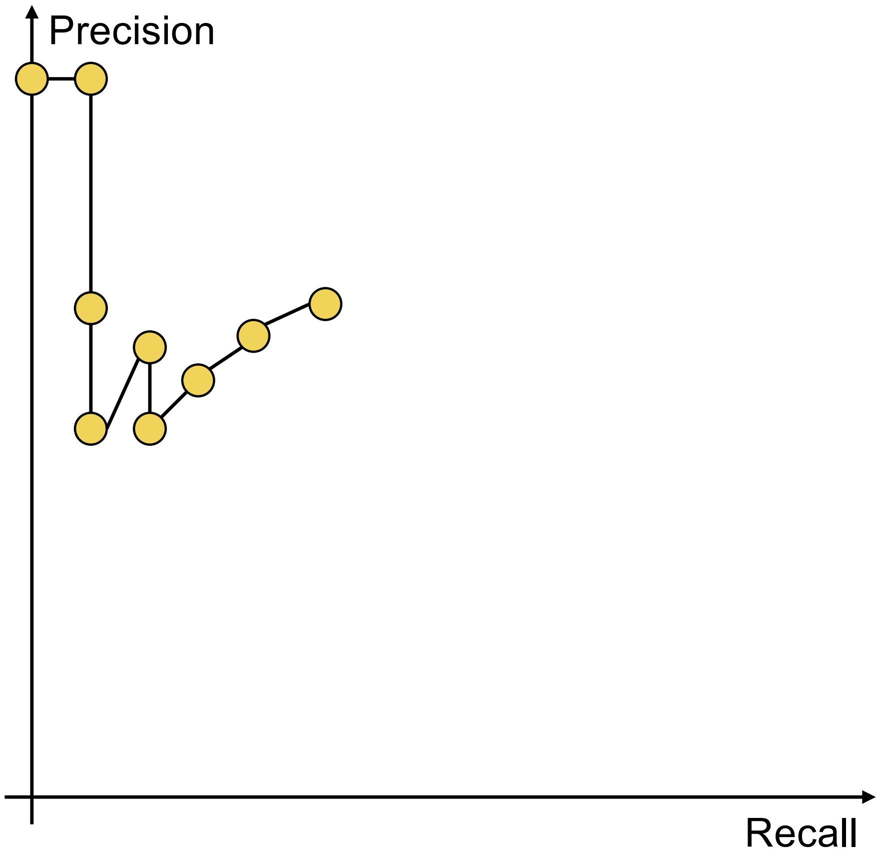


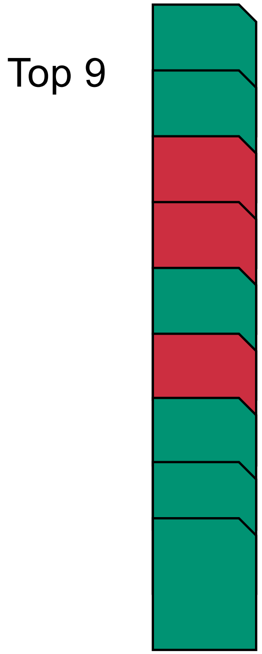


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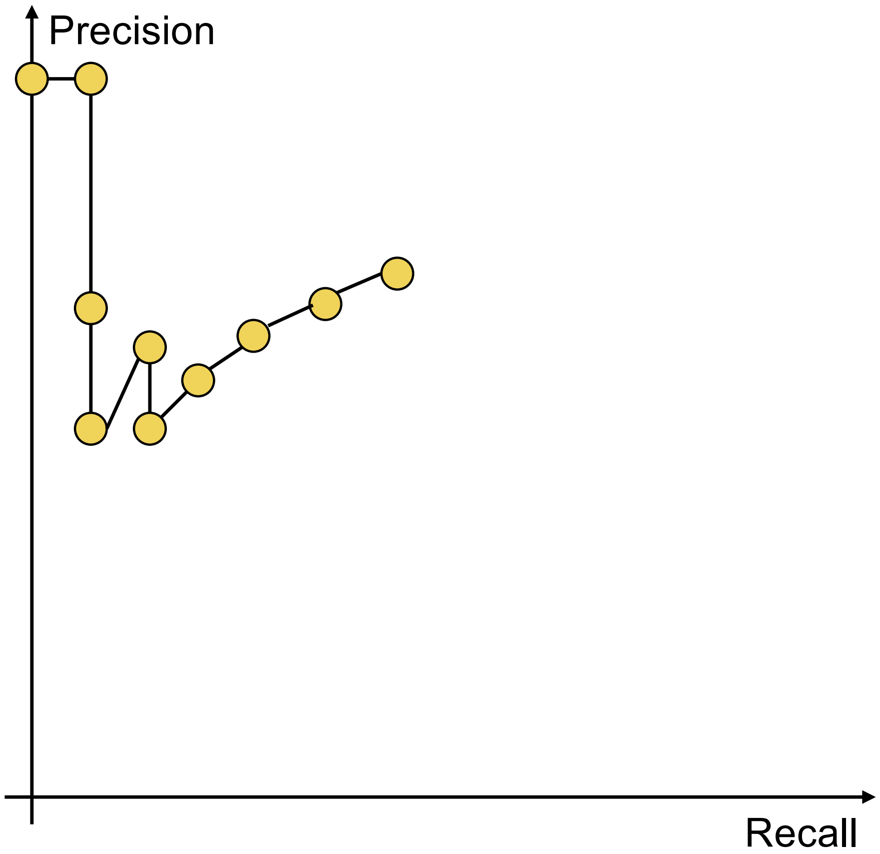
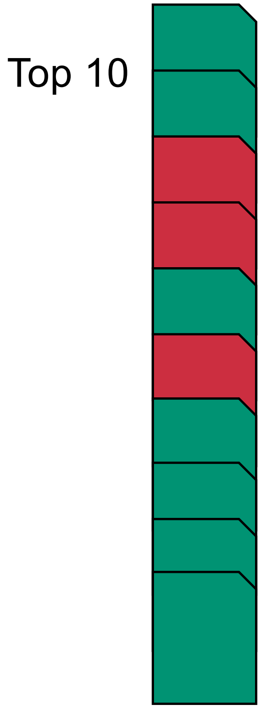
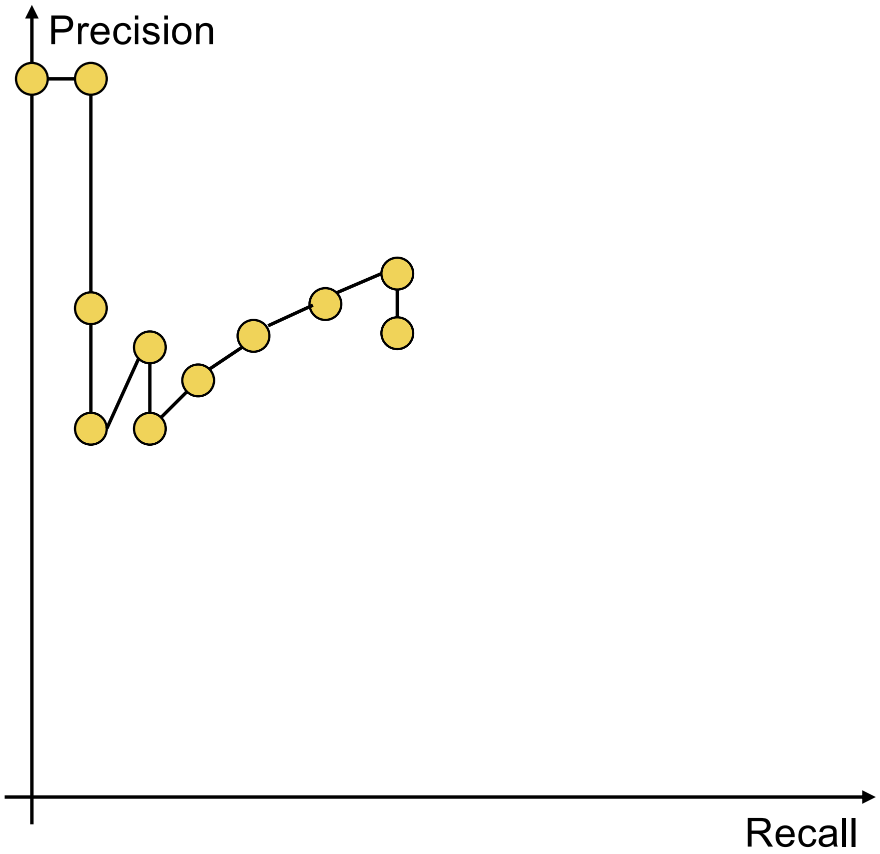
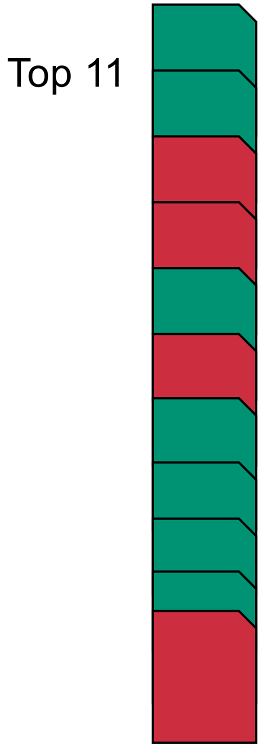


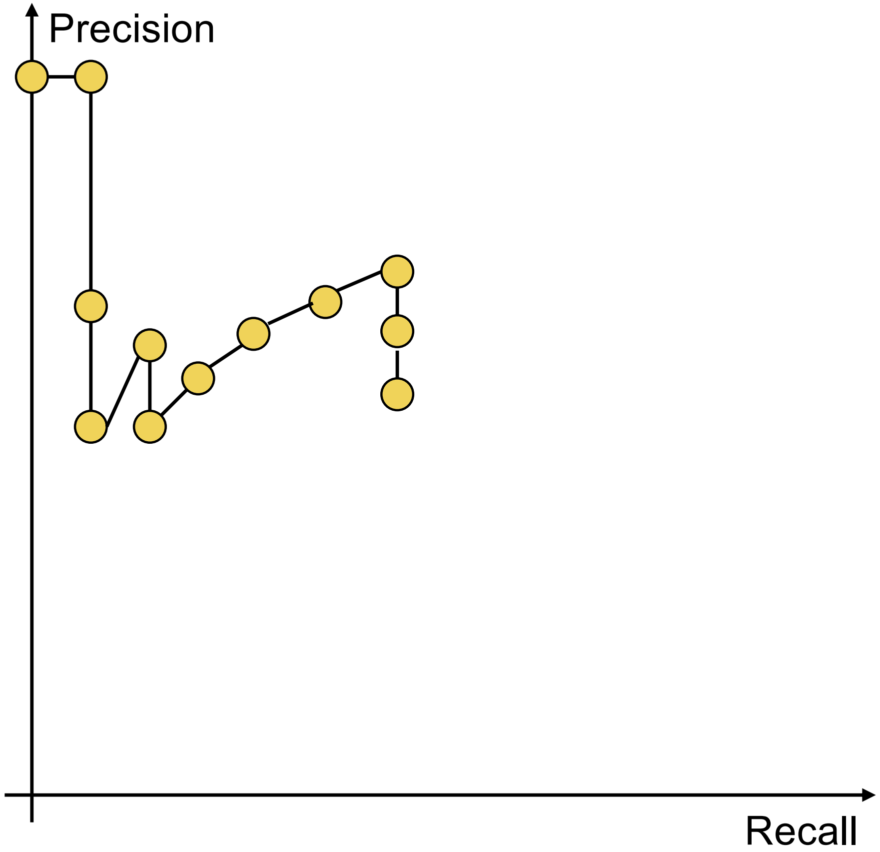



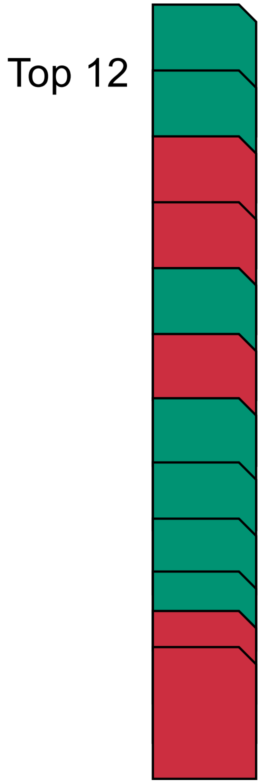

# Precision vs. Recall



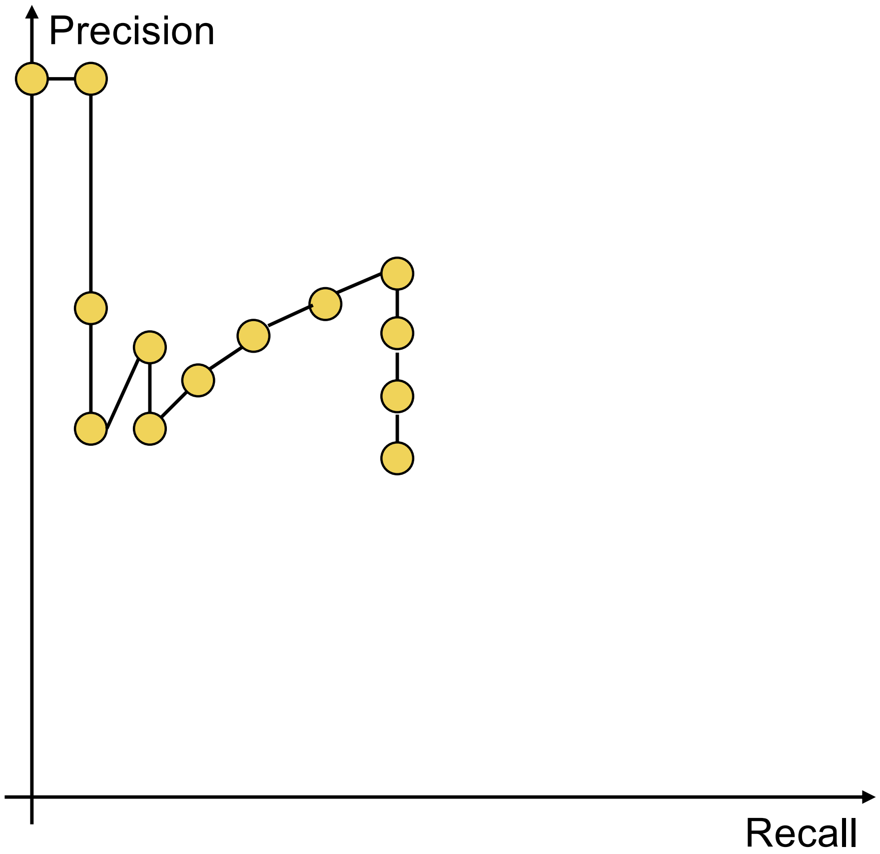



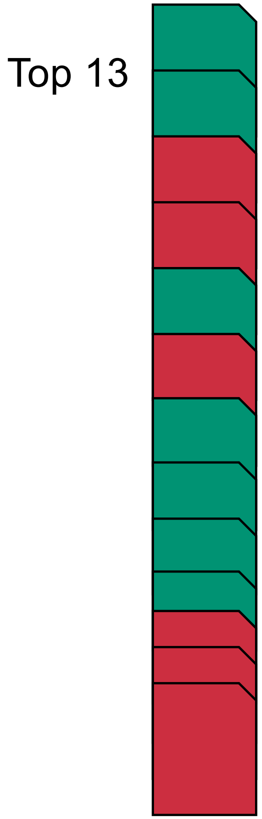

# Precision vs. Recall



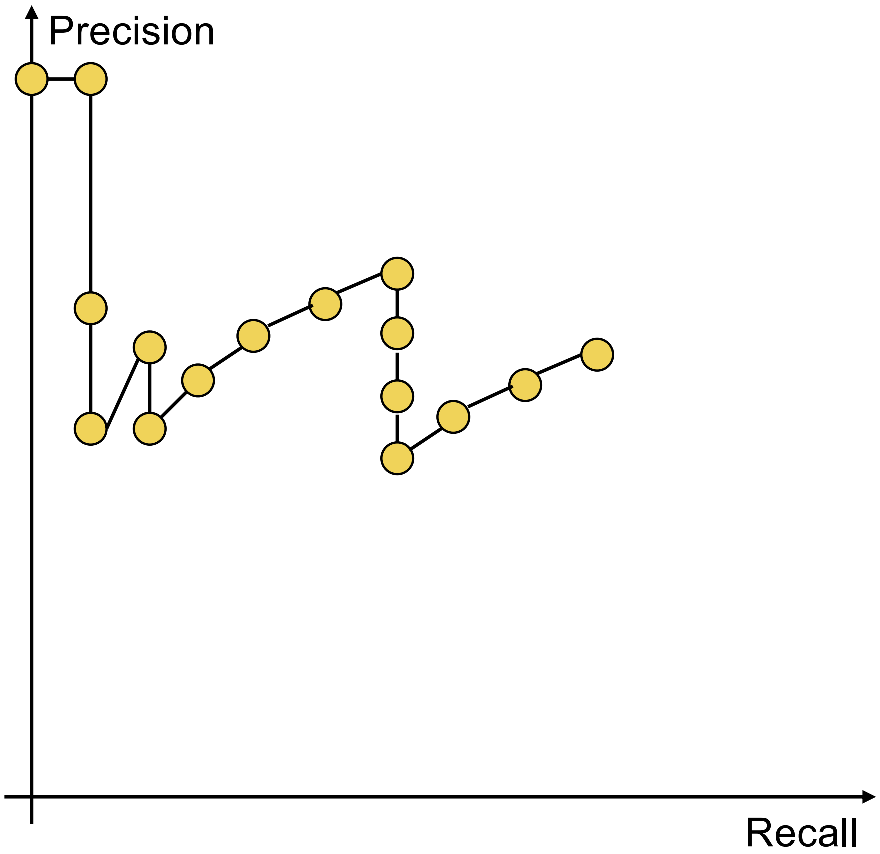




# Precision vs. Recall

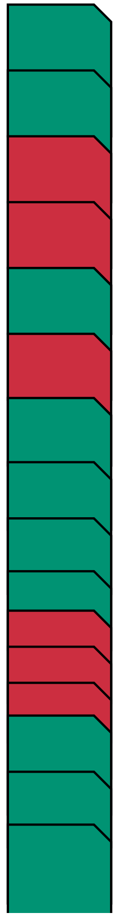


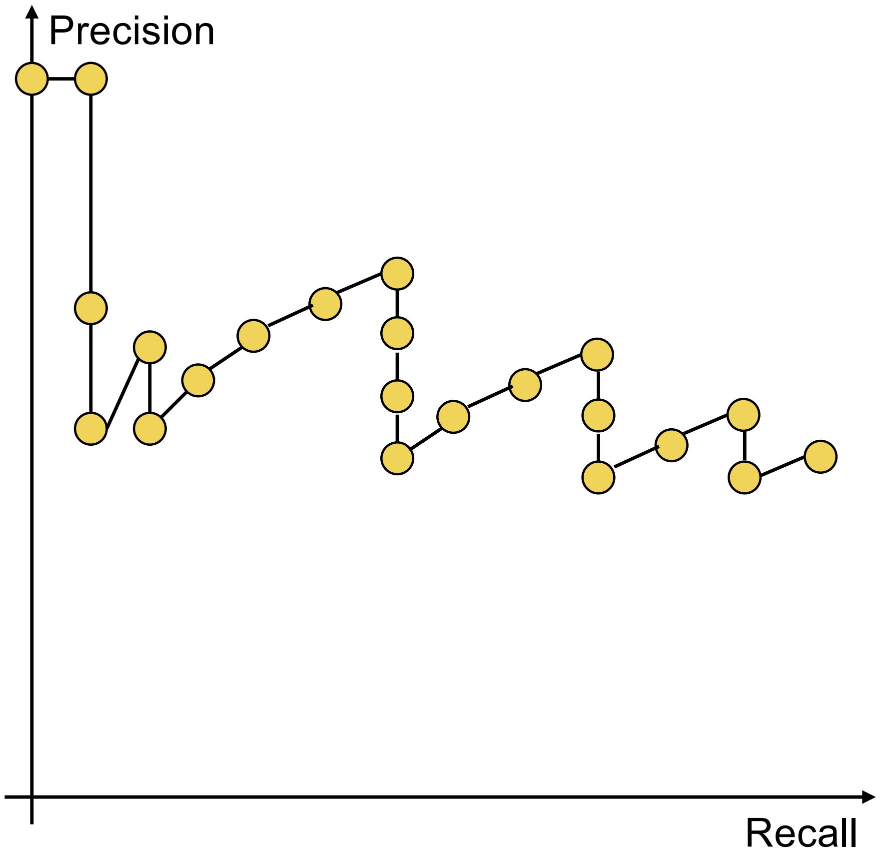

# Precision vs. Recall



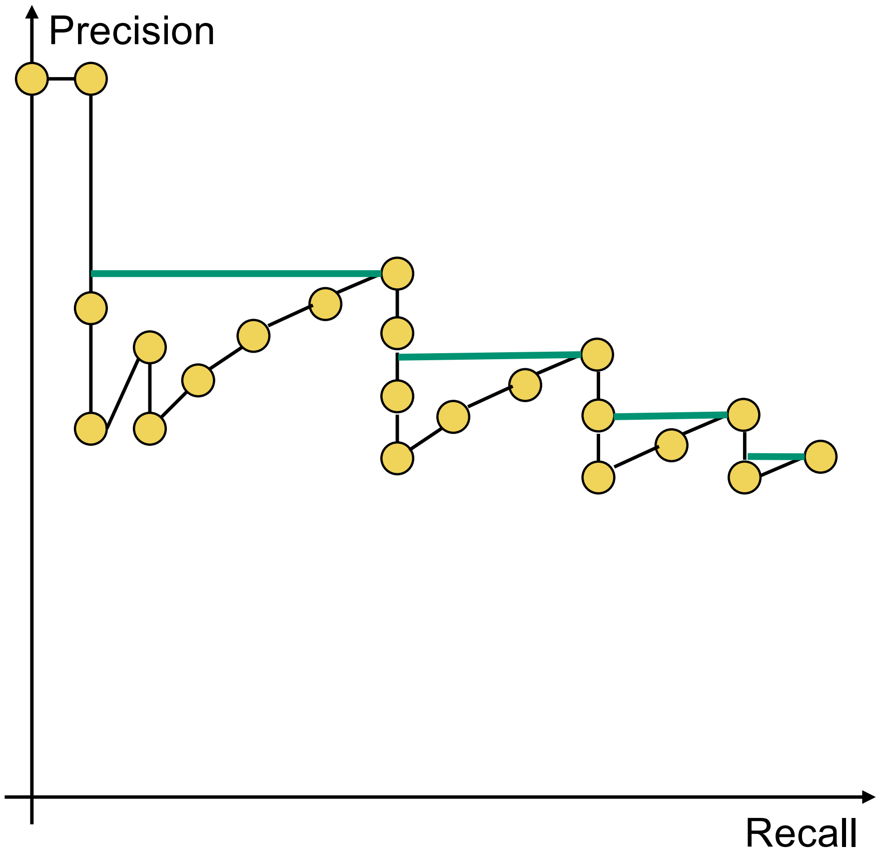

Top 15



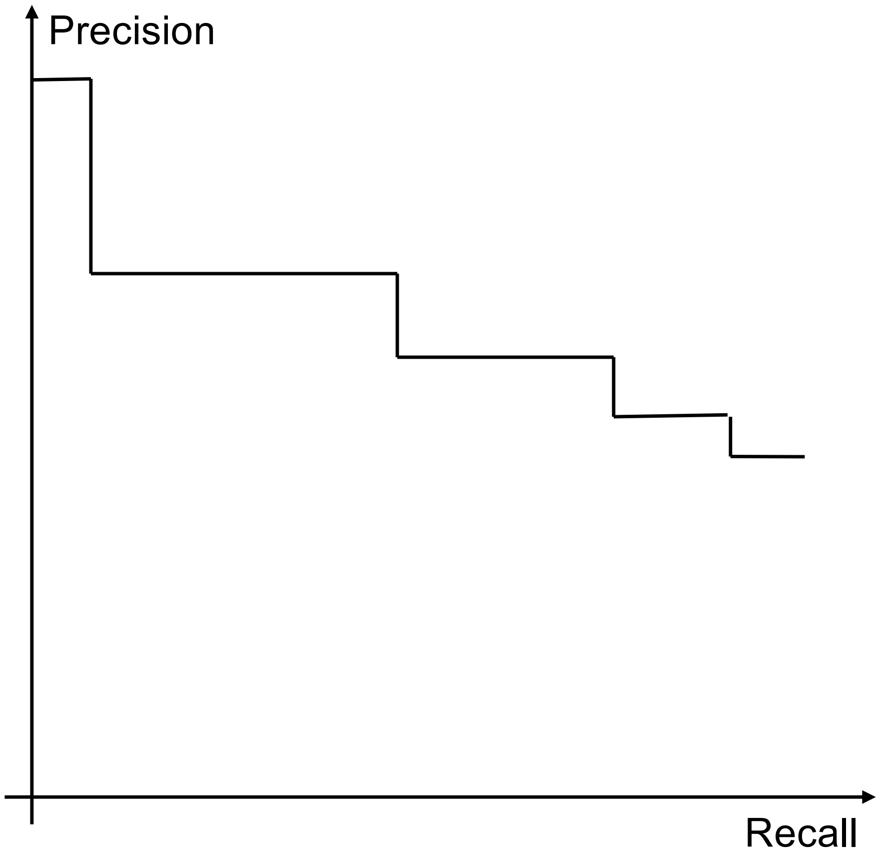
# Precision vs. Recall



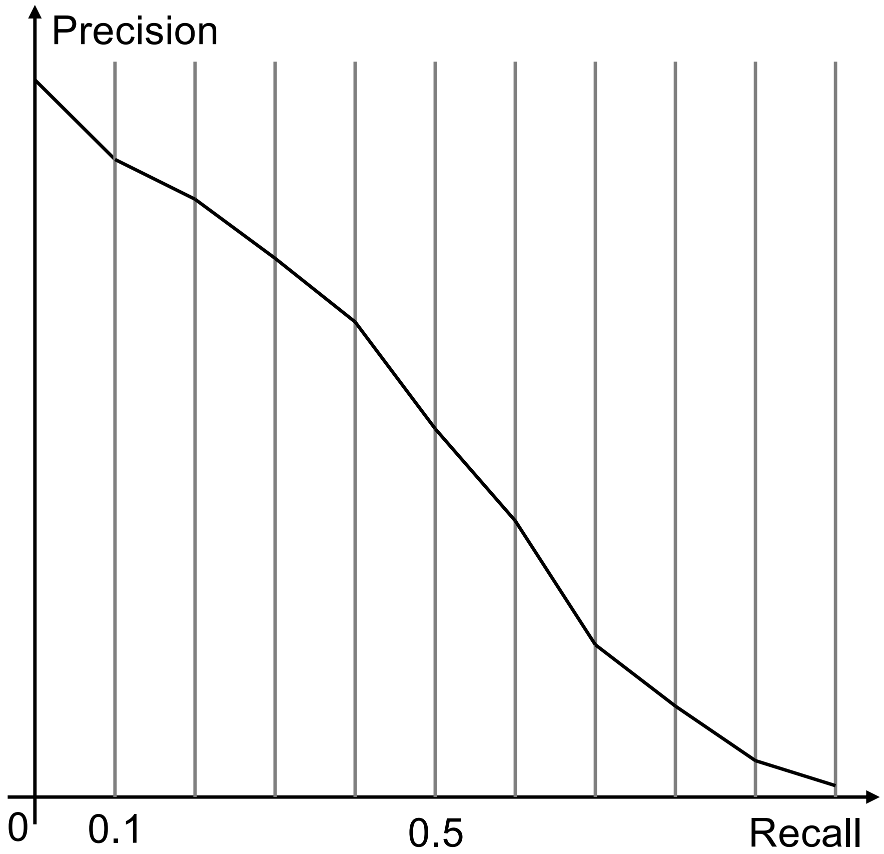

# Interpolated precision



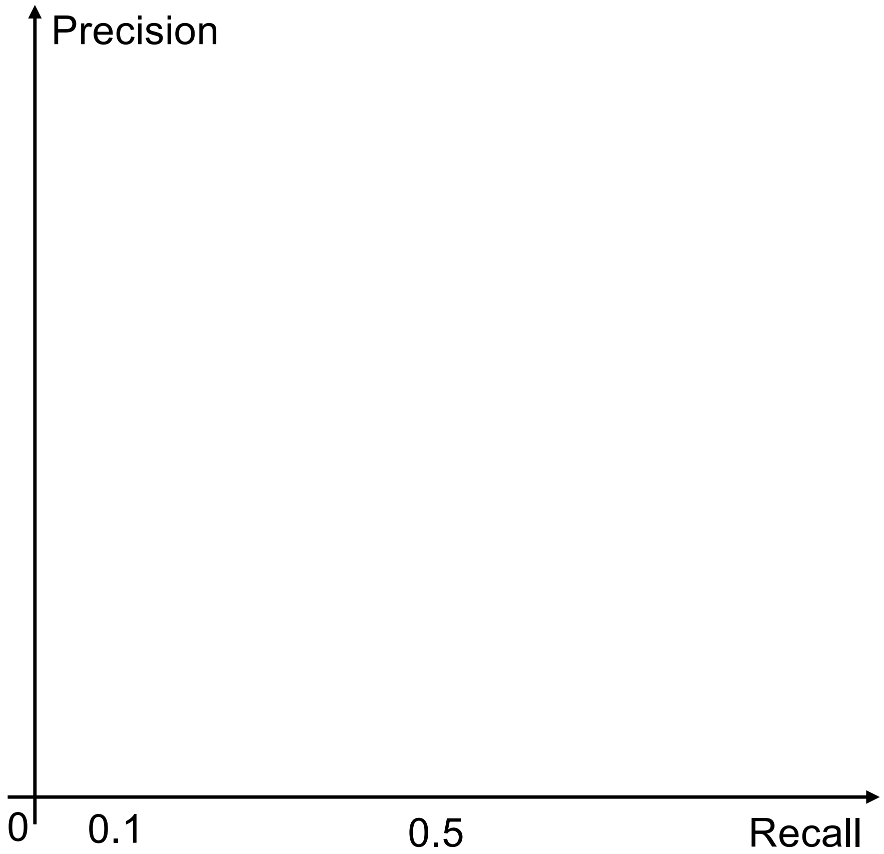

# Precision vs. Recall



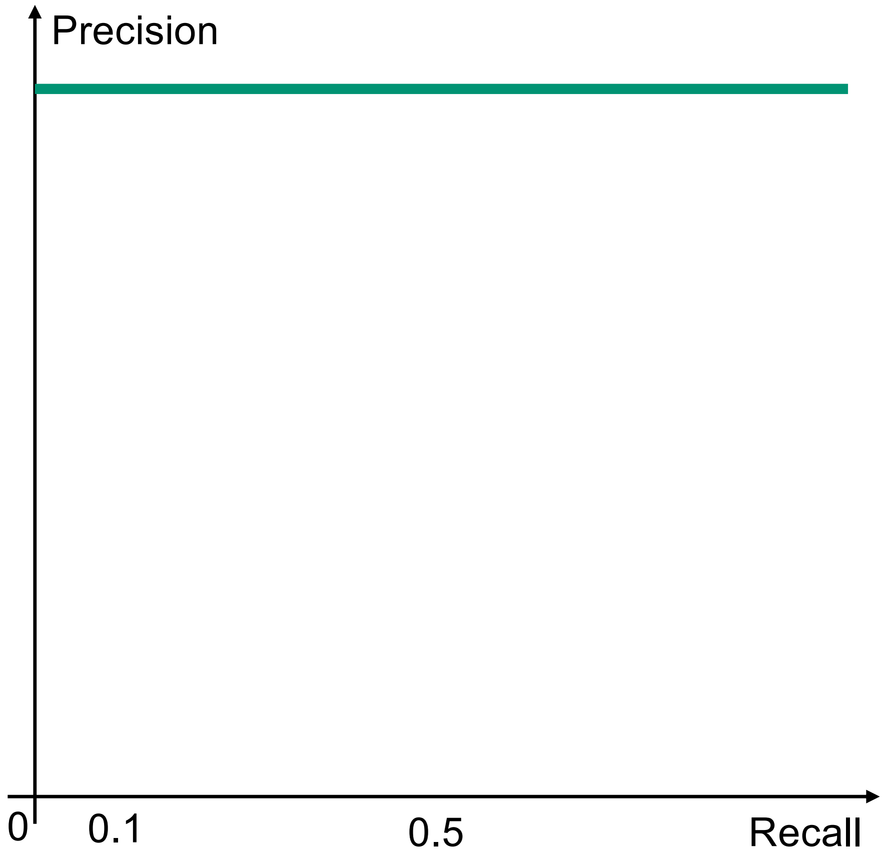

# Eleven-point Interpolated Average Precision



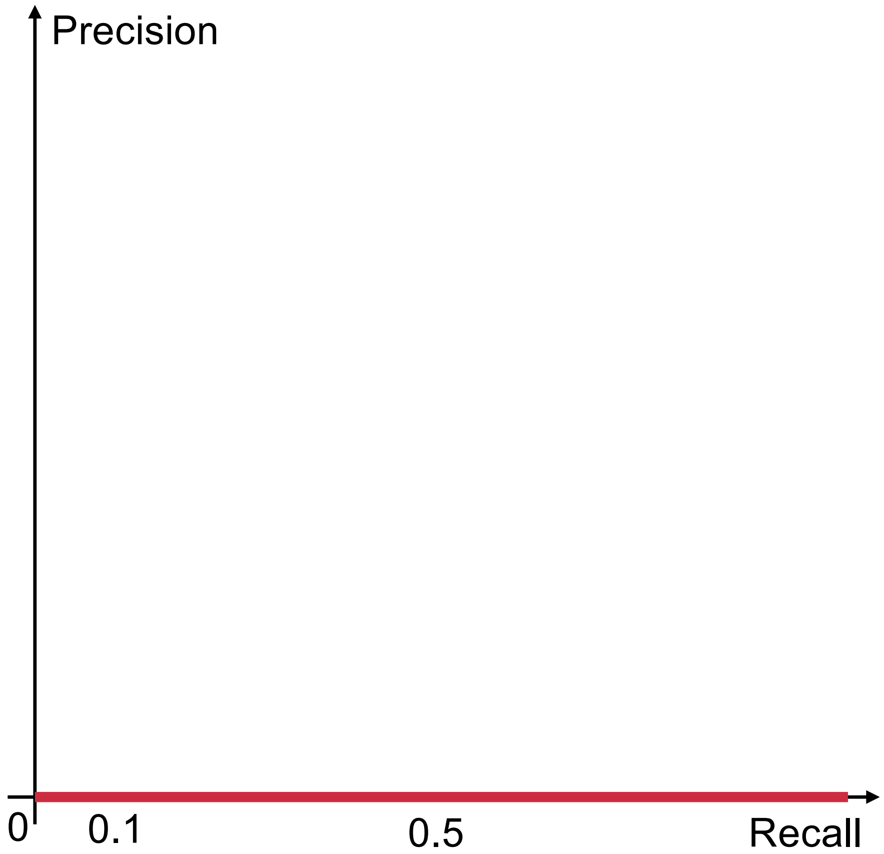
# Awesome Search Engine



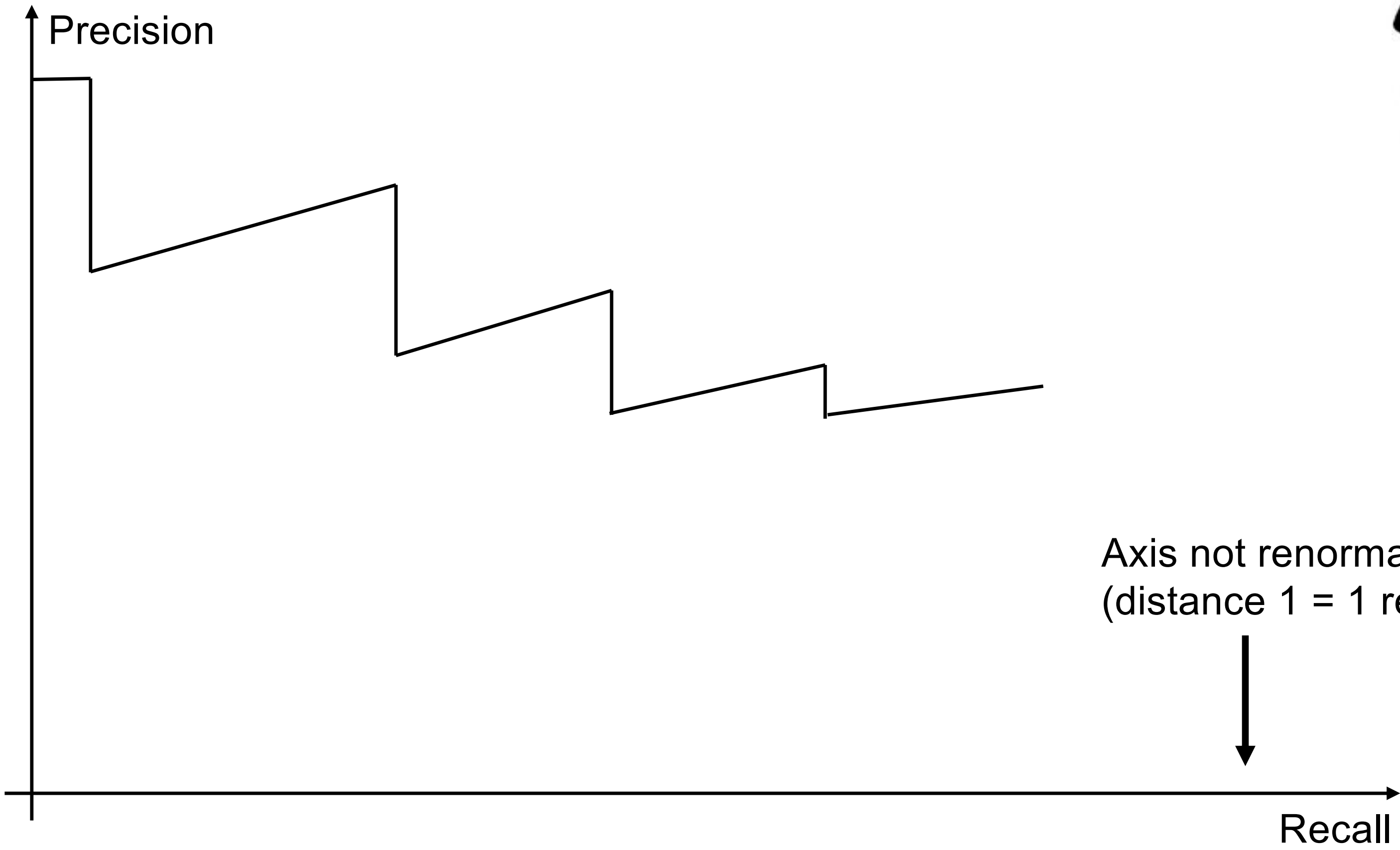
# Awesome Search Engine



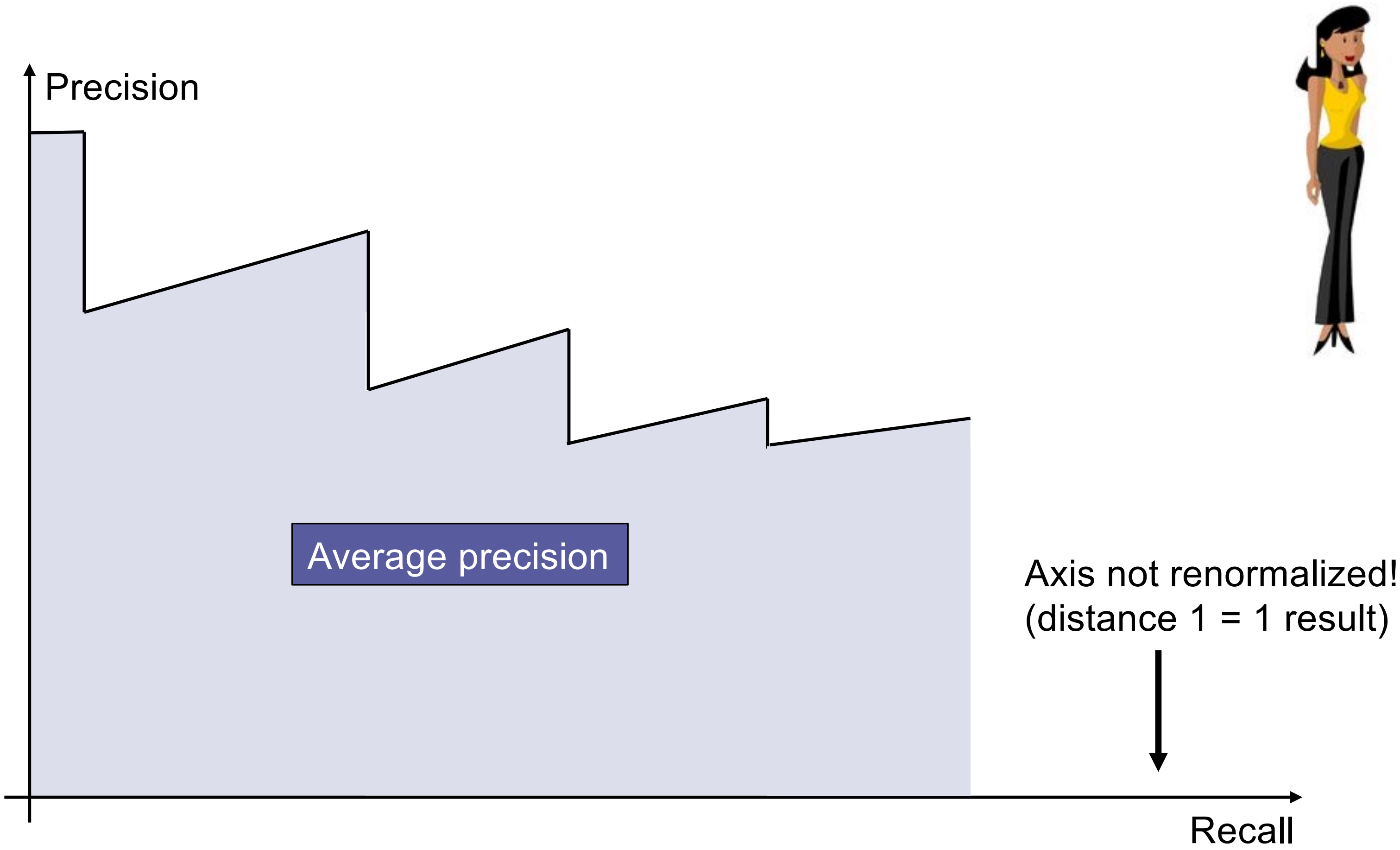
# Awful Search Engine



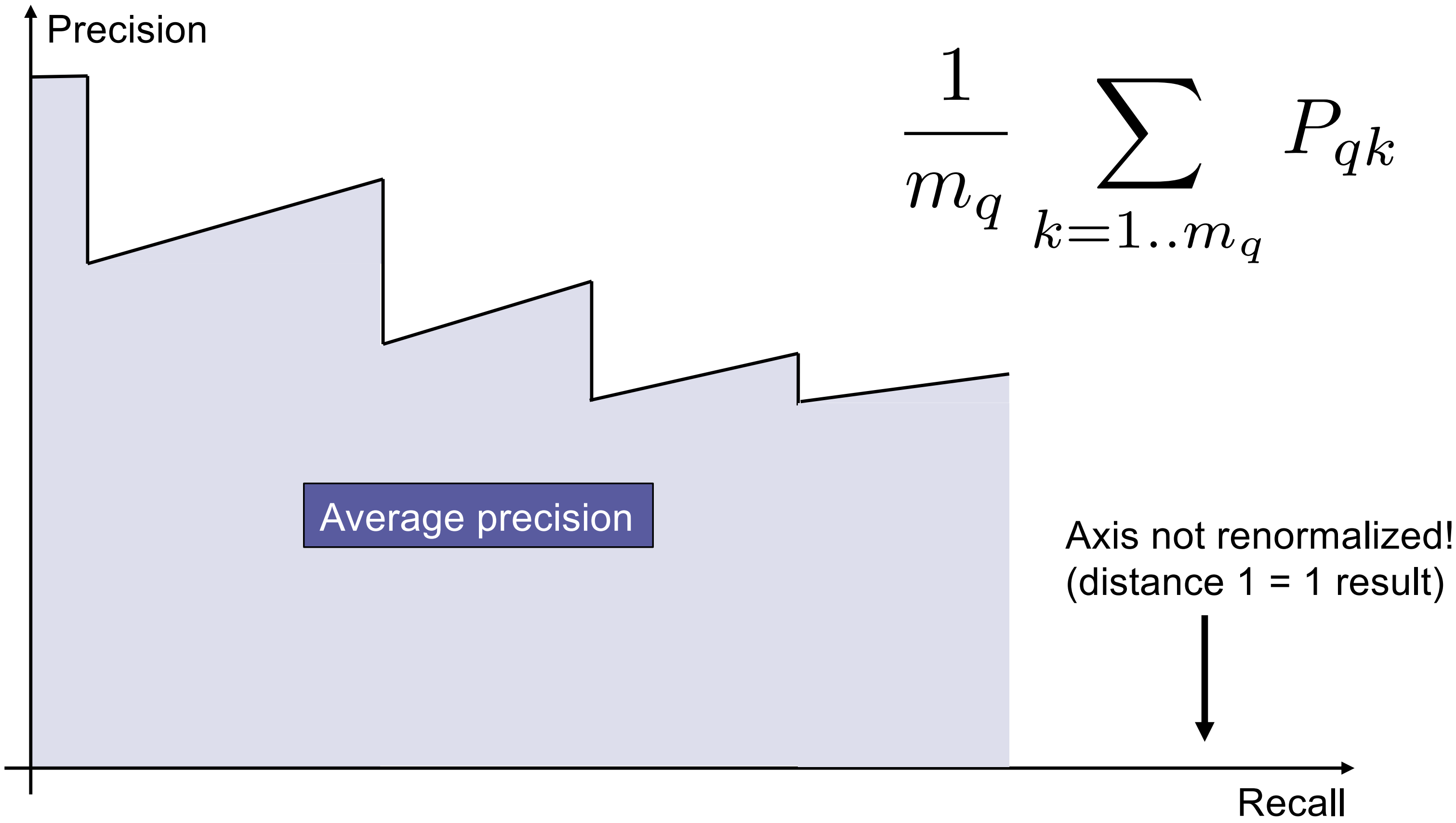
# Mean Average Precision (MAP)




# Mean Average Precision (MAP)

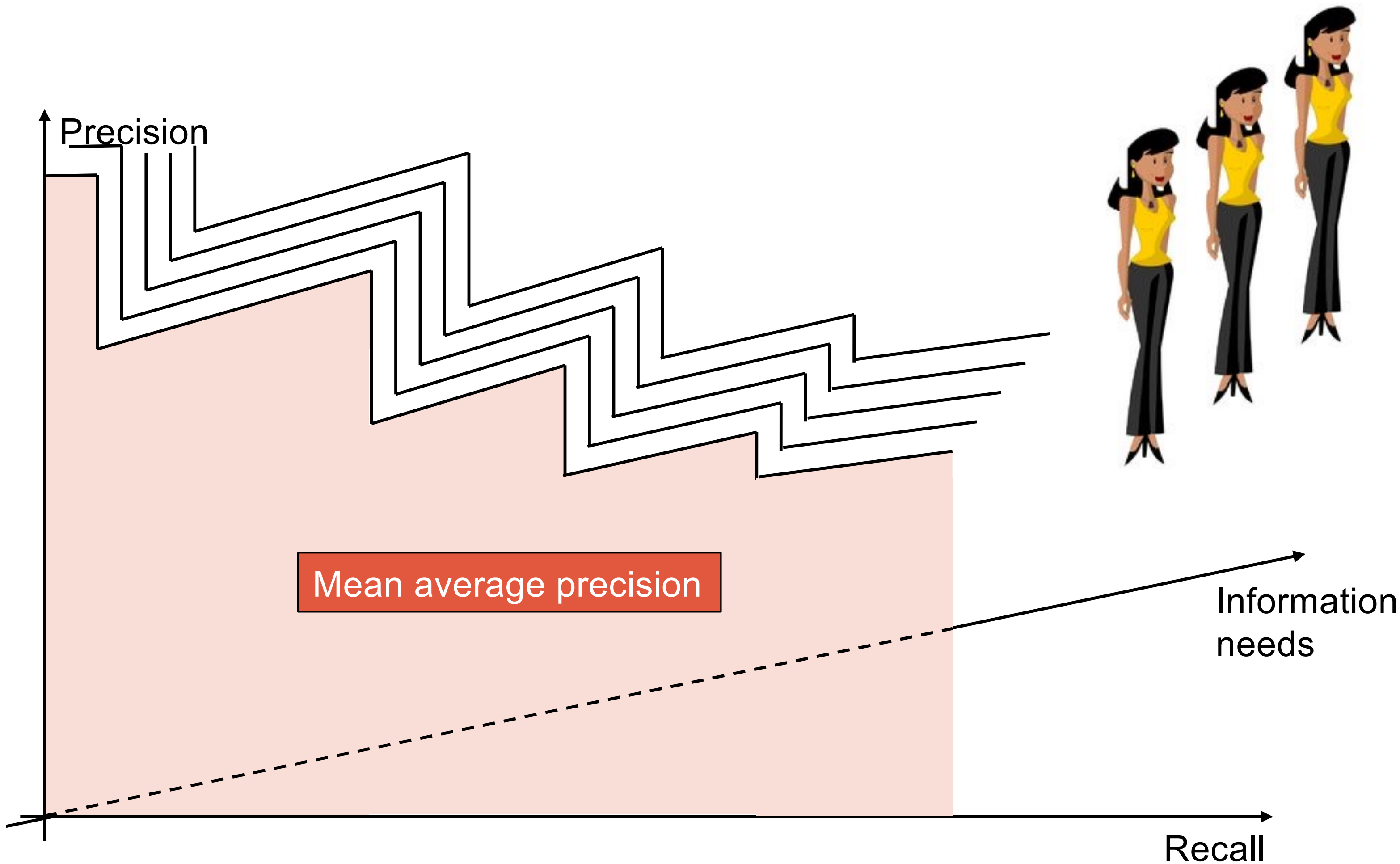


# Mean Average Precision (MAP)

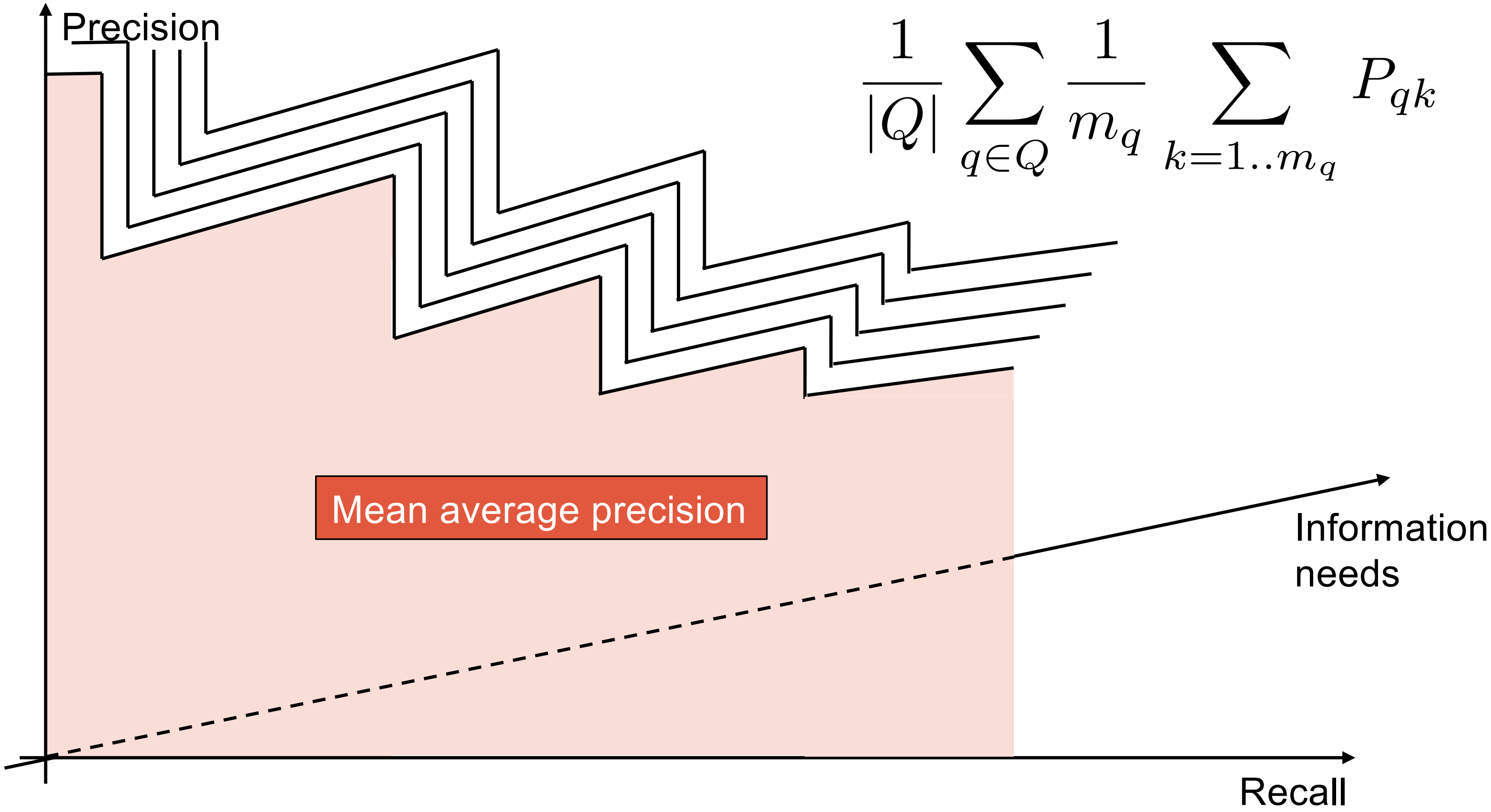



$$\frac{1}{m_q} \sum_{k=1..m_q} P_{qk}$$

# Mean Average Precision (MAP)

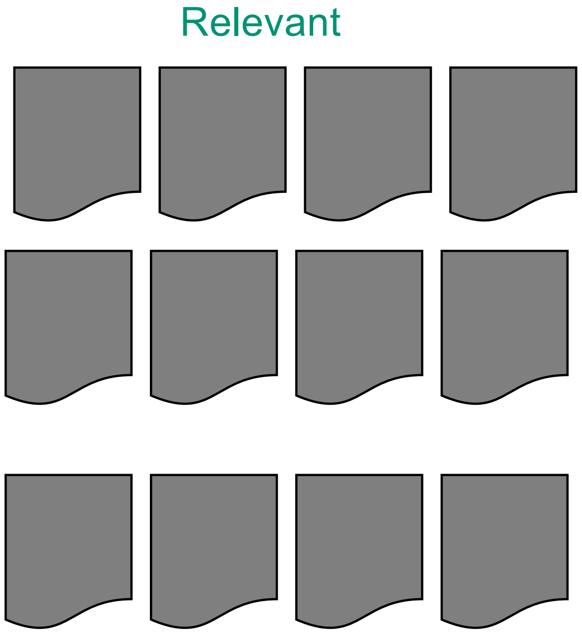


# Mean Average Precision (MAP)



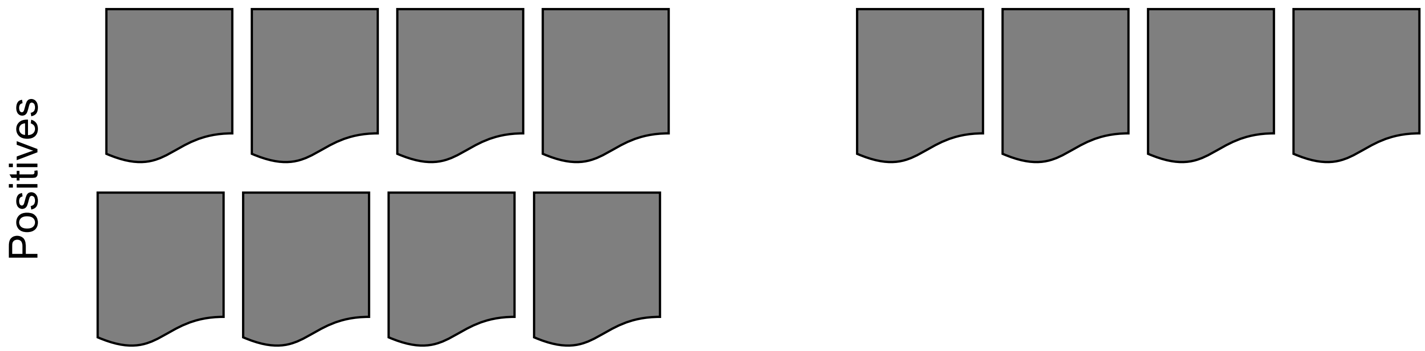

$$\frac{1}{|Q|}\sum_{q\in Q}\frac{1}{m_q}\sum_{k=1..m_q}P_{qk}$$

# R-Precision



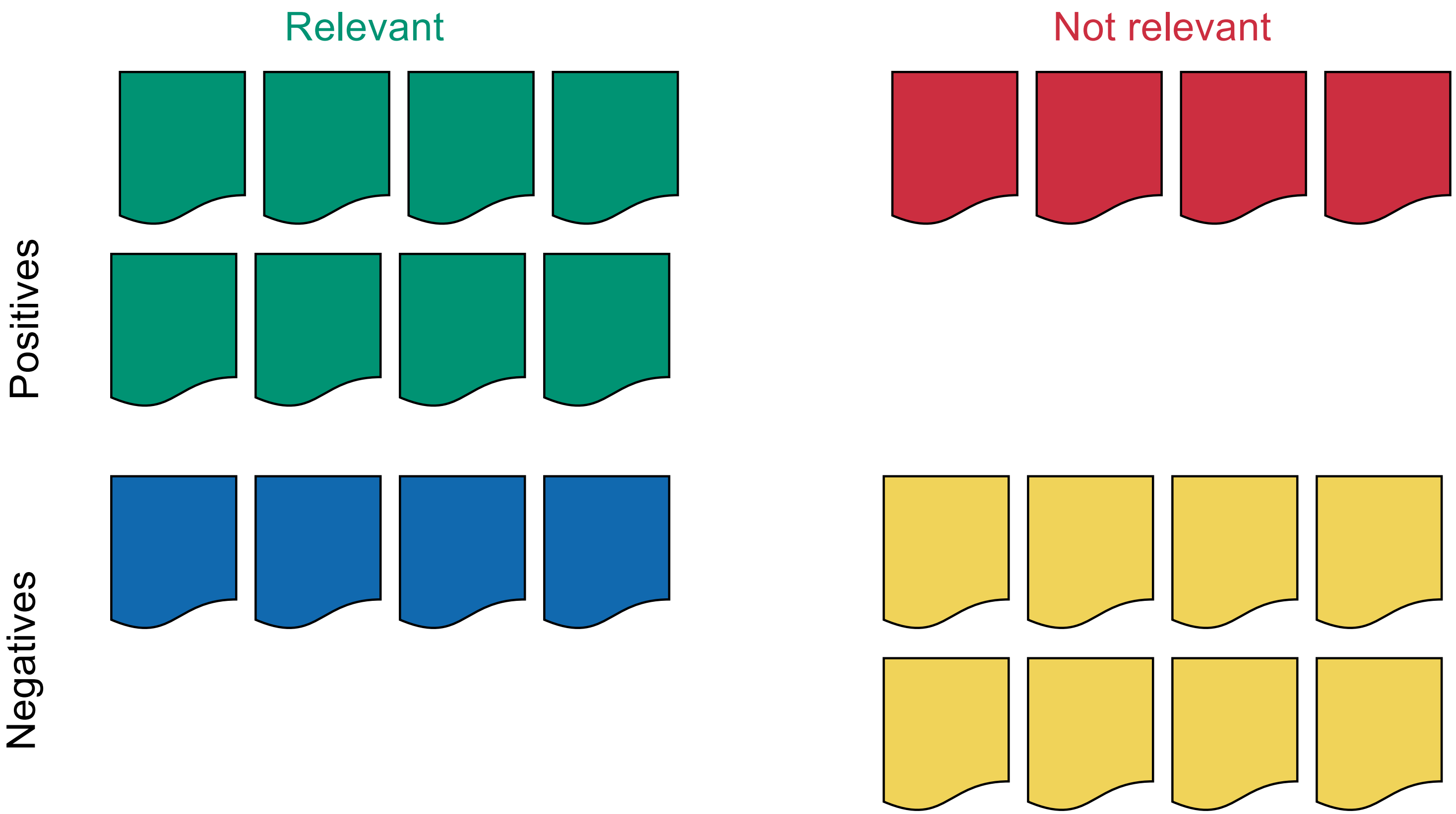

k relevant documents

# R-Precision

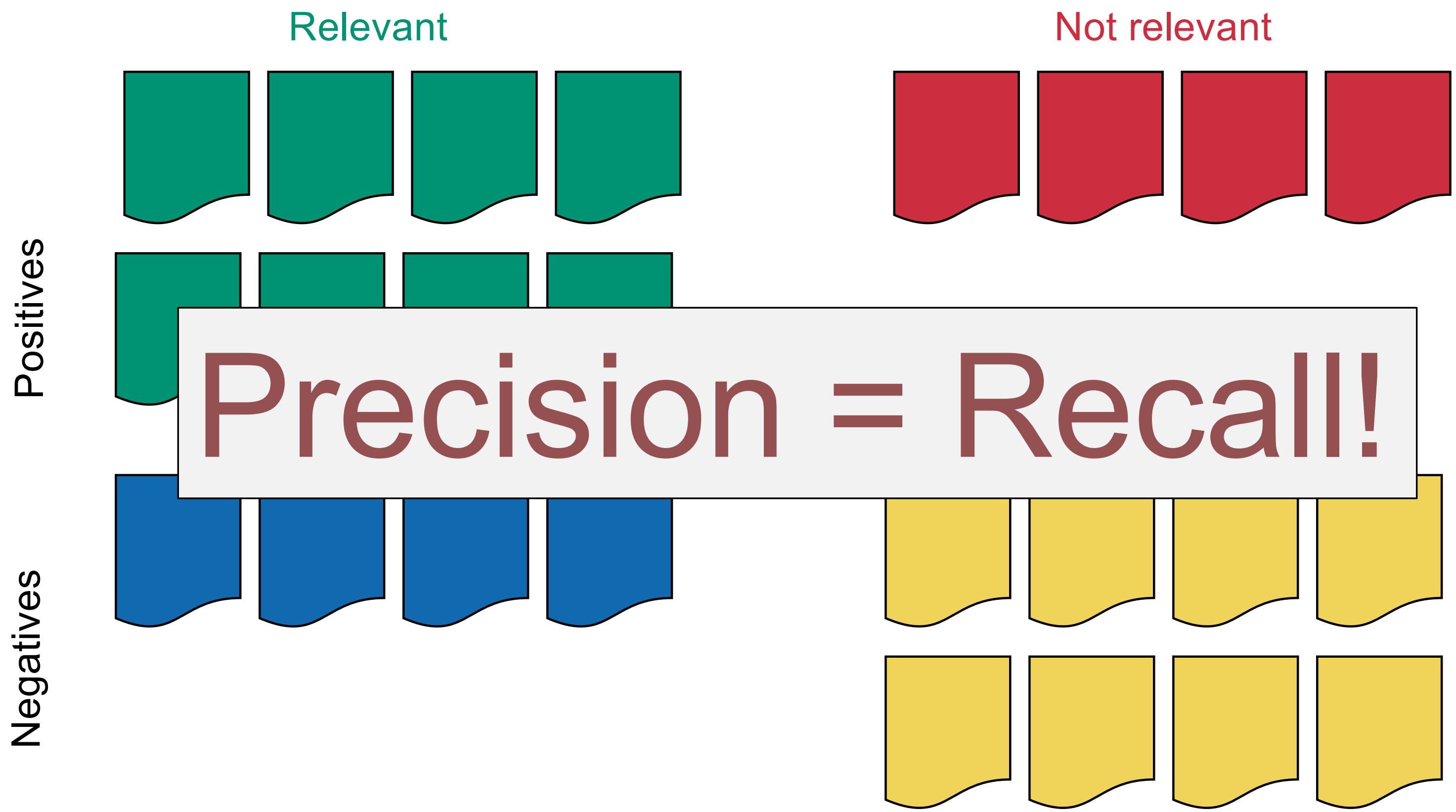



return top k

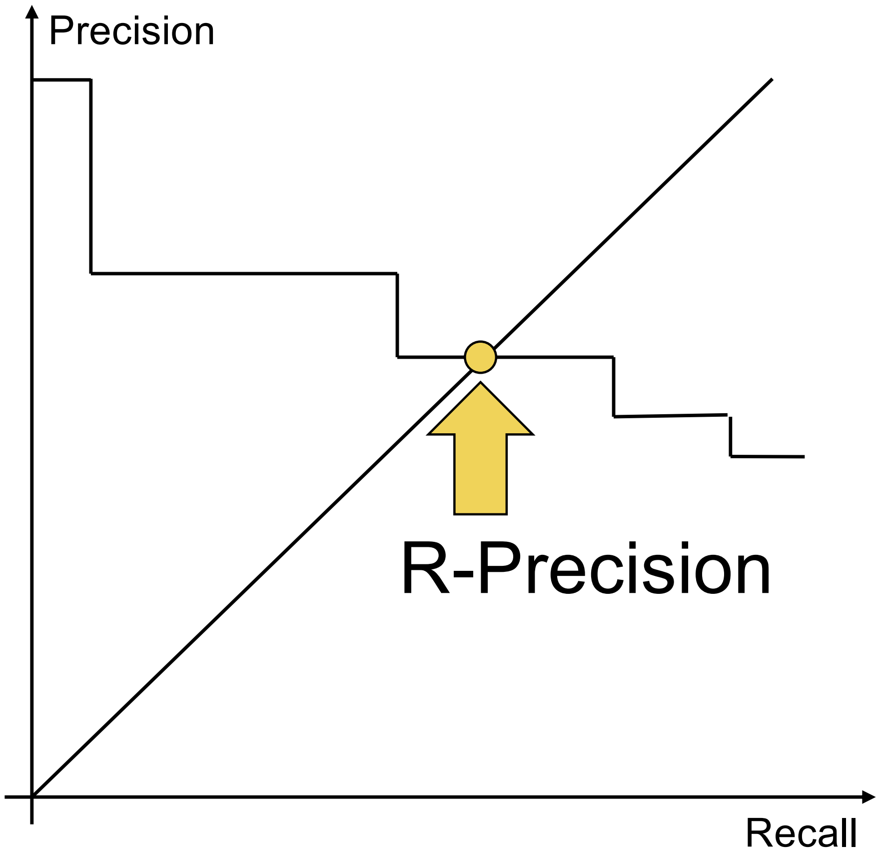
# R-Precision



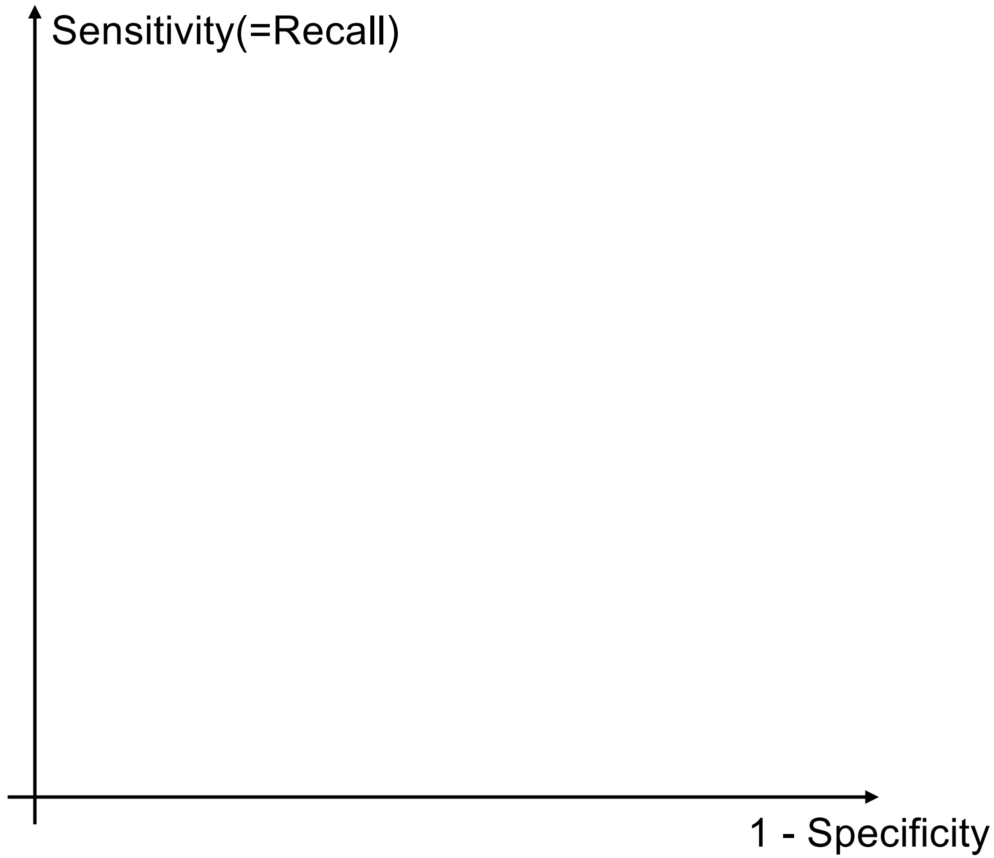
# R-Precision


Relevant
Not relevant
Positives
Negatives
Precision = Recall!

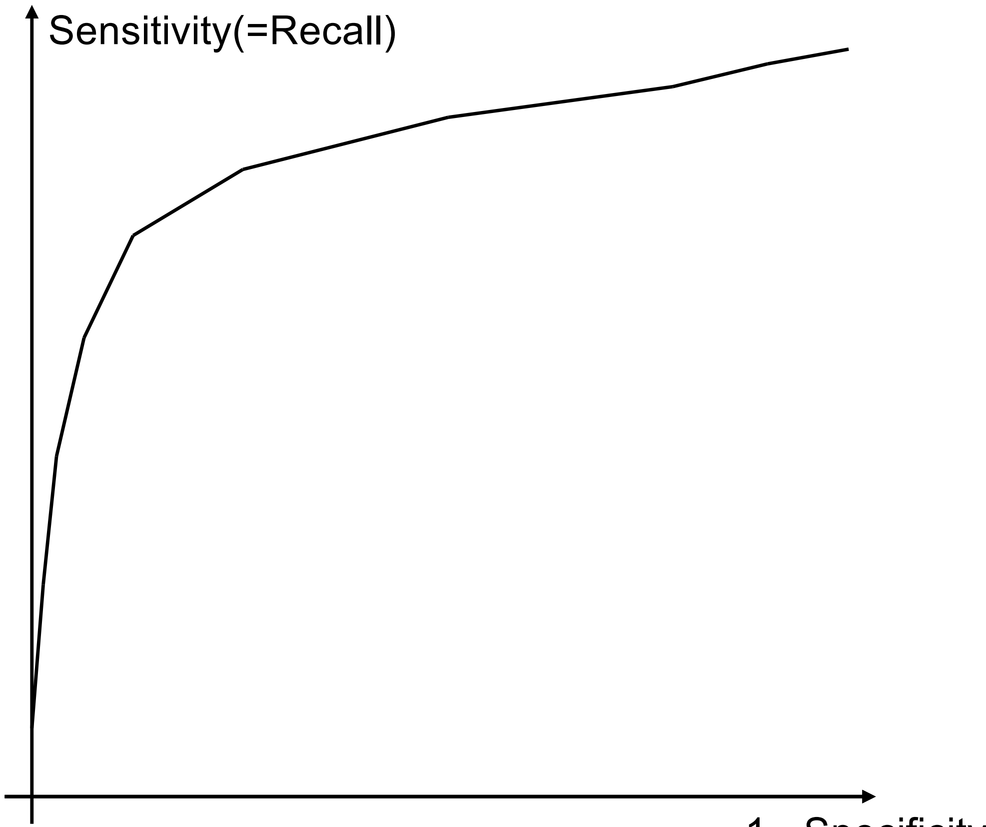
Break-even point (visual interpretation)



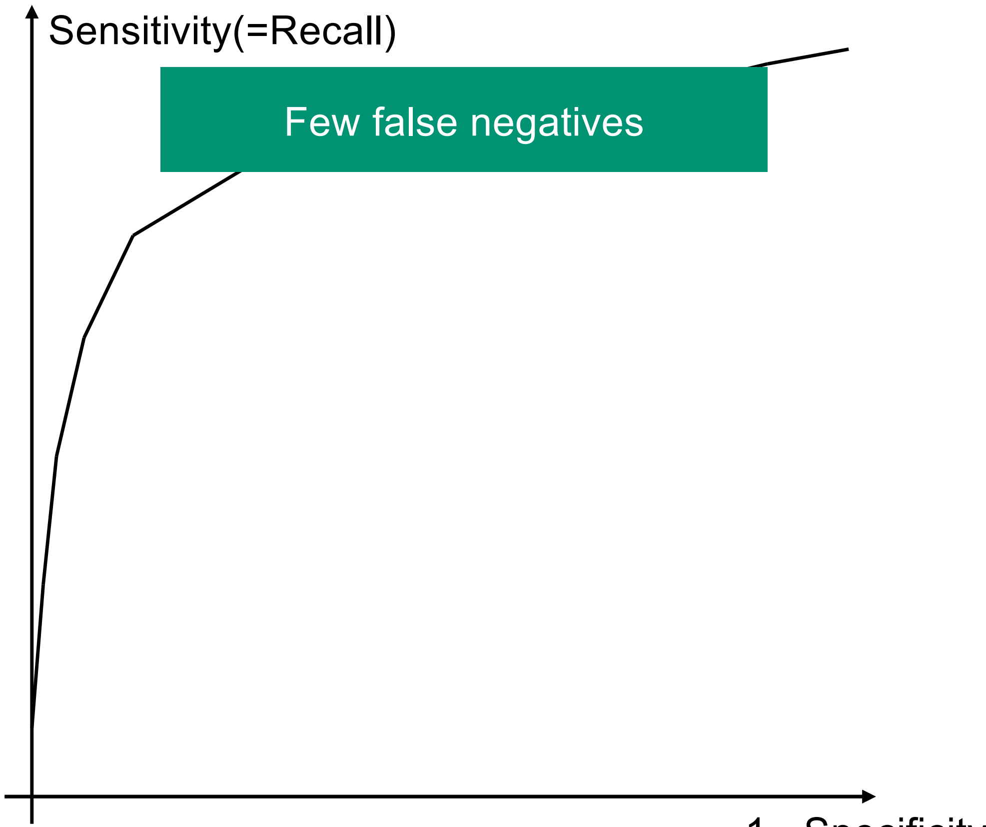

# Recall vs. Sensitivity (ROC curve)



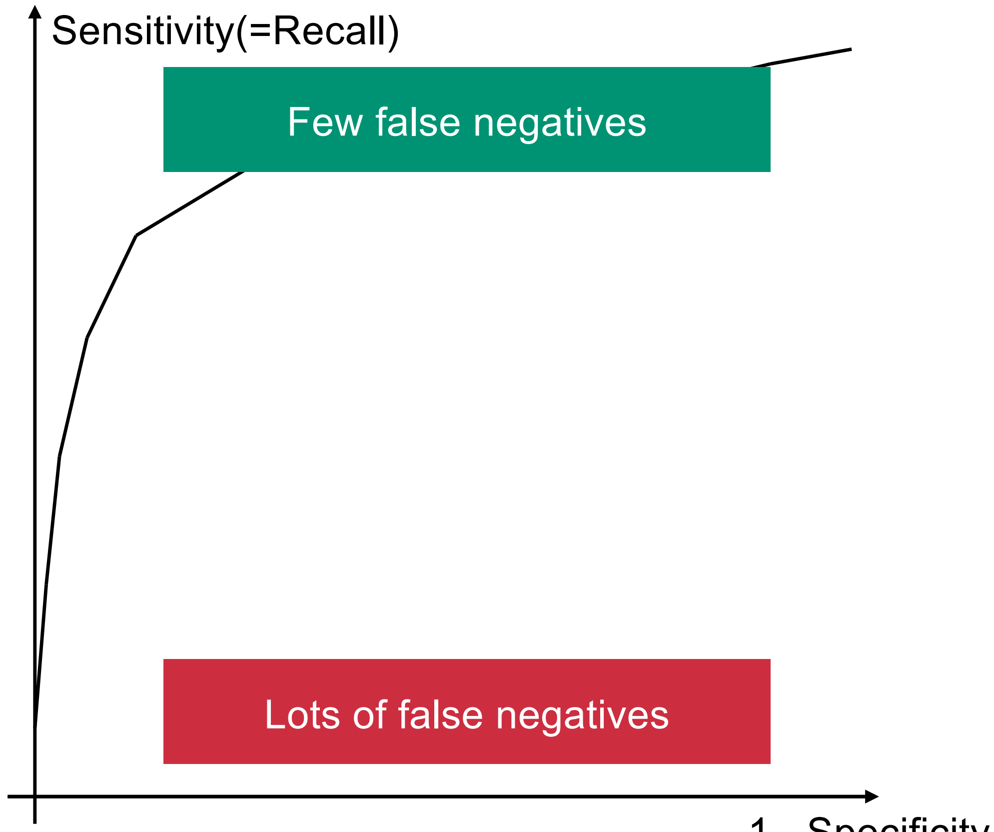
## Recall vs. Sensitivity (ROC curve)



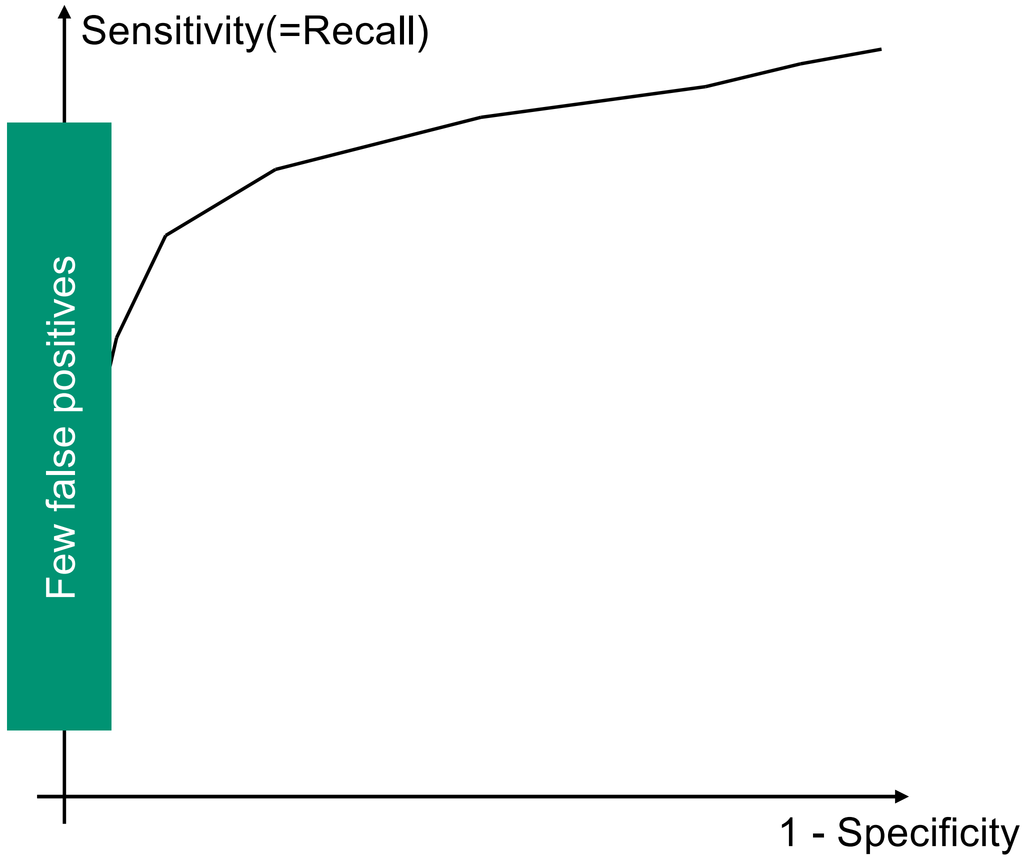
# Recall vs. Sensitivity (ROC curve)



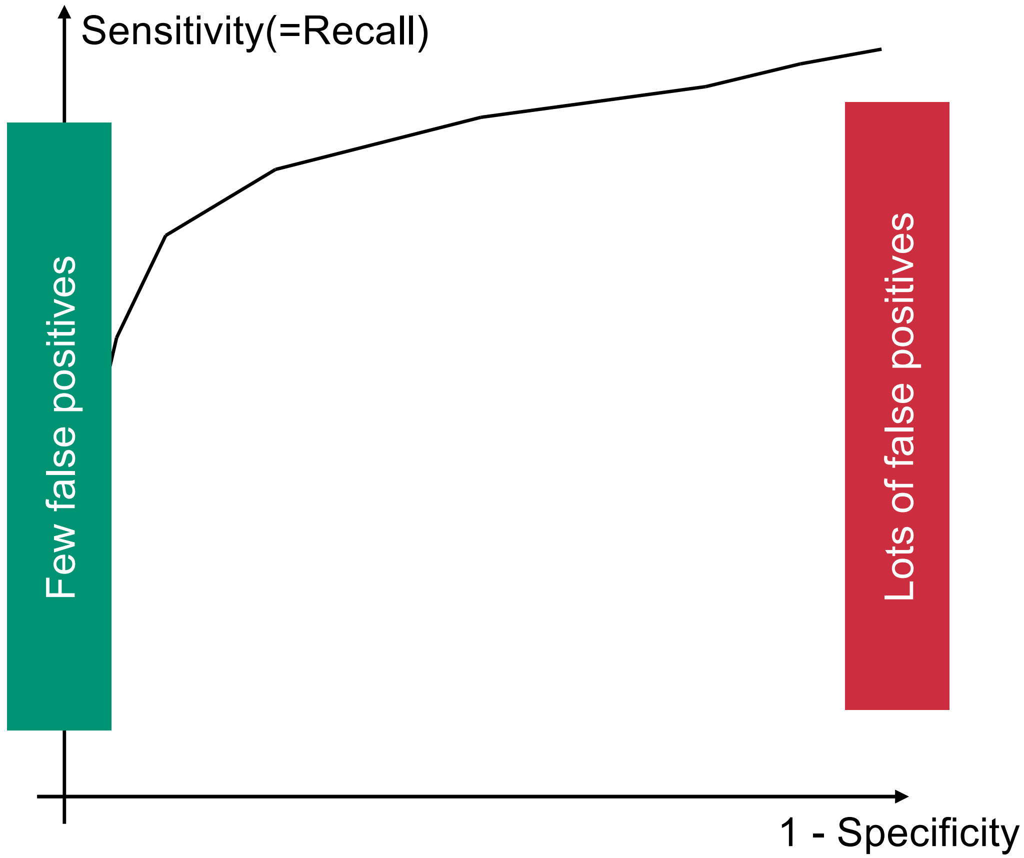
# Recall vs. Sensitivity (ROC curve)



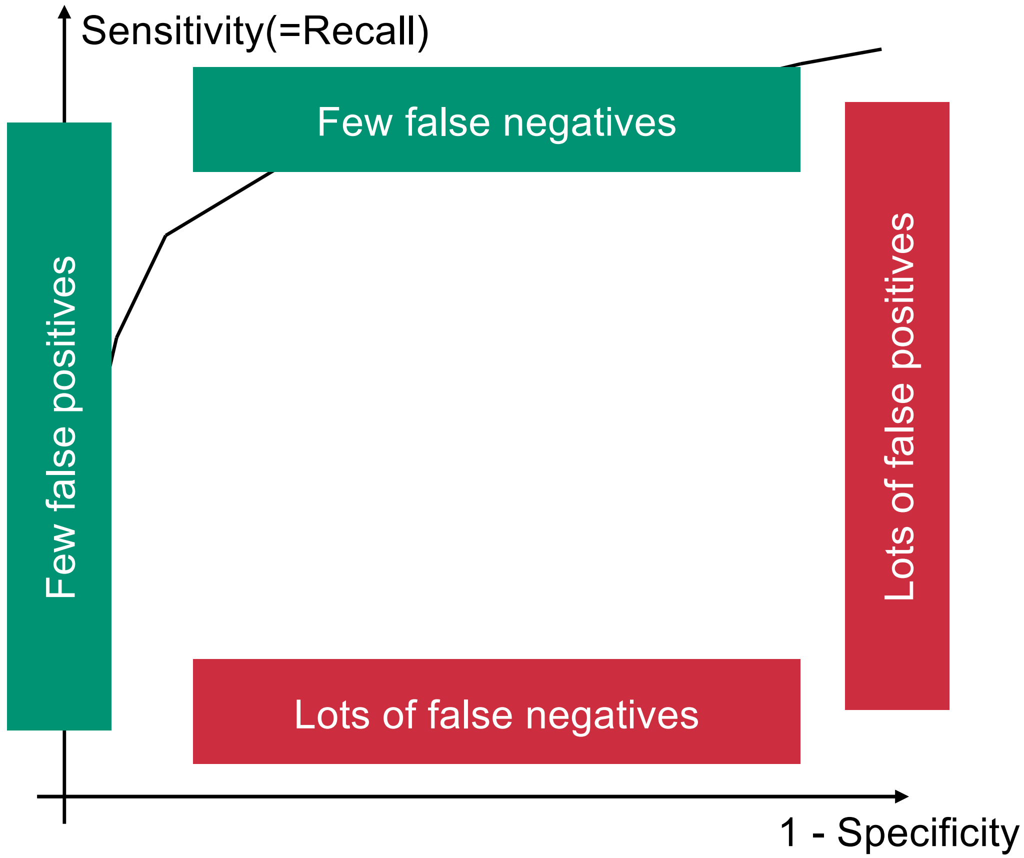
# Recall vs. Sensitivity (ROC curve)



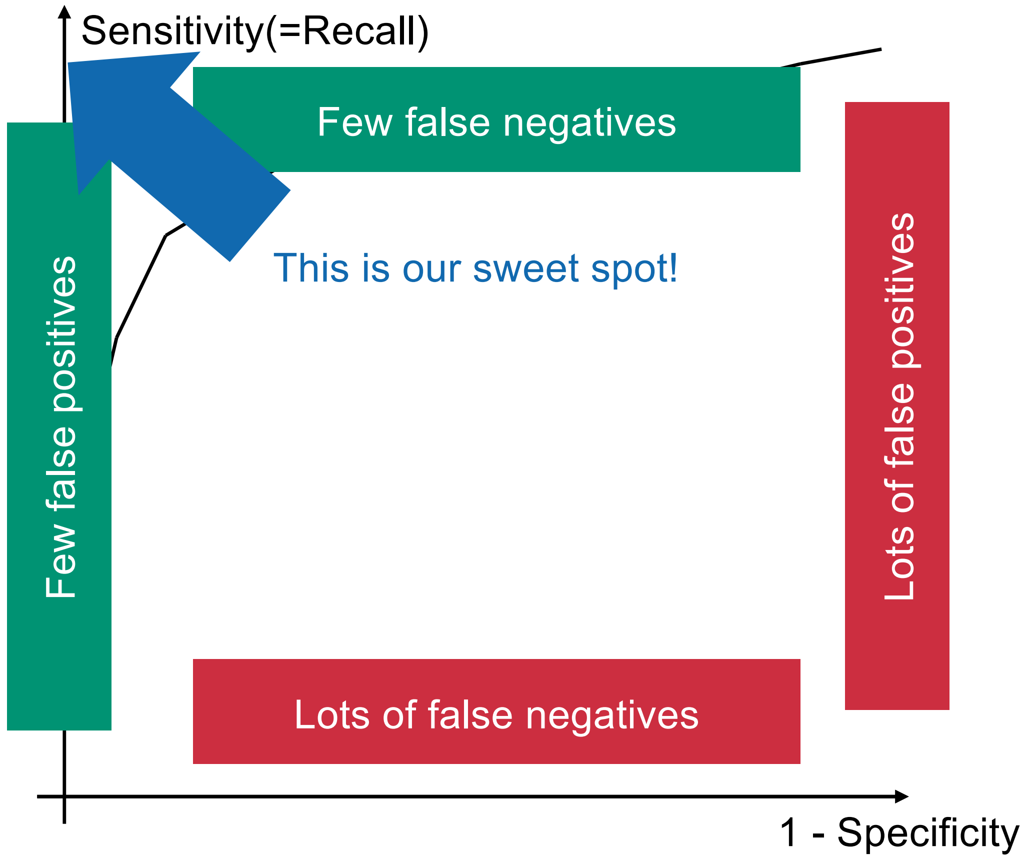
# Recall vs. Sensitivity (ROC curve)



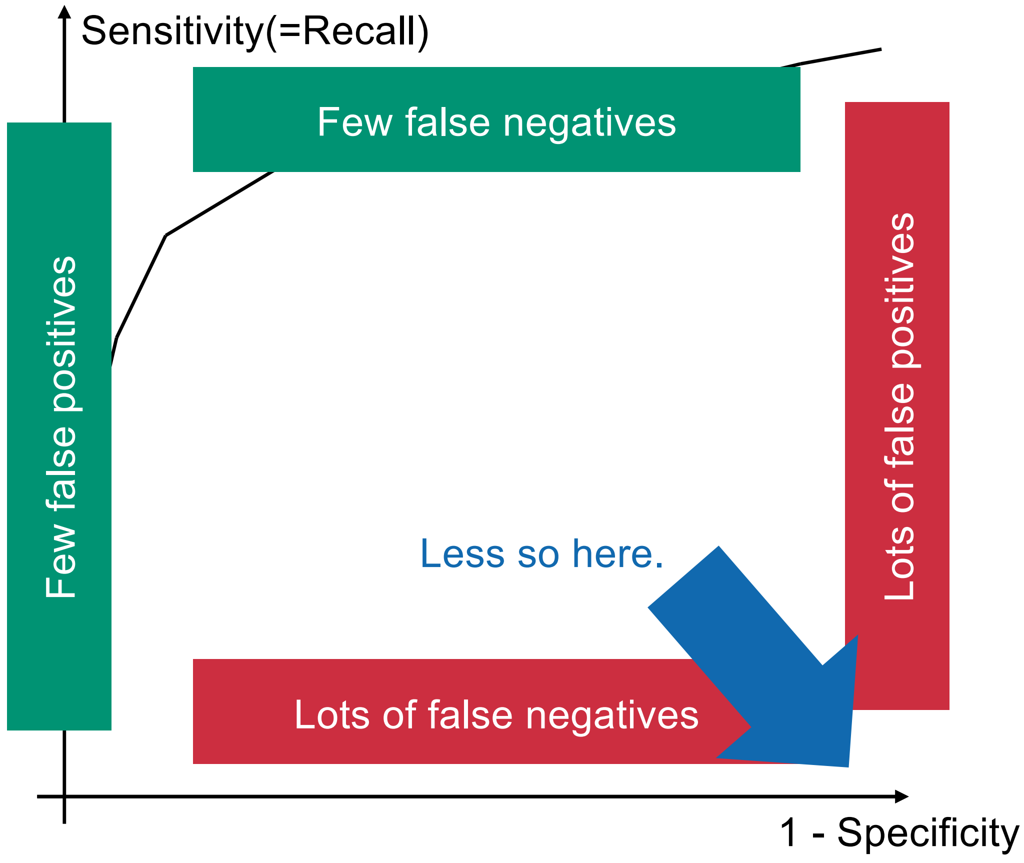
# Recall vs. Sensitivity (ROC curve)



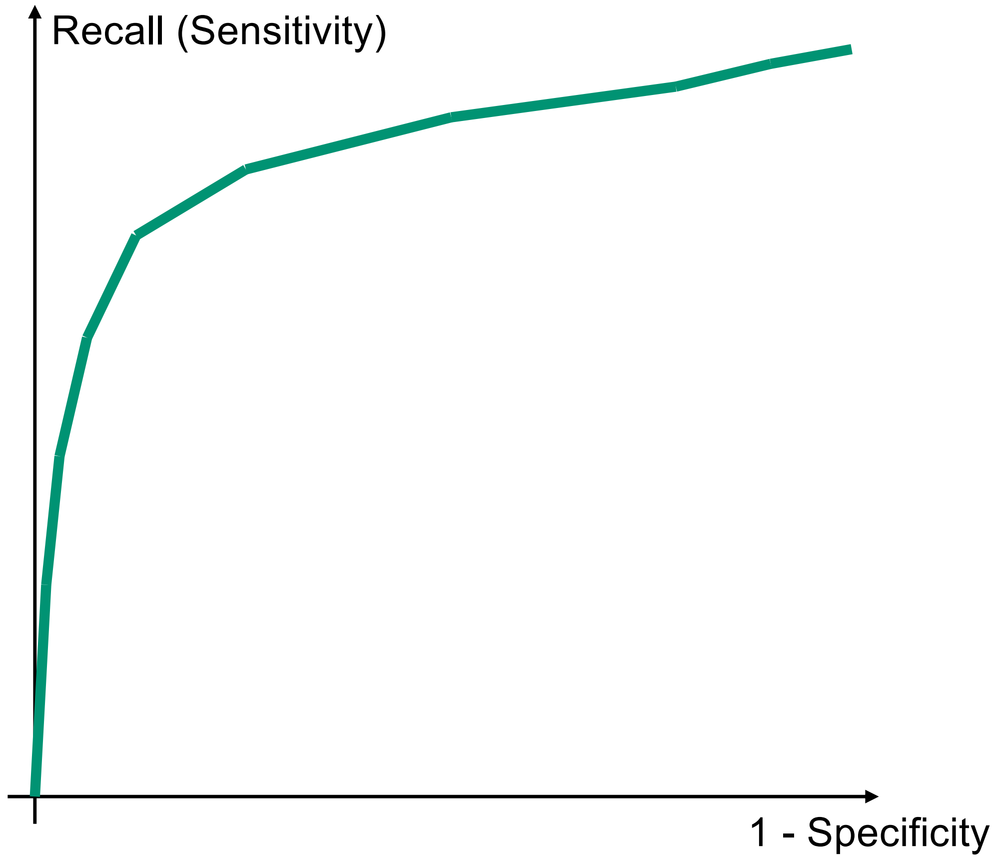
# Recall vs. Sensitivity (ROC curve)



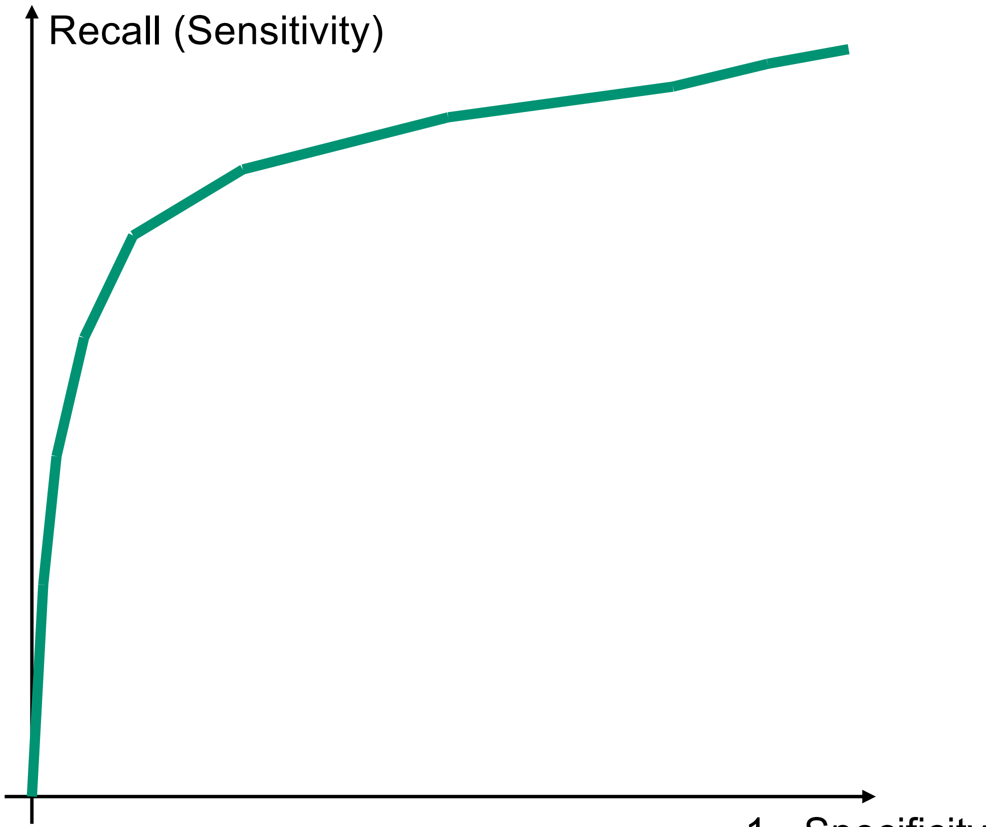
# Recall vs. Sensitivity (ROC curve)



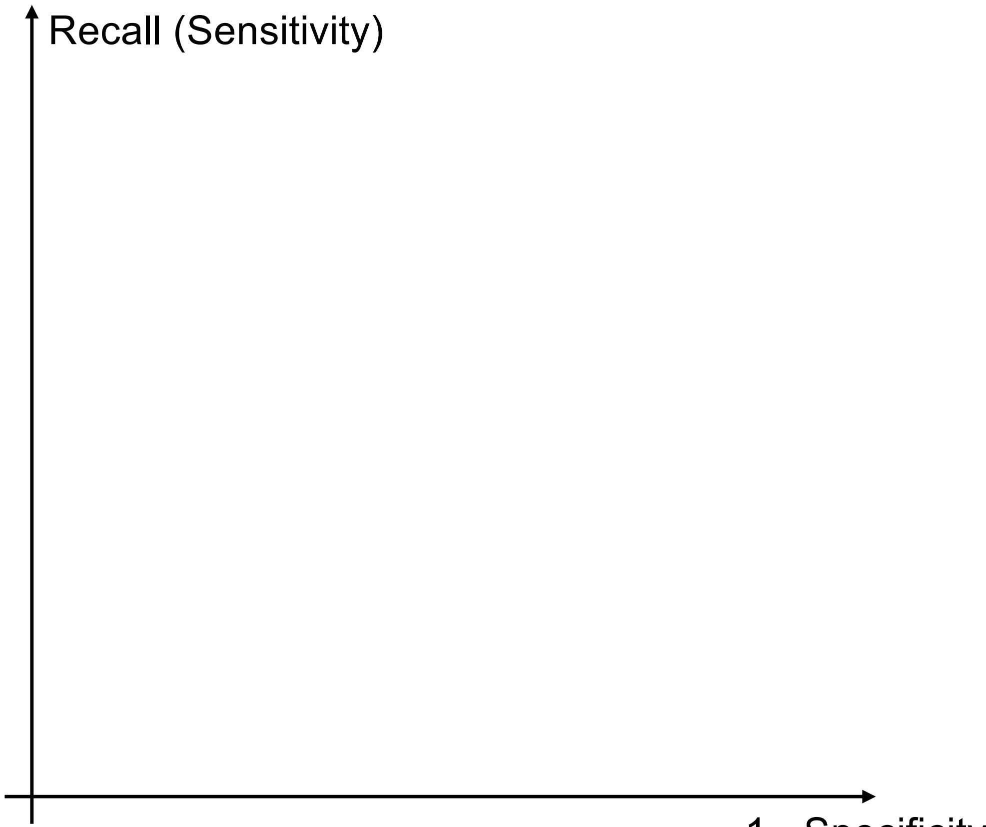
How good is this?



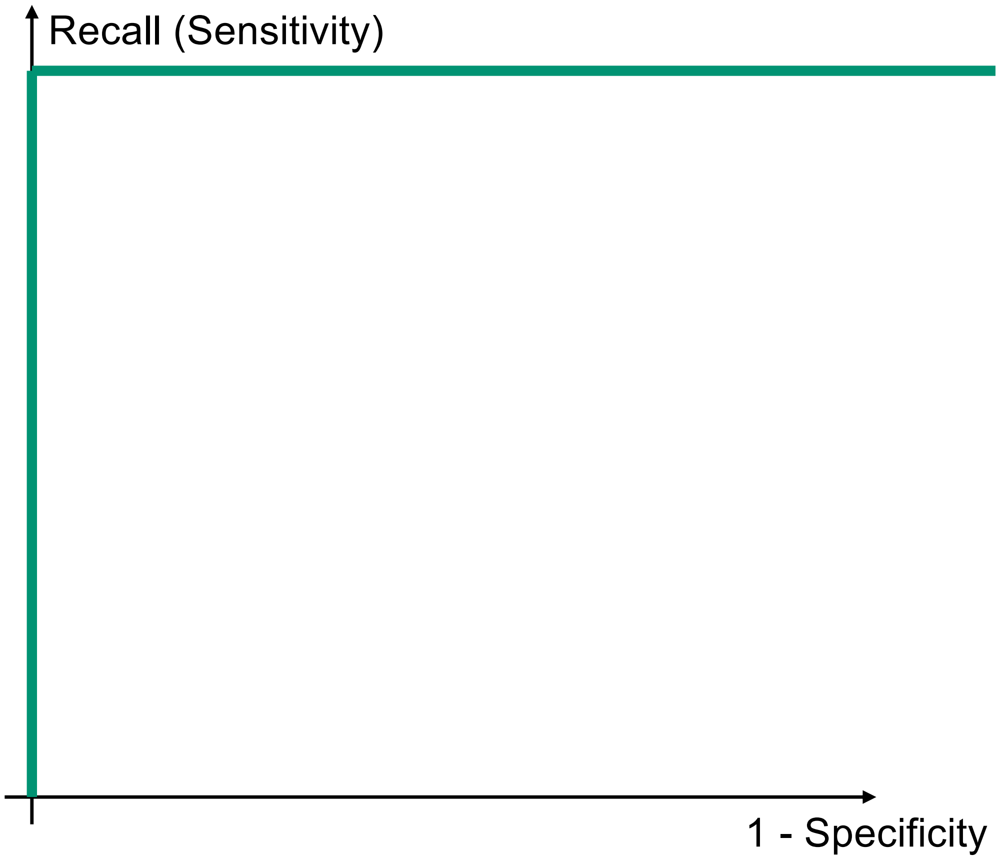
# "Fine" Search Engine



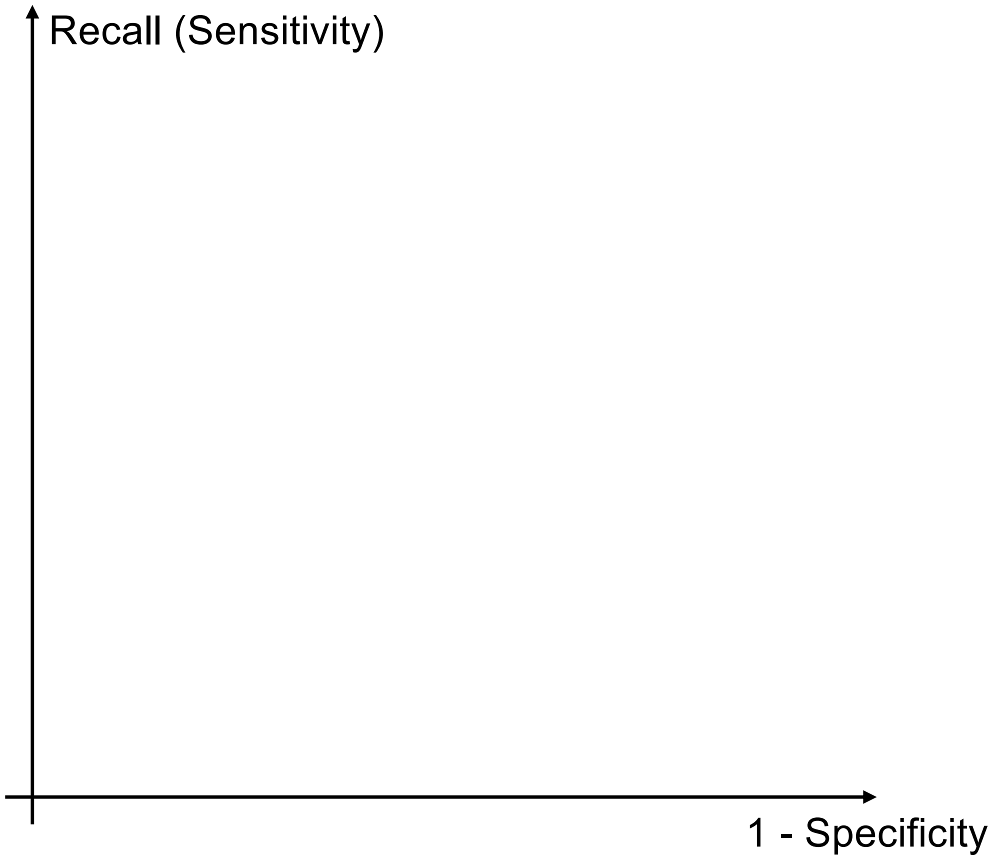
# Awesome search engine



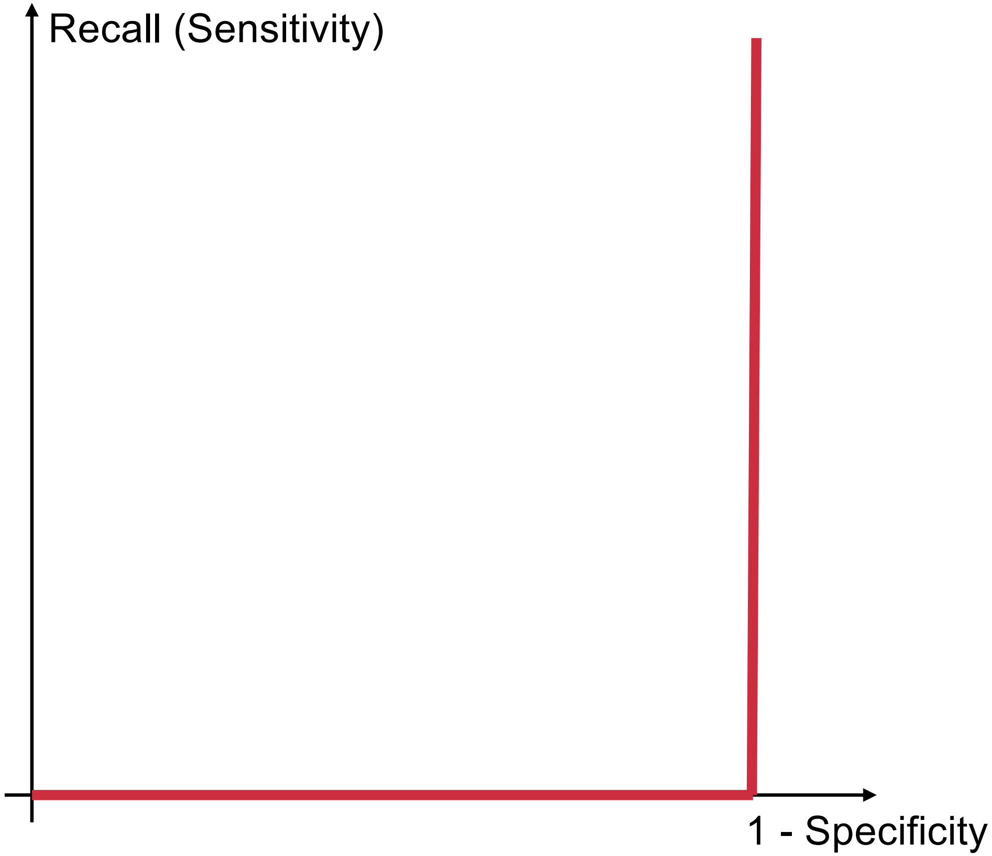
# Awesome search engine



# Awful search engine

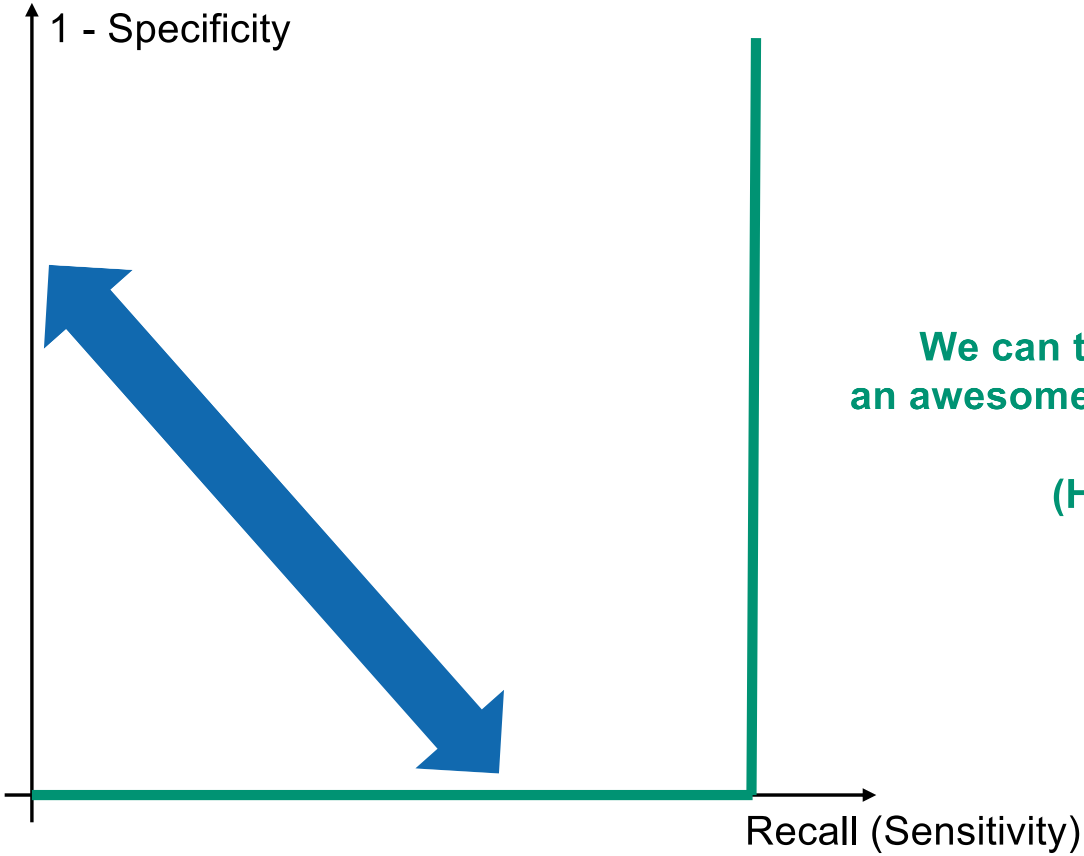


# Awful search engine




**Is it actually THAT awful?**

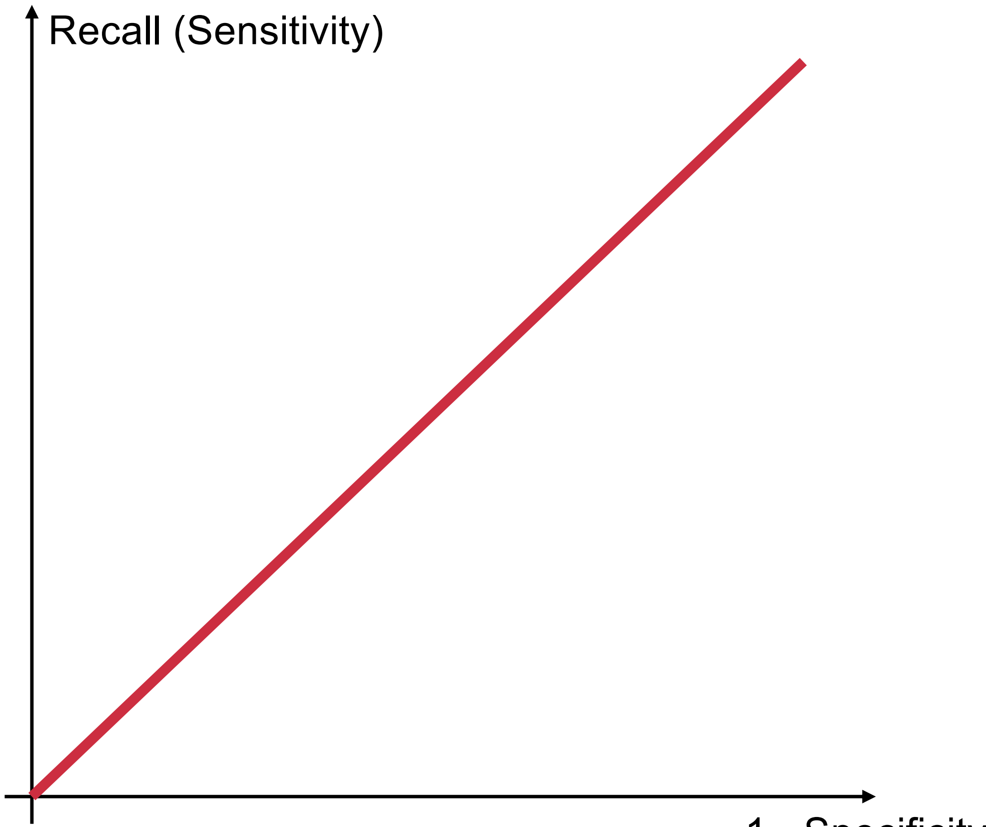
# Awful search engine




**We can turn this into an awesome search engine!**

**(How?)**

# (Really) awful search engine



Normalized Discouted Cumulative Gain (NDCG)

Relevance score (not just 0 or 1)

$$\frac{1}{|Q|}\sum_{q\in Q} Z_q \sum_{k=1..m_q} \frac{2^{P_{qk}}-1}{1+k}$$

(for Machine Learning aficionados)

# Other metrics

Speed of index construction 

Speed of search 

Expressiveness of the query language !

Size of collection